



Learning Notes

Module 1 — AI Fundamentals

Student • Interview • Practical Learning Edition

Prepared for structured AI/ML learning



Module 1 — AI Fundamentals

This module builds the **foundation of Artificial Intelligence, Machine Learning, Deep Learning, Generative AI, and LLMs**. The explanations below are designed for students and interviews.

1. What is Artificial Intelligence?

Concept

Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines and software that can perform tasks that normally require **human intelligence**.

These tasks include:

- Learning from data
- Understanding language
- Recognizing images
- Solving problems
- Making decisions
- Predicting outcomes
- Understanding speech
- Generating text, images, audio, and code

Simple Explanation

Think of AI as giving a computer the ability to perform an intelligent task.

Traditional programming:

Rules + Data

↓

Program

↓

Output

Machine Learning:

Data + Expected Output

↓

ML Algorithm

↓

Model

↓

New Data → Prediction

Real-World Examples

Application	AI Task
Google Maps	Route optimization
Netflix	Recommendation
Face Unlock	Face recognition
Siri/Alexa	Speech understanding
ChatGPT	Language generation
Fraud detection	Suspicious transaction detection
Self-driving systems	Environment perception and decision-making

Key Point

AI is the broader field; ML, Deep Learning, and Generative AI are approaches/subfields within the broader AI ecosystem.

Interview Questions

1. What is Artificial Intelligence?
2. What are the major applications of AI?
3. What is the difference between AI and traditional programming?
4. Give three real-world examples of AI.

Practice Questions

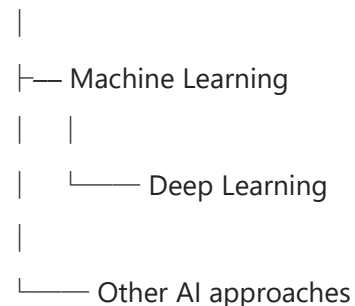
1. Define AI in your own words.
2. Identify five AI applications you use every day.
3. Explain the difference between rule-based software and AI-based systems.

2. AI vs ML vs Deep Learning

Concept

These terms are related but **not identical**.

Artificial Intelligence



Artificial Intelligence

AI is the **largest concept**.

It aims to make machines perform tasks that require intelligence.

Machine Learning

Machine Learning (ML) is a method of AI where systems **learn patterns from data** instead of being explicitly programmed with every rule.

Example:

House Data

↓

ML Algorithm

↓

Trained Model

↓

House Price Prediction

Deep Learning

Deep Learning is a subset of ML that uses **multi-layer neural networks** to learn complex patterns.

It is particularly useful for:

- Images
- Video
- Speech
- Natural language
- Complex pattern recognition

Comparison

Feature	AI	ML	Deep Learning
Meaning	Broad field	AI learning approach	ML using deep neural networks
Data requirement	Varies	Usually required	Often large amounts
Human feature engineering	May be high/low	Often important	Often reduced
Example	Expert system	Spam classifier	Image recognition
Technology	Broad	Algorithms	Neural networks

Real-World Example

Suppose we want to identify cats in images.

AI:

The overall goal is intelligent image understanding.

ML:

Train a model using examples of cats and non-cats.

Deep Learning:

Use a CNN to automatically learn visual features such as edges, shapes, and patterns.

Interview Question

Q: Is Deep Learning a part of AI?

Yes.

AI

↓

ML

↓

Deep Learning

3. History of AI

AI has developed through several important stages.

1940s–1950s: Early Foundations

Researchers began exploring whether machines could imitate human reasoning.

In 1950, **Alan Turing** proposed the famous **Turing Test**, which explored whether a machine could exhibit behavior indistinguishable from human intelligence in conversation.

1956: Birth of AI as a Field

The **Dartmouth workshop** in 1956 is commonly associated with the formal beginning of AI as an academic field.

The term **Artificial Intelligence** was associated with **John McCarthy**.

1960s–1970s

Researchers developed:

- Rule-based systems
- Search algorithms
- Early natural-language systems
- Problem-solving programs

1980s

Expert systems became popular.

They used:

Knowledge + Rules

↓

Expert System

↓

Decision

1990s–2000s

Machine Learning became increasingly important.

Systems began learning statistical patterns from data rather than relying entirely on manually written rules.

2010s

Deep Learning experienced major growth because of:

- Large datasets
- GPUs
- Improved algorithms
- Increased computing power

Deep learning achieved major advances in:

- Computer vision
- Speech recognition
- NLP

2020s

Generative AI and large language models became major areas of AI development.

Examples include systems capable of generating:

- Text
- Images
- Code
- Audio
- Video

Interview Question

Q: Why did AI grow rapidly after 2010?

Major factors included **more data, powerful GPUs, improved algorithms, cloud computing, and advances in deep learning.**

4. Types of AI

AI can be classified in different ways. A common classification based on capability is:

1. Narrow AI / ANI

Artificial Narrow Intelligence (ANI) is designed to perform a specific task or a limited range of tasks.

Examples:

- Spam detection
- Recommendation systems
- Face recognition
- Voice assistants
- Fraud detection

Most practical AI systems today fall into this category.

2. General AI / AGI

Artificial General Intelligence (AGI) refers to a hypothetical system capable of performing a broad range of intellectual tasks with general-purpose ability comparable to humans.

AGI remains a research goal rather than an established real-world technology.

3. Super AI / ASI

Artificial Superintelligence (ASI) refers to a hypothetical intelligence that would surpass human intelligence across essentially all intellectual domains.

It remains speculative.

Important Distinction

ANI → Exists today

AGI → Hypothetical / research goal

ASI → Hypothetical

5. AI Capabilities

AI systems can perform many different types of tasks.

1. Learning

AI can learn patterns from data.

Example:

Historical Sales Data

↓

ML

↓

Future Sales Prediction

2. Reasoning

AI can use information and rules to reach conclusions.

3. Perception

AI can process information from:

- Images
- Video
- Audio
- Sensors

4. Natural Language Understanding

AI can process human language.

Examples:

- Chatbots
- Translation
- Sentiment analysis
- Question answering

5. Prediction

AI can estimate future or unknown outcomes.

Examples:

- Weather prediction
- Customer churn
- Demand forecasting
- Credit risk

6. Generation

Generative AI can create new content.

Examples:

- Text
- Images
- Code
- Audio
- Video

6. Real-World Applications of AI

Healthcare

AI can assist with:

- Medical image analysis
- Drug discovery
- Patient-risk prediction
- Clinical decision support

Banking & Finance

AI is used for:

- Fraud detection
- Credit scoring

- Risk analysis
- Algorithmic trading

Education

Applications include:

- Personalized learning
- AI tutors
- Automated assessment
- Question generation

E-Commerce

AI powers:

- Product recommendations
- Search ranking
- Customer support
- Demand forecasting

Transportation

AI is used for:

- Route optimization
- Traffic prediction
- Driver-assistance systems
- Autonomous-driving research

Cybersecurity

AI can help detect:

- Suspicious behavior
- Malware
- Network anomalies
- Fraudulent activity

Entertainment

Examples:

- Netflix recommendations
- Music recommendations
- Content generation
- Game AI

7. Advantages & Limitations of AI

Advantages

1. Automation

AI can automate repetitive tasks.

2. Speed

Computers can process large amounts of information quickly.

3. Pattern Detection

AI can discover patterns that may be difficult to identify manually.

4. 24/7 Availability

Software systems can operate continuously.

5. Personalization

AI can provide customized recommendations and experiences.

6. Scalability

One AI system can potentially serve a very large number of users.

Limitations

1. Data Dependency

Many ML systems depend heavily on the quality and quantity of training data.

2. Bias

Biased training data can lead to biased predictions.

3. Lack of Explainability

Some complex models can be difficult to interpret.

4. Hallucinations

Generative AI systems can sometimes produce information that sounds convincing but is incorrect.

5. Privacy

AI systems may involve sensitive or personal data.

6. Cost

Large AI models can require significant computing resources.

7. Human Oversight

Important decisions may still require human review and judgment.

Interview Question

Q: Can AI completely replace humans?

A better answer is:

AI can automate many tasks, but complete replacement depends on the task. Human judgment, domain expertise, accountability, creativity, communication, and ethical decision-making remain important in many situations.

8. Generative AI

Concept

Generative AI refers to AI systems that can generate new content based on patterns learned from data.

Traditional AI

Often focuses on:

Input

↓

Analysis / Prediction

↓

Output

Example:

Email → Spam / Not Spam

Generative AI

Produces new content:

Prompt

↓

Generative AI Model

↓

Generated Content

Examples

Type	Generated Content
Text generation	Articles, answers
Image generation	Images, illustrations
Code generation	Programs
Audio generation	Speech/music
Video generation	Videos

Generative AI Workflow

User Prompt

↓

AI Model

↓

Pattern Processing

↓

Generated Output

Important Concepts

Generative AI commonly involves:

- Neural networks
- Transformers
- Large datasets
- Foundation models
- LLMs
- Diffusion models
- Embeddings

Real-World Example

A user asks:

"Explain machine learning to a beginner."

A generative AI model generates a new response based on patterns learned during training.

Interview Questions

1. What is Generative AI?
2. How is Generative AI different from traditional predictive AI?
3. What can Generative AI generate?
4. What is a foundation model?
5. What are hallucinations?

9. LLM Basics

What is an LLM?

LLM = Large Language Model

An LLM is a machine-learning model trained on large amounts of text and other data to perform language-related tasks.

LLMs can be used for:

- Question answering

- Text generation
- Summarization
- Translation
- Classification
- Coding assistance
- Information extraction
- Conversational applications

Basic LLM Flow

User Prompt

↓

Tokenization

↓

Language Model

↓

Next-token prediction

↓

Generated Response

What is a Token?

A **token** is a unit of text processed by a language model.

Depending on the tokenizer, a token may represent:

- A word
- Part of a word
- Punctuation
- Other text units

For example:

"I love AI"

may be divided into several tokens depending on the tokenizer being used.

How Does an LLM Generate Text?

At a simplified level, the model predicts likely next tokens based on the preceding context.

The sky is

↓

blue

The model assigns probabilities to possible next tokens and generates a sequence according to its decoding strategy.

Transformer Architecture

Modern LLMs are largely based on the **Transformer architecture**.

Important concepts include:

- Attention
- Self-attention
- Positional information
- Encoder/decoder architectures
- Transformer blocks

These will be covered in detail in **Module 13 — Transformers & Generative AI**.

Training vs Inference

Training:

Large Dataset

↓

Training

↓

Model Parameters

Inference:

User Prompt

↓

Trained Model

↓

Generated Response

LLM Limitations

LLMs can:

- Generate incorrect information
- Misinterpret prompts
- Produce biased responses
- Have limited knowledge depending on training/data access
- Generate plausible but unsupported statements

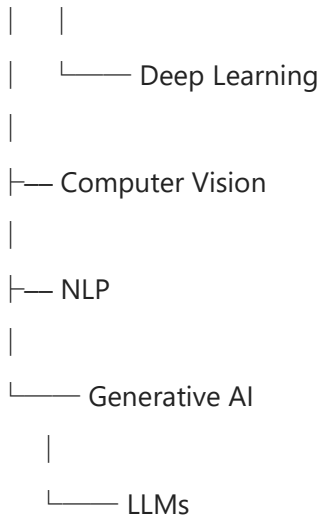
Therefore, important applications may require **retrieval, validation, monitoring, and human oversight**.

Module 1 Quick Revision

AI

|

└─ Machine Learning



Important Definitions

Term	One-Line Definition
AI	Field of creating systems capable of intelligent behavior
ML	AI approach where models learn patterns from data
Deep Learning	ML based on multi-layer neural networks
Generative AI	AI that generates new content
LLM	Large model trained to process and generate language
NLP	AI field focused on human language
Computer Vision	AI field focused on images and video
AGI	Hypothetical general-purpose artificial intelligence
ASI	Hypothetical intelligence exceeding humans broadly

Module 1 Interview Questions

1. What is Artificial Intelligence?
2. What is the difference between AI, ML, and Deep Learning?
3. What are the different types of AI?
4. What is Narrow AI?
5. What is AGI?
6. What is Generative AI?
7. What is an LLM?
8. What is a token?
9. What is NLP?
10. What is Computer Vision?
11. What are the advantages of AI?
12. What are the limitations of AI?
13. How does Machine Learning differ from traditional programming?
14. Why has Deep Learning become so successful?

15. What is the difference between AI and Generative AI?

Practice Assignment

Beginner

1. Write 10 examples of AI applications around you.
2. Explain AI vs ML vs DL in your own words.
3. Create a table comparing Narrow AI, AGI, and ASI.

Intermediate

4. Explain how an AI chatbot works at a high level.
5. Research and explain five applications of Generative AI.
6. Explain the difference between training and inference in an LLM.

Mini Project

Build an AI Application Concept

Choose one domain—**Education, Healthcare, Banking, E-commerce, or Customer Support**—and document:

Problem → Data → AI Approach → Expected Output → Benefits → Limitations → Ethical Considerations.



Module 2 — Mathematics for AI/ML

Mathematics is the foundation of Machine Learning. You do **not** need advanced mathematics initially, but you should understand the concepts well enough to interpret algorithms, loss functions, model predictions, and optimization.

For every topic, we will follow:

Concept → Simple Explanation → Real-World Example → Formula → Python Code → Output → Interview Questions → Practice Questions

1. Number Systems

Concept

A **number system** is a way of representing numbers.

For AI/ML, the most important number types are:

- Natural numbers
- Integers
- Rational numbers
- Real numbers
- Binary numbers

- Decimal numbers

1. Natural Numbers

1, 2, 3, 4, 5, ...

Used for counting.

2. Integers

..., -3, -2, -1, 0, 1, 2, 3, ...

3. Rational Numbers

Numbers that can be represented as:

[
 $\frac{p}{q}, \quad q \neq 0$
]

Example:

[
 $\frac{1}{2} = 0.5$
]

4. Real Numbers

Real numbers include both rational and irrational numbers.

Examples:

5

-2

0.75

$\sqrt{2}$

π

5. Binary Numbers

Computers fundamentally represent information using **0 and 1**.

Example:

Decimal 5 = Binary 101

Why is this important in AI/ML?

Computers represent data numerically.

For example:

Image

↓

Pixels

↓

Numbers

↓

ML Model

A grayscale pixel can commonly be represented using a value from **0 to 255**.

2. Linear Algebra

Concept

Linear algebra is the mathematics of vectors, matrices, and linear transformations.

It is extremely important in ML because datasets and model parameters are commonly represented using vectors and matrices.

Simple Example

Suppose we have information about three students:

Student Hours Studied Attendance

A	5	80
B	7	90
C	3	70

We can represent the features as a matrix:

[
X =
 $\begin{bmatrix}$
5 & 80\
7 & 90\
3 & 70
 $\end{bmatrix}$
]

Machine-learning algorithms operate on these numerical representations.

Applications

Linear algebra is used in:

- Linear Regression
 - Neural Networks
 - Computer Vision
 - PCA
 - Recommendation systems
 - Transformers
 - Deep Learning
-

3. Vectors & Matrices

Vector

A **vector** is an ordered collection of numbers.

Example:

```
[
v =
\begin{bmatrix}
10\
20\
30
\end{bmatrix}
]
```

This is a 3-dimensional vector.

Real-World Example

A house could be represented as:

```
[
X =
[1500,\ 3,\ 2]
]
```

where:

1500 → area

3 → bedrooms

2 → bathrooms

The vector represents the house using numerical features.

Matrix

A **matrix** is a rectangular arrangement of numbers.

Example:

```
[
A =
\begin{bmatrix}
1 & 2 & 3\
4 & 5 & 6
\end{bmatrix}
]
```

This matrix has:

- 2 rows
- 3 columns

Therefore its shape is:

```
[
2 \times 3
]
```

ML Example

A dataset can be represented as:

```
[
X =
\begin{bmatrix}
x_{11} & x_{12} & x_{13} \\
x_{21} & x_{22} & x_{23} \\
x_{31} & x_{32} & x_{33}
\end{bmatrix}
]
```

Usually:

- Rows → observations/samples
- Columns → features

4. Matrix Operations

Matrix operations are heavily used in ML.

A. Matrix Addition

Matrices must have the same dimensions.

```
[
A=
\begin{bmatrix}
1&2 \\
3&4
\end{bmatrix}
]
```

```
[
B=
\begin{bmatrix}
5&6 \\
7&8
\end{bmatrix}
]
```

Then:

```
[
A+B=
\begin{bmatrix}
6&8 \\
10&12
\end{bmatrix}
]
```

$$\end{bmatrix}$$

B. Matrix Subtraction

$$A - B = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}$$

C. Scalar Multiplication

Multiply every element by a number.

$$2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

D. Matrix Multiplication

If:

$$A_{m \times n}$$

and

$$B_{n \times p}$$

then:

$$AB_{m \times p}$$

For example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

```
\end{bmatrix}
]
[
B=
\begin{bmatrix}
5&6\
7&8
\end{bmatrix}
]
```

Then:

```
[
AB=
\begin{bmatrix}
19&22\
43&50
\end{bmatrix}
]
```

Why important in ML?

Neural networks perform many operations that can be represented as:

```
[
Z = WX+b
]
```

where:

- (X) = input
- (W) = weights
- (b) = bias
- (Z) = resulting value

Python Example

```
import numpy as np

A = np.array([[1, 2],
              [3, 4]])

B = np.array([[5, 6],
              [7, 8]])

print("Addition:")
print(A + B)

print("Multiplication:")
print(A @ B)
```

Output

Addition:

[[6 8]

[10 12]]

Multiplication:

[[19 22]

[43 50]]

Interview Questions

1. What is a vector?
2. What is a matrix?
3. What is the difference between a vector and matrix?
4. What is matrix multiplication?
5. What is the shape of a matrix?
6. Why is linear algebra important in ML?

5. Statistics

Concept

Statistics is the process of collecting, organizing, analyzing, and interpreting data.

In ML, statistics helps us understand datasets before building models.

Statistics is used for:

- Understanding data
- Finding averages
- Measuring variation
- Detecting outliers
- Understanding distributions
- Evaluating models
- Making predictions

Two Major Types

Descriptive Statistics

Describes existing data.

Examples:

- Mean
- Median
- Mode

- Variance
- Standard deviation

Inferential Statistics

Uses a sample to make conclusions about a larger population.

6. Mean, Median & Mode

Consider:

10, 20, 20, 30, 40

Mean

Mean is the average.

```
[
Mean=\frac{\sum x}{n}
]
```

Here:

```
[
Mean=\frac{10+20+20+30+40}{5}=24
]
```

```
genui {"descriptive_statistics_sampling":{"type_id":"ARITHMETIC_MEAN","locale_override":"en-IN"}}
```

Median

Median is the middle value after sorting.

10, 20, 20, 30, 40

↑

Median

Therefore:

```
[
Median=20
]
```

Mode

Mode is the most frequently occurring value.

Here:

```
[
Mode=20
]
```

Real-World Example

Suppose salaries are:

20k, 22k, 25k, 27k, 2 lakh

The very high salary can significantly increase the **mean**, while the median may better represent the typical salary.

Interview Question

Q: Which is more resistant to outliers, mean or median?

Answer: Median.

7. Variance & Standard Deviation

These measure how spread out the data is.

Consider:

10, 20, 30

Mean:

[
 $\mu = 20$
]

Differences from mean:

$10 - 20 = -10$

$20 - 20 = 0$

$30 - 20 = 10$

We square them:

100, 0, 100

Variance

Population variance:

[
 $\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$
]

For the example:

[
 $\sigma^2 = \frac{100 + 0 + 100}{3}$
]

[
 $\sigma^2 = 66.67$
]

Standard Deviation

Standard deviation is the square root of variance.

```
[
\sigma=\sqrt{\sigma^2}
]
```

So:

```
[
\sigma\approx 8.16
]
```

```
genui {"descriptive_statistics_sampling":{"type_id":"STANDARD_DEVIATION","locale_override":"en-IN"}}
```

Simple Understanding

Small standard deviation

→ Data points are close together

Large standard deviation

→ Data points are more spread out

ML Importance

Standard deviation is important for:

- Feature scaling
- Standardization
- Detecting unusual observations
- Understanding distributions

8. Probability

Concept

Probability measures how likely an event is to happen.

```
[
0 \leq P(A) \leq 1
]
```

where:

0 → impossible

1 → certain

Formula

For equally likely outcomes:

```
[
P(A)=\frac{\text{Favorable outcomes}}{\text{Total outcomes}}
]
```

Example

For a fair coin:

Heads

Tails

Probability of heads:

```
[
P(H)=\frac{1}{2}=0.5
]
```

or:

```
[
50%
]
```

ML Applications

Probability is important in:

- Naive Bayes
- Logistic Regression
- Classification
- Generative models
- Uncertainty estimation
- Recommendation systems

9. Conditional Probability

Concept

Conditional probability is the probability of event **A occurring given that B has already occurred**.

Notation:

```
[
P(A|B)
]
```

Read as:

Probability of A given B.

Formula

```
[
P(A|B)=
\frac{P(A\cap B)}
{P(B)}
]
```

```
genui  {"probability_combinatorics":{"type_id":"CONDITIONAL_PROBABILITY_DEFINITION","locale_ove
rride":"en-IN"}}
```

Real-World Example

Suppose:

- A = student passed
- B = student attended classes regularly

Then:

[
P(A|B)
]

means:

Probability that a student passes given that the student attended classes regularly.

ML Example

In classification:

[
P(Class|Features)
]

means the probability of a particular class given the observed features.

10. Bayes Theorem

Concept

Bayes theorem helps us update the probability of an event when new evidence becomes available.

Formula

[
P(A|B)=
 $\frac{P(B|A)P(A)}{P(B)}$
]

genui {"probability_combinatorics":{"type_id":"BAYES_THEOREM","locale_override":"en-IN"}}

Where:

- (P(A|B)) → posterior probability
- (P(B|A)) → likelihood
- (P(A)) → prior probability
- (P(B)) → evidence

Simple Example

Suppose an email contains certain words commonly associated with spam.

A spam classifier can estimate:

[
P(Spam|Words)
]

Using information about:

- How common spam is
- How frequently those words appear in spam
- How frequently they appear in normal emails

Applications

Bayes theorem is important in:

- Naive Bayes
- Spam detection
- Medical diagnosis
- Risk analysis
- Recommendation systems
- Probabilistic inference

Interview Question

Q: What is the difference between $P(A|B)$ and $P(B|A)$?

They are generally **not equal**.

For example:

[
 $P(\text{Disease}|\text{Positive})$
]

is different from:

[
 $P(\text{Positive}|\text{Disease})$
]

This distinction is extremely important when interpreting diagnostic tests.

11. Derivatives & Gradients

This is one of the most important mathematical concepts for **Deep Learning and optimization**.

Derivative

A derivative tells us how quickly a function changes with respect to its input.

For example:

[
 $y=x^2$
]

The derivative is:

[
 $\frac{dy}{dx}=2x$
]

At:

[
 $x=3$
]

the derivative is:

[
 $2(3)=6$
]

Simple Understanding

Think of a derivative as the **slope** of a function at a particular point.

Positive derivative

→ Function increasing

Negative derivative

→ Function decreasing

Zero derivative

→ Flat point

Why Derivatives Matter in ML

Machine Learning models have parameters/weights.

We want to find parameters that minimize the model's error.

Model

↓

Prediction

↓

Loss

↓

Derivative

↓

Update weights

↓

Lower loss

Gradient

A **gradient** is a vector of derivatives with respect to multiple variables.

Suppose:

```
[
f(x,y)
]
```

Then its gradient is:

```
[
\nabla f =
\begin{bmatrix}
\frac{\partial f}{\partial x}\
\frac{\partial f}{\partial y}
\end{bmatrix}
]
```

Simple Example

If:

```
[
f(x,y)=x^2+y^2
]
```

then:

```
[
\frac{\partial f}{\partial x}=2x
]
```

and:

```
[
\frac{\partial f}{\partial y}=2y
]
```

Therefore:

```
[
\nabla f =
\begin{bmatrix}
2x\
2y
\end{bmatrix}
]
```

Gradient Descent

Machine Learning commonly uses gradient-based optimization.

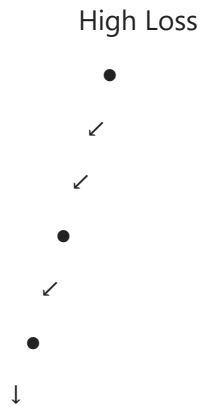
A simplified update rule is:

```
[
\theta_{new}
\theta_{old}
\eta \nabla J(\theta)
]
```

where:

- θ = model parameters
- η = learning rate
- $J(\theta)$ = loss/cost function
- $\nabla J(\theta)$ = gradient

Intuition



The algorithm tries to move toward a region where the loss is smaller.

Real-World ML Example

During neural-network training:

Input



Neural Network



Prediction



Loss



Backpropagation



Gradients



Update Weights



Repeat

This process allows the neural network to learn.

Python Practice

Mean, Median & Mode


```
import numpy as np
from statistics import mode

data = [10, 20, 20, 30, 40]

print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Mode:", mode(data))
```

Output

Mean: 24.0

Median: 20.0

Mode: 20

Variance & Standard Deviation

```
import numpy as np

data = [10, 20, 30]

print("Variance:", np.var(data))
print("Standard Deviation:", np.std(data))
```

Output

Variance: 66.66666666666667

Standard Deviation: 8.16496580927726

Matrix Operations

```
import numpy as np

A = np.array([
    [1, 2],
    [3, 4]
])

B = np.array([
    [5, 6],
    [7, 8]
])

print("A + B:")
print(A + B)
```

```
print("A x B:")
print(A @ B)
```

Module 2 Quick Revision

Topic	Key Idea
Number Systems	Ways of representing numbers
Linear Algebra	Mathematics of vectors and matrices
Vector	Ordered collection of values
Matrix	Rectangular arrangement of values
Matrix Multiplication	Fundamental operation in ML/DL
Statistics	Understanding and analyzing data
Mean	Average
Median	Middle value
Mode	Most frequent value
Variance	Measures squared spread
Standard Deviation	Measures spread in original units
Probability	Measures likelihood
Conditional Probability	Probability given some condition
Bayes Theorem	Updates probability using evidence
Derivative	Rate of change
Gradient	Vector of partial derivatives
Gradient Descent	Optimization method used to minimize loss

Module 2 Interview Questions

Basic

1. What is linear algebra?
2. What is a vector?
3. What is a matrix?
4. What is the difference between a scalar, vector, and matrix?
5. What is mean?
6. What is median?
7. What is mode?
8. What is variance?
9. What is standard deviation?
10. What is probability?

Intermediate

11. Why is linear algebra important in Machine Learning?
12. Why is median less affected by outliers?

13. What is the difference between population variance and sample variance?
14. What is conditional probability?
15. Explain Bayes theorem.
16. What is a derivative?
17. What is a gradient?
18. What is gradient descent?
19. What is the learning rate?
20. Why are derivatives used in neural-network training?

Advanced

21. Why must matrix dimensions be compatible for matrix multiplication?
22. Why is feature scaling important for gradient-based algorithms?
23. What happens if the learning rate is too large?
24. What happens if the learning rate is too small?
25. How are vectors and matrices used in neural networks?
26. How does gradient descent minimize a loss function?
27. What is the relationship between derivatives, gradients, and backpropagation?

Practice Questions

Level 1 — Basic

1. Calculate the mean, median, and mode of:
5, 10, 10, 15, 20
2. Calculate the variance and standard deviation of:
2, 4, 6, 8, 10
3. Explain vectors and matrices with a student-dataset example.

Level 2 — ML

4. Create two NumPy matrices and perform:
 - Addition
 - Subtraction
 - Multiplication
 - Transpose
5. Calculate the probability of getting an even number when rolling a fair die.
6. Explain:

[
P(A|B)
]

using a real-world example.

Level 3 — AI/ML

7. Explain how Bayes theorem can be used for spam detection.
8. Explain why standard deviation is useful in feature scaling.
9. Explain why derivatives are required during model optimization.
10. Explain the complete relationship:

Loss Function

↓

Derivative

↓

Gradient

↓

Gradient Descent

↓

Parameter Update

↓

Better Model

Mini Project

Statistical Analysis of Student Performance

Take a dataset containing:

Student_ID

Study_Hours

Attendance

Assignment_Score

Exam_Score

Perform:

1. Mean
2. Median
3. Mode
4. Variance
5. Standard deviation
6. Correlation
7. Identify possible outliers
8. Create visualizations
9. Write 5 observations about the dataset

This project will prepare students for **Module 4 (Data Preprocessing)** and **Module 5 (EDA)**.

Module 3 — Python for AI/ML

Python is one of the most widely used programming languages for **Artificial Intelligence, Machine Learning, Data Science, Deep Learning, and automation**.

In this module, students will move from **Python fundamentals** → **data structures** → **OOP** → **files** → **NumPy** → **Pandas** → **visualization** → **Jupyter Notebook**.

1. Python Basics

Concept

Python is a high-level, interpreted, general-purpose programming language known for its simple syntax and large ecosystem of libraries.

Python is widely used in AI/ML because it provides libraries such as:

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- TensorFlow
- PyTorch
- OpenCV



Simple Example

```
print("Hello, AI/ML!")
```

Output

Hello, AI/ML!

Python Characteristics

- Easy to learn
- Dynamically typed
- Interpreted
- Object-oriented
- Large ecosystem
- Cross-platform
- Strong community support

Real-World Example

A company might use:

Python

↓

Pandas → Data Processing

↓

Scikit-learn → ML

↓

FastAPI → API

↓

Docker → Deployment

Interview Questions

1. What is Python?
2. Why is Python popular in AI/ML?
3. What does interpreted language mean?
4. Is Python statically or dynamically typed?
5. What are Python's major applications?

2. Variables & Data Types

Variables

A variable stores a value.

```
name = "Atasi"
age = 21
score = 85.5
```

Python automatically determines the type.

```
print(type(name))
print(type(age))
print(type(score))
```

Output

```
<class 'str'>
```

```
<class 'int'>
```

```
<class 'float'>
```

Important Data Types

Data Type Example

int	10
float	10.5
str	"Python"
bool	True
list	[1, 2, 3]
tuple	(1, 2, 3)

Data Type Example

```
set      {1, 2, 3}
dict     {"name": "John"}
NoneType None
```

Type Conversion

```
x = "100"

x = int(x)

print(x + 50)
```

Output:

150

Interview Question

Q: What is dynamic typing?

Python determines a variable's type at runtime, and the same variable name can later refer to an object of another type.

3. Operators

Operators are symbols used to perform operations.

Arithmetic Operators

```
a = 10
b = 3

print(a + b)
print(a - b)
print(a * b)
print(a / b)
print(a // b)
print(a % b)
print(a ** b)
```

Important operators:

- + Addition
- Subtraction
- * Multiplication
- / Division
- // Floor division
- % Modulus
- ** Exponentiation

Comparison Operators

```
a = 10
```

```
b = 20
```

```
print(a == b)
print(a != b)
print(a > b)
print(a < b)
print(a >= b)
print(a <= b)
```

Logical Operators

```
age = 22
```

```
print(age > 18 and age < 30)
print(age < 18 or age > 60)
print(not age > 18)
```

Membership Operators

```
languages = ["Python", "Java", "C++"]
```

```
print("Python" in languages)
```

Output:

True

4. Conditions

Conditions allow a program to make decisions.

if

```
age = 20
```

```
if age >= 18:
    print("Eligible")
```

if-else

```
marks = 75
```

```
if marks >= 40:
    print("Pass")
else:
    print("Fail")
```

if-elif-else

```
marks = 85
```

```
if marks >= 90:
    grade = "A+"
elif marks >= 75:
    grade = "A"
elif marks >= 60:
    grade = "B"
else:
```



```
grade = "C"
```

```
print(grade)
```

Output:

A

ML Connection

Conditions are useful when implementing simple business rules around ML predictions.

```
probability = 0.82
```

```
if probability >= 0.5:
    print("Positive")
else:
    print("Negative")
```

5. Loops

Loops allow us to repeat operations.

```
for Loop
for i in range(5):
    print(i)
```

Output:

0

1

2

3

4

Example

```
scores = [80, 75, 90, 85]
```

```
for score in scores:
    print(score)
while Loop
count = 1
```

```
while count <= 5:
    print(count)
    count += 1
```

break

Stops the loop.

```
for i in range(10):
    if i == 5:
        break
```

```
print(i)
```

continue

Skips the current iteration.

```
for i in range(5):
    if i == 2:
        continue
    print(i)
```

Interview Questions

1. Difference between for and while?
2. What does break do?
3. What does continue do?
4. What is range()?

6. Functions

Concept

A function is a reusable block of code.

```
def greet():
    print("Hello!")
```

```
greet()
```

Parameters

```
def add(a, b):
    return a + b
```

```
result = add(10, 20)
```

```
print(result)
```

Output:

30

Default Arguments

```
def greet(name="Student"):
    print("Hello", name)
```

```
greet()
```

```
greet("Atasi")
```

Lambda Function

A lambda is a small anonymous function.

```
square = lambda x: x * x
```

```
print(square(5))
```

Output:

25

Why Functions Matter in ML

ML projects contain reusable operations such as:

```
load_data()
```

↓

```
clean_data()
```

↓

```
preprocess_data()
```

↓

```
train_model()
```

↓

```
evaluate_model()
```

7. Lists, Tuples, Sets & Dictionaries

These are important Python data structures.

List

A list is:

- Ordered
- Mutable
- Allows duplicate values

```
students = ["A", "B", "C"]
```

```
students.append("D")
```

```
print(students)
```

Tuple

A tuple is:

- Ordered
- Immutable

```
coordinates = (10, 20)
```

```
print(coordinates[0])
```

You cannot normally modify its elements after creation.

Set

A set stores unique elements.

```
numbers = {1, 2, 2, 3, 3}
```

```
print(numbers)
```

Output:

```
{1, 2, 3}
```

Useful for removing duplicates.

Dictionary

A dictionary stores key-value pairs.

```
student = {
    "name": "Rahul",
    "age": 21,
    "score": 85
}
```

```
print(student["name"])
```

Output:

```
Rahul
```

ML Connection

JSON-like data is commonly represented in Python using dictionaries.

```
data = {
    "age": 25,
    "experience": 2,
    "salary": 45000
}
```

8. OOP Basics

Concept

OOP = Object-Oriented Programming

OOP organizes programs around **objects and classes**.

Class

A class is a blueprint.

```
class Student:
```

```
    def __init__(self, name, marks):
```

```
self.name = name
```

```
self.marks = marks
```

```
def display(self):
    print(self.name, self.marks)
```

Object

```
student1 = Student("Rahul", 85)
```

```
student1.display()
```

Output:

Rahul 85

Four Important OOP Concepts

1. Encapsulation

Combining data and methods within a class.

2. Inheritance

One class can inherit properties/methods from another.

3. Polymorphism

The same interface can behave differently for different objects.

4. Abstraction

Hiding unnecessary implementation details.

ML Connection

Many ML libraries use classes and objects.

For example:

```
from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()
```

Here LinearRegression is a class and model is an object.

9. Exception Handling

Errors can occur while a program is running.

Python provides:

```
try
```

```
except
```

```
else
```

finally

Example

try:

```
number = int(input("Enter number: "))
print(100 / number)
```

except ValueError:

```
print("Please enter a valid number.")
```

except ZeroDivisionError:

```
print("Cannot divide by zero.")
```

finally

try:

```
print("Processing...")
```

except Exception:

```
print("Error occurred")
```

finally:

```
print("Process completed")
```

Why Important in AI/ML?

Production ML applications need to handle:

- Invalid input
- Missing files
- API failures
- Database errors
- Invalid user requests
- Model loading failures

10. File Handling

Python can read and write files.

Writing a File

```
with open("data.txt", "w") as file:
    file.write("AI/ML Learning")
```

Reading a File

```
with open("data.txt", "r") as file:
    content = file.read()
```

```
print(content)
```

Output:

AI/ML Learning

CSV Files

```
import csv

with open("students.csv", "r") as file:
    reader = csv.reader(file)

    for row in reader:
        print(row)
```

In practical ML work, **Pandas** is commonly used for CSV and tabular data.

11. NumPy

Concept

NumPy = Numerical Python

NumPy provides efficient numerical arrays and mathematical operations.

It is a fundamental library for scientific computing in Python.

```
import numpy as np
```

Creating an Array

```
numbers = np.array([10, 20, 30, 40])
```

```
print(numbers)
```

Mathematical Operations

```
print(numbers.mean())
print(numbers.sum())
print(numbers.max())
print(numbers.min())
```

Output

25.0

100

40

10

2D Array

```
matrix = np.array([
    [1, 2, 3],
    [4, 5, 6]
])
```

```
print(matrix.shape)
```

Output:

(2, 3)

Useful NumPy Functions

```

np.array()
np.zeros()
np.ones()
np.arange()
np.linspace()
np.mean()
np.median()
np.std()
np.sum()
np.min()
np.max()

```

Why NumPy is Important

It provides efficient operations on numerical data and forms a foundation for many data-science and ML libraries.

12. Pandas

Concept

Pandas is a Python library for working with structured and tabular data.

```
import pandas as pd
```

Series

```
scores = pd.Series([80, 90, 75, 88])
```

```
print(scores)
```

DataFrame

A DataFrame is a table-like data structure.

```

data = {
    "Name": ["A", "B", "C"],
    "Age": [20, 21, 22],
    "Score": [80, 90, 85]
}

```

```
df = pd.DataFrame(data)
```

```
print(df)
```

Output

```

Name Age Score
0  A  20   80
1  B  21   90
2  C  22   85

```

Reading CSV


```
df = pd.read_csv("students.csv")
```

Important Functions

```
df.head()
```

```
df.tail()
```

```
df.info()
```

```
df.describe()
```

```
df.shape
```

```
df.columns
```

```
df.isnull().sum()
```

Selecting a Column

```
print(df["Score"])
```

Filtering

```
high_scores = df[df["Score"] > 85]
```

```
print(high_scores)
```

ML Connection

Pandas is frequently used for:

CSV/Database

↓

Pandas

↓

Cleaning

↓

EDA

↓

Features + Target

↓

ML Model

13. Matplotlib

Concept

Matplotlib is a Python library for creating visualizations.

```
import matplotlib.pyplot as plt
```

Line Chart

```
x = [1, 2, 3, 4, 5]
```

```
y = [10, 20, 15, 30, 25]
```

```
plt.plot(x, y)

plt.xlabel("Day")
plt.ylabel("Sales")
plt.title("Sales Trend")

plt.show()
```

Bar Chart

```
names = ["A", "B", "C"]
scores = [80, 90, 75]

plt.bar(names, scores)

plt.xlabel("Student")
plt.ylabel("Score")
plt.title("Student Scores")

plt.show()
```

Common Charts

- Line chart
- Bar chart
- Histogram
- Scatter plot
- Pie chart
- Box plot



Why Visualization Matters

Visualization helps identify:

- Trends
- Relationships
- Distributions
- Outliers
- Patterns

14. Seaborn

Concept

Seaborn is a statistical data-visualization library built on top of Matplotlib.

```
import seaborn as sns
```

It provides convenient functions for statistical visualization.

Example

```
import seaborn as sns
import matplotlib.pyplot as plt

data = sns.load_dataset("tips")

sns.scatterplot(
    data=data,
    x="total_bill",
    y="tip"
)

plt.show()
```

Important Seaborn Charts

```
sns.scatterplot()
sns.lineplot()
sns.barplot()
sns.histplot()
sns.boxplot()
sns.violinplot()
sns.heatmap()
sns.pairplot()
```

Correlation Heatmap

```
import seaborn as sns
import matplotlib.pyplot as plt

correlation = data.select_dtypes("number").corr()

sns.heatmap(correlation, annot=True)

plt.show()
```

This is particularly useful during **EDA**.

15. Jupyter Notebook

Concept

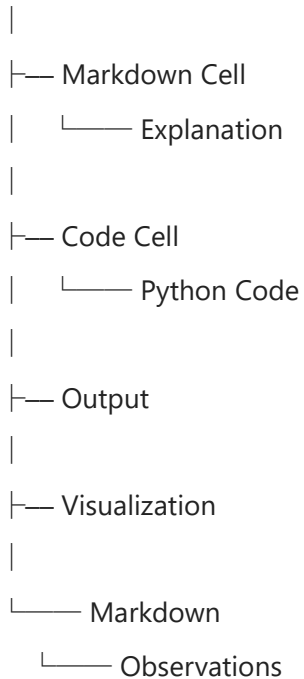
Jupyter Notebook is an interactive environment where you can combine:

- Python code
- **Output**
- Visualizations
- Markdown
- Explanations
- Mathematical equations

It is widely used for learning, experimentation, data analysis, and ML prototyping.

Basic Structure

Jupyter Notebook



Example

Code Cell:

```
import numpy as np

data = np.array([10, 20, 30, 40, 50])

print(data.mean())
```

Output:

30.0

Why Jupyter is Useful for ML

You can execute the project step-by-step:

Load Dataset

↓

Explore Data

↓

Clean Data

↓

Visualize

↓

Feature Engineering

↓

Train Model

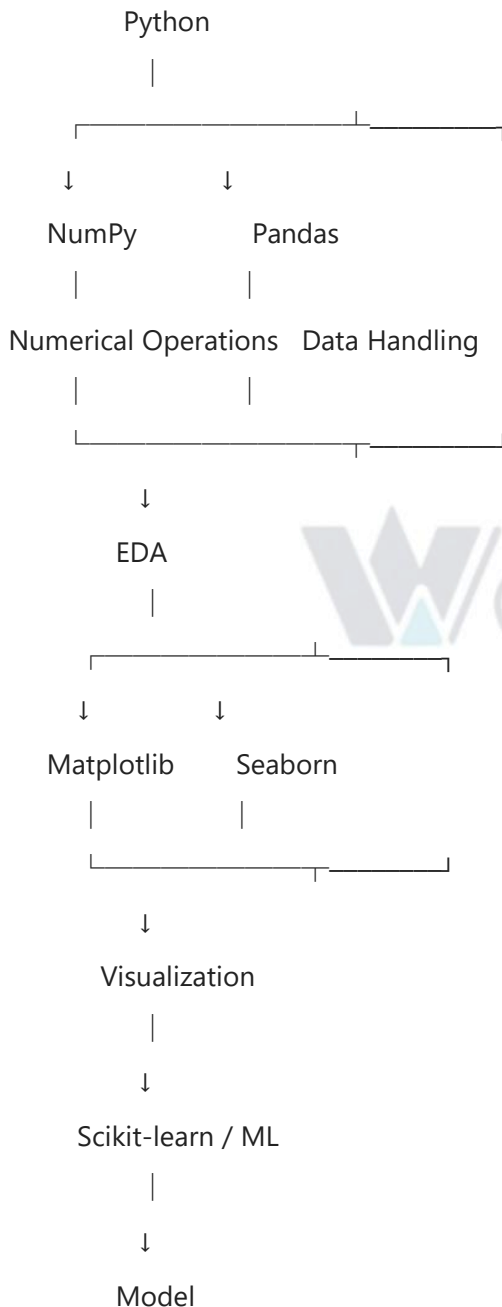


Evaluate

without repeatedly running the entire program.

How Python Tools Connect in an ML Project

This is one of the most important concepts for students:



Mini End-to-End Python + Data Analysis Example

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
  
```

```
# Create dataset
data = {
    "Study_Hours": [2, 4, 5, 6, 8],
    "Exam_Score": [50, 60, 65, 75, 90]
}

df = pd.DataFrame(data)

# Display data
print(df)

# Statistics
print("\nStatistics:")
print(df.describe())

# Visualization
sns.scatterplot(
    data=df,
    x="Study_Hours",
    y="Exam_Score"
)

plt.title("Study Hours vs Exam Score")
plt.show()
```

Expected Observation

Generally, in this small example:

As study hours increase, exam scores also tend to increase.

Important: This is only an illustrative dataset and does not establish causation.

Module 3 Quick Revision

Topic	Key Idea
Python Basics	Core programming language concepts
Variables	Store references to values/objects
Data Types	Define the kind of data being represented
Operators	Perform calculations and comparisons
Conditions	Make decisions
Loops	Repeat operations
Functions	Reusable blocks of logic
Lists	Ordered, mutable collection

Topic	Key Idea
Tuples	Ordered, immutable collection
Sets	Unique elements
Dictionary	Key-value data structure
OOP	Object-oriented programming
Exception Handling	Handle runtime errors
File Handling	Read/write files
NumPy	Numerical computing
Pandas	Data manipulation and analysis
Matplotlib	Data visualization
Seaborn	Statistical visualization
Jupyter	Interactive development environment

Module 3 Interview Questions

Python

1. What is Python?
2. Why is Python widely used in AI/ML?
3. What is dynamic typing?
4. What is the difference between `==` and `is`?
5. What is the difference between mutable and immutable objects?
6. What is the difference between a list and tuple?
7. What is the difference between a list and set?
8. What is a dictionary?
9. What are `*args` and `**kwargs`?
10. What is a lambda function?

OOP

11. What is a class?
12. What is an object?
13. Explain inheritance.
14. What is encapsulation?
15. What is polymorphism?
16. What is abstraction?

NumPy & Pandas

17. What is NumPy?
18. Why is NumPy preferred for numerical operations?
19. What is a NumPy array?

20. What is Pandas?
21. What is a DataFrame?
22. Difference between Series and DataFrame?
23. How do you detect missing values in Pandas?
24. How do you read a CSV file using Pandas?
25. What does `df.describe()` do?

Visualization

26. What is Matplotlib?
27. What is Seaborn?
28. Difference between Matplotlib and Seaborn?
29. What is a correlation heatmap?
30. When would you use a scatter plot?

Jupyter

31. What is Jupyter Notebook?
32. Why is Jupyter useful for ML?
33. Difference between a Python script and Jupyter Notebook?

Practice Questions

Level 1 — Python

1. Write a program to check whether a number is even or odd.
2. Find the largest of three numbers.
3. Print numbers from 1 to 100 using a loop.
4. Write a function to calculate factorial.
5. Count vowels in a string.
6. Remove duplicates from a list.
7. Create a dictionary containing student information.

Level 2 — NumPy

8. Create a NumPy array containing 10 numbers.
9. Calculate its:
 - Mean
 - Median
 - Standard deviation
 - Minimum
 - Maximum
10. Create a 3×3 matrix and calculate its transpose.

Level 3 — Pandas

11. Create a DataFrame containing:

Name

Age

Department

Marks

12. Find students whose marks are greater than 75.

13. Find missing values.

14. Calculate average marks.

15. Sort students by marks.

Level 4 — Visualization

16. Create a line chart of monthly sales.

17. Create a bar chart of student scores.

18. Create a histogram of ages.

19. Create a scatter plot of study hours vs marks.

20. Create a correlation heatmap.

Module 3 Mini Project

Student Performance Data Analysis

Create a dataset:

Student_ID

Name

Age

Study_Hours

Attendance

Assignment_Score

Exam_Score

Perform:

Step 1 — Data Creation

Create the dataset using Pandas.

Step 2 — Data Inspection

Use:

`df.head()`

`df.info()`

`df.describe()`

`df.shape`

Step 3 — Statistical Analysis

Calculate:

- Mean
- Median
- Standard deviation
- Minimum
- Maximum

Step 4 — Visualization

Create:

- Study Hours vs Exam Score → Scatter plot
- Student vs Exam Score → Bar chart
- Exam Score distribution → Histogram
- Feature relationships → Heatmap

Step 5 — Findings

Write at least five observations.

Step 6 — Prepare for ML

Identify:

Features:

Study_Hours

Attendance

Assignment_Score

Target:

Exam_Score

This mini-project creates the bridge from **Python** → **Pandas** → **EDA** → **Machine Learning**, which students will use directly in **Modules 4–6**.

Module 4 — Data Preprocessing

Data Preprocessing is one of the most important stages of a Machine Learning project. Real-world data is rarely perfect. It may contain missing values, duplicate records, inconsistent formats, outliers, categorical values, and features with very different scales.

A typical ML workflow is:

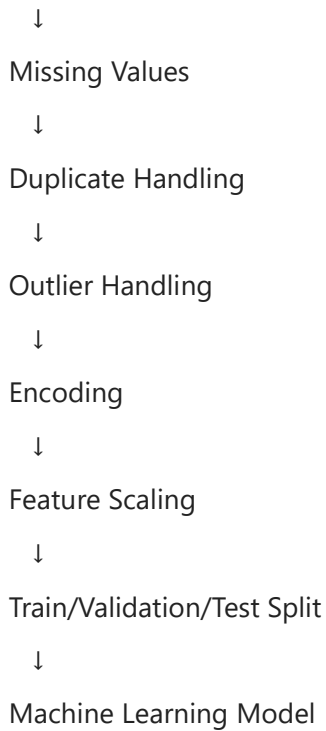
Raw Data

↓

Data Collection

↓

Data Cleaning



1. Dataset

Concept

A **dataset** is a collection of data used for analysis or machine learning.

For example, a student-performance dataset may contain:

Student Age Study Hours Attendance Exam Score

A	20	5	85	78
B	21	7	90	88
C	20	3	70	60
D	22	8	95	92

Each row represents an **observation/sample**, while each column represents a **variable/feature**.

Dataset Components

Dataset

|

├— Rows → Samples / Observations

|

└— Columns

├— Features

└— Target

Common Dataset Formats

- CSV

- Excel
- JSON
- SQL database
- Images
- Audio
- Text
- Sensor data

Python

```
import pandas as pd
```

```
df = pd.read_csv("students.csv")
```

```
print(df.head())
```

```
print(df.shape)
```

Interview Questions

1. What is a dataset?
2. What is a sample?
3. What is a feature?
4. What is a target variable?
5. What are structured and unstructured datasets?

2. Features & Target

Feature

A **feature** is an input variable used by a machine-learning model to make a prediction.

Example:

Study Hours

Attendance

Assignment Score

Target

The **target** is the output that we want to predict.

For example:

Features	Target
----------	--------

Study Hours

Attendance → Exam Score

Assignment Score

Example

Suppose:

Study Hours = 6

Attendance = 90%

Assignment Score = 85

The model predicts:

Exam Score = 82

Therefore:

$X \rightarrow$ Features

$y \rightarrow$ Target

Python

```
X = df[["Study_Hours", "Attendance", "Assignment_Score"]]
```

```
y = df["Exam_Score"]
```

Interview Question

Q: What is X and y in ML?

Usually:

- $X \rightarrow$ input features
- $y \rightarrow$ target/output

3. Data Collection

Concept

Data collection is the process of obtaining data required for an ML problem.

Sources

Data can come from:

- Databases
- APIs
- CSV files
- Websites
- Sensors
- Mobile applications
- Business systems
- Surveys
- Public datasets
- Logs

Example

For a **Customer Churn Prediction** project, data might contain:

Customer_ID

Age

Monthly_Charges

Contract_Type

Tenure

Support_Calls

Churn

Data Collection Process

Define Problem



Identify Required Data



Choose Data Sources



Collect Data



Store Data



Validate Data

Important Considerations

When collecting data, consider:

- Data quality
- Relevance
- Completeness
- Privacy
- Security
- Bias
- Legal requirements

4. Data Cleaning

Concept

Data cleaning means identifying and correcting problems in a dataset.

Common problems include:

- Missing values

- Duplicate records
- Incorrect data types
- Invalid values
- Inconsistent formats
- Outliers
- Typographical errors

Example

Raw data:

Name	Age	City	Salary
Rahul	25	Pune	40000
Priya	—	Pune	45000
Rahul	25	Pune	40000
Amit	-5	Mumbai	50000

Problems:

- Missing age
- Duplicate Rahul record
- Invalid age -5

Basic Inspection

```
print(df.head())  
print(df.info())  
print(df.describe())  
print(df.isnull().sum())  
print(df.duplicated().sum())
```

Goal

Dirty Data

↓

Cleaning

↓

Reliable Data

5. Missing Values

Concept

A missing value means information is unavailable for a particular observation.

Example:

Name	Age	Salary
A	25	40000
B	NaN	50000

Name Age Salary

C 30 NaN

NaN means **Not a Number / missing numerical value** in many Python data workflows.

Detect Missing Values

```
print(df.isnull().sum())
```

Example:

Age 1

Salary 1

Method 1 — Remove Rows

```
df.dropna(inplace=True)
```

Use this carefully because you may lose useful data.

Method 2 — Fill with Mean

```
df["Age"] = df["Age"].fillna(df["Age"].mean())
```

Useful for some numerical variables when mean imputation is appropriate.

Method 3 — Fill with Median

```
df["Salary"] = df["Salary"].fillna(df["Salary"].median())
```

Median can be preferable when the variable is skewed or contains outliers.

Method 4 — Categorical Data

For categorical values, mode is often used:

```
df["City"] = df["City"].fillna(df["City"].mode()[0])
```

Important

There is no single best missing-value strategy.

The choice depends on:

- Data type
- Amount of missing data
- Distribution
- Meaning of the missingness
- ML algorithm
- Business context

Interview Questions

1. How do you handle missing values?

2. When would you use mean vs median?
3. When should you avoid deleting missing rows?
4. What is NaN?

6. Duplicate Data

Concept

A duplicate is a record that appears more than once when it should represent the same observation.

Example:

ID Name Score

101 Rahul 85

102 Priya 90

101 Rahul 85

The first and third records may be duplicates.

Detect Duplicates

```
print(df.duplicated().sum())
```

View Duplicates

```
print(df[df.duplicated()])
```

Remove Duplicates

```
df.drop_duplicates(inplace=True)
```

Why Remove Duplicates?

Duplicates can:

- Bias statistics
- Affect model training
- Artificially increase the importance of repeated observations
- Produce misleading results

Important

Do not automatically remove every repeated row. Sometimes repeated records represent **real repeated events** rather than accidental duplicates.

7. Outliers

Concept

An **outlier** is an observation that is unusually far from the typical values in a dataset.

Example:

20, 22, 21, 23, 25, 24, 150

150 may be an outlier depending on the context.

Real-World Example

Suppose employee ages are:

22, 25, 27, 30, 31, 29, 250

An age of 250 is clearly invalid.

But an unusually high salary may be a **real observation**, not an error.

Detecting Outliers Using IQR

The **Interquartile Range (IQR)** is:

```
[
IQR=Q3-Q1
]
```

Common outlier boundaries are:

```
[
Lower=Q1-1.5(IQR)
]
```

```
[
Upper=Q3+1.5(IQR)
]
```

Values outside these boundaries are often flagged as potential outliers.

Python

```
Q1 = df["Salary"].quantile(0.25)
Q3 = df["Salary"].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
```

```
outliers = df[
(df["Salary"] < lower) |
(df["Salary"] > upper)
]
```

```
print(outliers)
```

What Should We Do With Outliers?

Possible approaches:

- Investigate them
- Correct data-entry errors
- Remove them if justified
- Cap/winsorize them

- Transform the feature
- Use models less sensitive to outliers

Important

An outlier is not automatically an error.

8. Encoding Categorical Data

Concept

Machine-learning algorithms generally require numerical representations, so categorical values often need to be encoded.

Example:

City

Pune

Mumbai

Delhi

Pune

The model needs a numerical representation.

A. Label Encoding

Example:

Beginner → 0

Intermediate → 1

Advanced → 2

This can be appropriate for **ordinal categories**, where order is meaningful.

Python

```
from sklearn.preprocessing import LabelEncoder
```

```
encoder = LabelEncoder()
```

```
df["Level"] = encoder.fit_transform(df["Level"])
```

Caution

For nominal categories such as:

Pune

Mumbai

Delhi

encoding them as:

```
Pune = 0
Mumbai = 1
Delhi = 2
```

can incorrectly suggest an order.

B. One-Hot Encoding

For nominal categories:

City

Pune

Mumbai

Delhi

One-hot encoding creates columns such as:

City_Delhi

City_Mumbai

City_Pune

Python

```
df = pd.get_dummies(
    df,
    columns=["City"],
    dtype=int
)
```

Example:

City	City_Delhi	City_Mumbai	City_Pune
Pune	0	0	1
Mumbai	0	1	0
Delhi	1	0	0

9. Feature Scaling

Concept

Feature scaling brings numerical features to comparable scales.

Suppose:

Age → 18–60

Salary → 20,000–200,000

Experience → 0–20

Salary has a much larger numerical scale.

Some algorithms can be affected by these differences.

Scaling is particularly important for algorithms based on:

- Distance
- Gradients
- Similarity

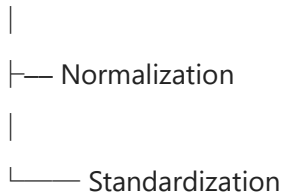
Examples:

- KNN
- K-Means
- SVM
- Logistic Regression
- Neural Networks
- Gradient-based models

Tree-based models generally do not require scaling in the same way.

Main Techniques

Feature Scaling



10. Normalization

Concept

Normalization commonly transforms values into a fixed range, often **0 to 1**.

Min-Max Scaling Formula

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Example

Suppose:

```

Minimum = 10
Maximum = 50
Value = 30
  
```

Then:

$$x' = \frac{30 - 10}{50 - 10}$$

```
[
x'=0.5
]
```

Python

```
from sklearn.preprocessing import MinMaxScaler
```

```
scaler = MinMaxScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

When Useful?

Normalization can be useful when you want features mapped to a bounded range and when the algorithm benefits from comparable scales.

11. Standardization

Concept

Standardization transforms data so that the feature is centered around a mean of approximately 0 with a standard deviation of approximately 1.

Formula

```
[
z=
\frac{x-\mu}{\sigma}
]
```

where:

- (x) = original value
- (μ) = mean
- (σ) = standard deviation

```
genui {"descriptive_statistics_sampling":{"type_id":"STANDARD_SCORE_Z","locale_override":"en-IN"}}
```

Example

Suppose:

Mean = 50

Standard deviation = 10

Value = 70

Then:

```
[
z=\frac{70-50}{10}=2
]
```

The value is **2 standard deviations above the mean.**

Python

```
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

Normalization vs Standardization

Feature	Normalization	Standardization
Common method	Min-Max	Z-score
Typical range	0 to 1	No fixed range
Uses	Min & Max	Mean & SD
Formula	$(x - \min) / (\max - \min)$	$(x - \text{mean}) / \text{std}$
Sensitive to outliers	Yes	Can also be affected

Interview Question

Q: Which scaling technique should I always use?

There is no universal answer. The choice depends on:

- Dataset
- Algorithm
- Feature distribution
- Outliers
- Business/application requirements

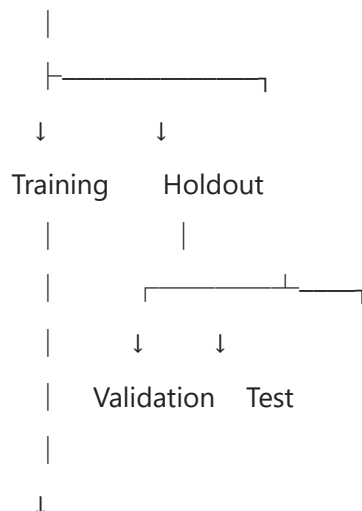
12. Train / Validation / Test Split

Concept

We divide data into separate subsets so that we can train, tune, and evaluate a model fairly.

Three Sets

Complete Dataset



Train Model

Training Set

Used to **learn model parameters**.

Validation Set

Used during model development for:

- Model selection
- Hyperparameter tuning
- Comparing approaches

Test Set

Used for the final evaluation after development decisions have been made.

Example Split

A common example is:

70% → Training

15% → Validation

15% → Test

Other splits such as 80/10/10 or 80/20 can also be appropriate.

There is **no universal split ratio**.

Python Example

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_temp, y_train, y_temp = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=42
)
```

```
X_val, X_test, y_val, y_test = train_test_split(
    X_temp,
    y_temp,
    test_size=0.50,
    random_state=42
)
```

This gives approximately:

70% Training

15% Validation

15% Test

⚠️ **Very Important: Data Leakage**

Data leakage occurs when information from outside the training process improperly influences model training.

This can make model performance look unrealistically good.

Incorrect Approach

Entire Dataset

↓

Scale Everything

↓

Train/Test Split

Better Approach

Dataset

↓

Train/Test Split

↓

Fit preprocessing on Training Data

↓

Transform Training Data

↓

Transform Validation/Test Data

↓

Train Model

For example:

`scaler.fit(X_train)`

```
X_train_scaled = scaler.transform(X_train)
```

```
X_test_scaled = scaler.transform(X_test)
```

Do not fit the scaler separately on the test set.

In production ML, using a Pipeline is often a safer approach because it keeps preprocessing and modeling steps together.

Complete Preprocessing Example

Let's create a small student dataset.

```
import pandas as pd
```

```
import numpy as np

data = {
    "Age": [20, 21, 22, np.nan, 20],
    "Study_Hours": [5, 7, 8, 6, 5],
    "City": ["Pune", "Mumbai", "Pune", "Delhi", "Pune"],
    "Exam_Score": [75, 88, 92, 80, 75]
}

df = pd.DataFrame(data)

print(df)
```

Step 1 — Check Missing Values

```
print(df.isnull().sum())
```

Step 2 — Fill Missing Values

```
df["Age"] = df["Age"].fillna(df["Age"].median())
```

Step 3 — Remove Duplicates

```
df.drop_duplicates(inplace=True)
```

Step 4 — Encode Categorical Data

```
df = pd.get_dummies(
    df,
    columns=["City"],
    dtype=int
)
```

Step 5 — Separate Features and Target

```
X = df.drop("Exam_Score", axis=1)
y = df["Exam_Score"]
```

Step 6 — Train/Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
```

```
random_state=42
)
```

Step 7 — Scale Numerical Features

```
from sklearn.preprocessing import StandardScaler
```

```
scaler = StandardScaler()
```

```
X_train_scaled = scaler.fit_transform(X_train)
```

```
X_test_scaled = scaler.transform(X_test)
```

Notice:

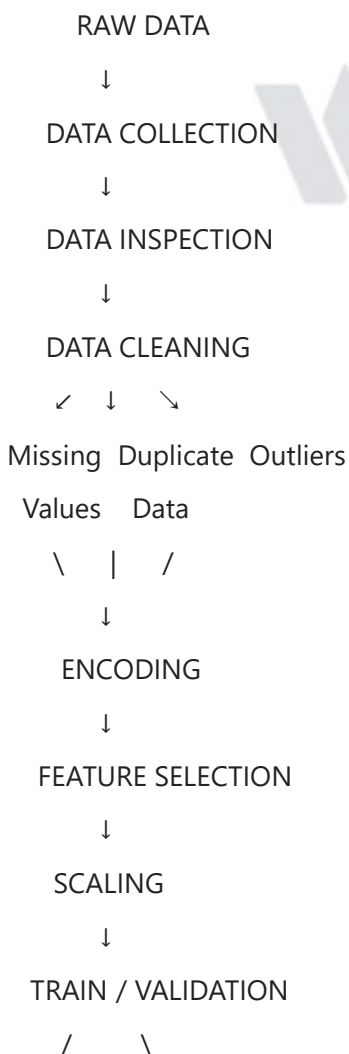
Training → fit + transform

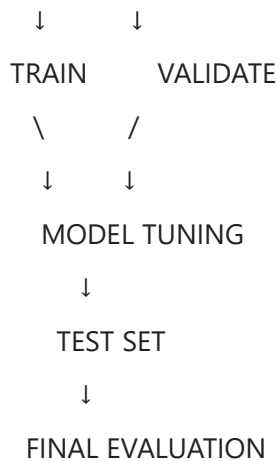
Test → transform only

This prevents information from the test set from being used to estimate the scaling parameters.

Complete Data Preprocessing Pipeline

Students should remember this workflow:





🔗 Module 4 Quick Revision

Topic	Key Idea
Dataset	Collection of observations and variables
Feature	Input variable
Target	Output to predict
Data Collection	Obtaining relevant data
Data Cleaning	Fixing data-quality problems
Missing Values	Unavailable observations
Duplicate Data	Repeated records
Outliers	Unusually distant observations
Encoding	Converting categories into numerical representations
Feature Scaling	Bringing numerical features to comparable scales
Normalization	Commonly maps values to a bounded range
Standardization	Centers and scales using mean and standard deviation
Training Set	Used to train the model
Validation Set	Used for development/tuning
Test Set	Used for final evaluation
Data Leakage	Unintended information flow into model training

🔗 Module 4 Interview Questions

Basic

1. What is data preprocessing?
2. Why is data preprocessing important?
3. What is a dataset?
4. What is a feature?
5. What is a target variable?
6. What is data cleaning?

7. What are missing values?
8. What are duplicate records?
9. What is an outlier?
10. What is categorical data?

Intermediate

11. How do you handle missing values?
12. When would you use mean vs median imputation?
13. How do you detect duplicate data?
14. What is the IQR method?
15. What is label encoding?
16. What is one-hot encoding?
17. When should you use one-hot encoding instead of ordinal/label encoding?
18. Why is feature scaling important?
19. What is the difference between normalization and standardization?
20. What is a z-score?

Advanced

21. Why should preprocessing be fitted only on training data?
22. What is data leakage?
23. Why can outliers be dangerous for some ML algorithms?
24. Why are tree-based algorithms generally less sensitive to feature scaling?
25. What is the purpose of a validation dataset?
26. Why shouldn't the test set be used for hyperparameter tuning?
27. How can a preprocessing pipeline reduce leakage and inconsistency?
28. What happens if categorical variables are encoded incorrectly?
29. How would you handle a dataset with 40% missing values?
30. How would you design a preprocessing pipeline for a production ML system?

Practice Questions

Level 1

1. Create a student dataset with 20 records.
2. Add at least five missing values.
3. Add duplicate records.
4. Add one or two suspicious/outlier values.
5. Identify all problems in the dataset.

Level 2

6. Handle missing numerical values using:

- Mean
 - Median
7. Handle missing categorical values using mode.
 8. Detect and remove duplicates.
 9. Detect potential outliers using IQR.
 10. Encode categorical columns using one-hot encoding.

Level 3

11. Separate your dataset into X and y.
12. Split the data into:

70% Training

15% Validation

15% Test

13. Apply StandardScaler.
14. Apply MinMaxScaler.
15. Compare the transformed values.

Level 4 — Interview/Industry

16. Explain why scaling the complete dataset before splitting can cause leakage.
17. Build a preprocessing pipeline containing:

Missing-value handling

↓

Encoding

↓

Scaling

↓

ML Model

18. Explain when you would **not** remove an outlier.
19. Explain why one-hot encoding is usually safer than assigning arbitrary numeric labels to nominal categories.
20. Design a preprocessing pipeline for a **Customer Churn Prediction** project.

Module 4 Mini Project — Customer Churn Data Preprocessing

Use a dataset containing:

Customer_ID

Age

Gender

Monthly_Charges

Tenure

Contract_Type

Support_Calls

Payment_Method

Churn

Tasks

Step 1: Load the dataset.

Step 2: Inspect:

`df.head()`

`df.info()`

`df.describe()`

`df.isnull().sum()`

Step 3: Handle missing values.

Step 4: Detect duplicates.

Step 5: Investigate outliers.

Step 6: Encode:

Gender

Contract_Type

Payment_Method

Step 7: Separate:

$X \rightarrow$ Customer features

$y \rightarrow$ Churn

Step 8: Split into training, validation, and test sets.

Step 9: Scale appropriate numerical features.

Step 10: Document every preprocessing decision.

Final Project Flow

Customer Dataset



Data Inspection



Cleaning



Missing Values



Duplicates



Outliers



Categorical Encoding



Feature Scaling



Train/Validation/Test



Ready for ML Model

This project provides the direct foundation for **Module 5 (EDA)** and then **Module 6 (Supervised Learning)**.

Module 5 — Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the process of understanding a dataset before building a Machine Learning model.

EDA helps us answer questions such as:

- What does the dataset contain?
- Are there missing values?
- Which features are important?
- What are the distributions of variables?
- Are there outliers?
- Which variables are related?
- Are there useful patterns?
- Are there suspicious or unexpected observations?

EDA Workflow

Raw Dataset



Understand Data



Clean Data



Univariate Analysis



Bivariate Analysis



Multivariate Analysis

↓

Correlation Analysis

↓

Visualization

↓

Find Patterns & Relationships

↓

Feature Engineering / ML

1. What is EDA?

Concept

EDA = Exploratory Data Analysis

EDA is the systematic process of exploring, summarizing, visualizing, and investigating data to understand its structure, quality, distributions, relationships, and unusual observations.

EDA is performed **before or alongside model development**.

Simple Explanation

Imagine you receive a dataset with:

Age

Salary

Experience

Education

City

Purchased

Before training a model, you should understand:

How many rows?

How many columns?

Any missing values?

Any duplicates?

What is the average salary?

What is the salary distribution?

Does experience relate to salary?

Does education relate to purchasing?

Are there outliers?

That investigation is EDA.

Example

Suppose we have:

Age Experience Salary Purchased

22	1	25000	No
25	3	35000	No
30	6	55000	Yes
35	10	75000	Yes
42	15	100000	Yes

EDA may reveal:

Salary tends to increase with experience in this small dataset.

This is an **observation**, not proof of causation.

Basic Python Inspection

```
import pandas as pd
```

```
df = pd.read_csv("data.csv")
```

```
print(df.head())
```

```
print(df.shape)
```

```
print(df.info())
```

```
print(df.describe())
```

```
print(df.isnull().sum())
```

Important Pandas Functions

Function	Purpose
head()	First rows
tail()	Last rows
shape	Rows and columns
info()	Data types and non-null values
describe()	Statistical summary
columns	Column names
isnull().sum()	Missing values
nunique()	Number of unique values
value_counts()	Frequency of categories

2. Univariate Analysis

Concept

Univariate analysis means analyzing **one variable at a time**.

One Variable

↓

Analyze Distribution

Examples:

- Age distribution
- Salary distribution
- Exam-score distribution
- Number of customers by city

Numerical Variables

For numerical data, examine:

- Mean
- Median
- Minimum
- Maximum
- Standard deviation
- Quartiles
- Distribution
- Outliers

Example

```
print(df["Age"].describe())
```

Histogram

A histogram shows the distribution of numerical data.

```
import matplotlib.pyplot as plt
```

```
plt.hist(df["Age"], bins=10)
```

```
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Age Distribution")
```

```
plt.show()
```

What Can We Learn?

A histogram can help identify:

- Center
- Spread
- Skewness
- Multiple peaks
- Potential outliers

Box Plot

```
plt.boxplot(df["Salary"])
```

```
plt.ylabel("Salary")
plt.title("Salary Distribution")
```

```
plt.show()
```

Box plots are useful for detecting potential outliers.

Box Plot Structure



Categorical Variables

For categorical variables, use frequency counts.

```
print(df["City"].value_counts())
```

Bar Chart

```
df["City"].value_counts().plot(kind="bar")
```

```
plt.xlabel("City")
plt.ylabel("Count")
plt.title("Customers by City")
```

```
plt.show()
```

Questions to Ask

- Which category occurs most frequently?
- Which category occurs least frequently?

- Is the dataset balanced?
- Are there unexpected categories?

3. Bivariate Analysis

Concept

Bivariate analysis studies the relationship between **two variables**.

Variable A



Variable B

Examples:

- Age vs Salary
- Study Hours vs Exam Score
- Experience vs Salary
- Gender vs Churn

Numerical + Numerical

A **scatter plot** is commonly used.

```
plt.scatter(
    df["Experience"],
    df["Salary"]
)

plt.xlabel("Experience")
plt.ylabel("Salary")
plt.title("Experience vs Salary")

plt.show()
```

Possible Patterns

Positive relationship:

Salary



As experience increases, salary tends to increase.

Numerical + Categorical

A box plot is useful.

```
import seaborn as sns

sns.boxplot(
    data=df,
    x="Education",
    y="Salary"
)
```

```
plt.show()
```

This allows us to compare salary distributions across education categories.

Categorical + Categorical

A contingency table can help.

```
pd.crosstab(
    df["Gender"],
    df["Purchased"]
)
```

Example:

No Yes

Male 40 30

Female 25 35

This can help explore differences between categories.

4. Multivariate Analysis

Concept

Multivariate analysis examines **three or more variables simultaneously**.

Feature 1 —

Feature 2 —|—→ Analyze Relationships

Feature 3 —|

Feature 4 —|

Example

Suppose we analyze:

Study Hours

Attendance

Assignment Score

Exam Score

Instead of examining only:

Study Hours → Exam Score

we can investigate all variables together.

Pair Plot

Seaborn provides pairplot():

```
sns.pairplot(df)
```

```
plt.show()
```

A pair plot shows relationships between multiple numerical variables.

It can help identify:

- Positive relationships
- Negative relationships
- Clusters
- Distributions
- Potential outliers

Example

```
data = df[
    "Study_Hours",
    "Attendance",
    "Assignment_Score",
    "Exam_Score"
]
```

```
sns.pairplot(data)
```

```
plt.show()
```

5. Correlation

Concept

Correlation measures the strength and direction of a relationship between two numerical variables.

The commonly used **Pearson correlation coefficient** ranges from:

[
 $-1 \leq r \leq 1$
]

Interpretation

Correlation General Interpretation

+1	Perfect positive linear relationship
0	No linear correlation
-1	Perfect negative linear relationship
0.7 to 1	Strong positive
0.3 to 0.7	Moderate positive
0 to 0.3	Weak positive
-0.3 to 0	Weak negative
-0.7 to -0.3	Moderate negative
-1 to -0.7	Strong negative

These ranges are **rules of thumb**, not universal scientific thresholds.

Positive Correlation

X increases

↓

Y generally increases

Example:

Study Hours ↑

↓

Exam Score ↑

Negative Correlation

X increases

↓

Y generally decreases

Example:

Price ↑

↓

Demand ↓

Correlation Formula

Pearson correlation can be expressed as:


```
[
r =
\frac{\operatorname{Cov}(X,Y)}
{\sigma_X\sigma_Y}
]
```

where:

- $\operatorname{Cov}(X,Y)$ = covariance
- σ_X = standard deviation of X
- σ_Y = standard deviation of Y

Calculate Correlation in Pandas

```
correlation = df.corr(numeric_only=True)

print(correlation)
```

Correlation Heatmap

```
correlation = df.corr(numeric_only=True)

sns.heatmap(
    correlation,
    annot=True
)

plt.title("Correlation Heatmap")
plt.show()
```

A heatmap makes relationships easier to inspect visually.

⚠ Correlation ≠ Causation

This is a very important interview concept.

If:

Ice Cream Sales ↑

↓

Swimming Incidents ↑

the two variables may be correlated, but ice cream sales do not necessarily cause swimming incidents.

A third variable, such as **hot weather**, may influence both.

Correlation shows association, not necessarily causation.

6. Data Visualization

Concept

Data visualization means representing data graphically to make patterns and relationships easier to understand.

Common Visualization Types

Chart	Common Use
Histogram	Distribution
Bar Chart	Category comparison
Line Chart	Trends over time
Scatter Plot	Relationship between numerical variables
Box Plot	Distribution + potential outliers
Heatmap	Correlation / matrix values
Violin Plot	Distribution comparison
Pair Plot	Multiple relationships

Histogram

```
sns.histplot(
    data=df,
    x="Salary"
)
```

```
plt.show()
```

Purpose: Understand distribution.

Bar Chart

```
sns.countplot(
    data=df,
    x="City"
)
```

```
plt.show()
```

Purpose: Compare categories.

Scatter Plot

```
sns.scatterplot(
    data=df,
    x="Experience",
    y="Salary"
)
```

```
plt.show()
```

Purpose: Study relationships.

Box Plot

```
sns.boxplot(
    data=df,
    x="Department",
    y="Salary"
)
```

```
plt.show()
```

Purpose: Compare distributions and inspect potential outliers.

Line Chart

For time-series data:

```
plt.plot(
    df["Date"],
    df["Sales"]
)
```

```
plt.xlabel("Date")
plt.ylabel("Sales")
```

```
plt.show()
```

Purpose: Identify trends over time.

7. Finding Patterns & Relationships

Concept

The main purpose of EDA is not simply to create graphs.

The goal is to **extract meaningful insights from data**.

Pattern 1 — Trend

Example:

Experience ↑

↓

Salary generally ↑

Possible observation:

Salary tends to increase with experience.

Pattern 2 — Distribution

Suppose most customers are between:

Age = 20–35

You might conclude:

The dataset is concentrated around younger customers.

Pattern 3 — Outliers

Suppose most salaries are:

₹30,000 – ₹80,000

but one record is:

₹10,00,000

Investigate whether it is:

- A genuine high salary
 - Data-entry error
 - Different employee category
 - Measurement issue
-

Pattern 4 — Class Imbalance

Suppose a churn dataset contains:

No Churn → 90%

Churn → 10%

This is an imbalanced target distribution.

A model predicting "No Churn" for everyone could appear highly accurate while being poor at detecting churners.

This connects directly to **Module 9 — Feature Engineering and imbalanced datasets**.

Pattern 5 — Strong Feature Relationship

Suppose:

Study Hours ↔ Exam Score

Correlation = 0.82

This suggests a strong positive linear association in the sample.

It may indicate that the feature could be useful for prediction, but correlation alone does not determine feature importance.

Complete EDA Example

Let's create a small student dataset.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    "Study_Hours": [2, 3, 4, 5, 6, 7, 8, 9],
    "Attendance": [65, 70, 72, 78, 82, 88, 92, 95],
    "Assignment_Score": [50, 55, 60, 65, 72, 80, 88, 92],
    "Exam_Score": [45, 50, 58, 65, 70, 78, 86, 90]
}

df = pd.DataFrame(data)

print(df.head())
print(df.describe())
```

Step 1 — Dataset Shape

```
print(df.shape)
```

Step 2 — Check Missing Values

```
print(df.isnull().sum())
```

Step 3 — Univariate Analysis

```
sns.histplot(
    data=df,
    x="Exam_Score"
)

plt.title("Exam Score Distribution")
plt.show()
```

Step 4 — Bivariate Analysis

```
sns.scatterplot(
    data=df,
    x="Study_Hours",
    y="Exam_Score"
)

plt.title("Study Hours vs Exam Score")
plt.show()
```

Step 5 — Multivariate Analysis

```
sns.pairplot(df)
```

```
plt.show()
```

Step 6 — Correlation

```
correlation = df.corr()
```

```
print(correlation)
```

Step 7 — Heatmap

```
sns.heatmap(  
correlation,  
    annot=True  
)
```

```
plt.title("Feature Correlation")  
plt.show()
```

🔍 How to Read EDA Results

Suppose your heatmap gives:

	Study Hours	Attendance	Exam Score
Study Hours	1.00	0.85	0.90
Attendance	0.85	1.00	0.88
Exam Score	0.90	0.88	1.00

You could report:

Finding 1

Study hours have a strong positive linear relationship with exam score.

Finding 2

Attendance also has a strong positive relationship with exam score.

Finding 3

Study hours and attendance are also strongly related to each other, so these variables may contain overlapping information.

Finding 4

Further analysis is required before concluding that either variable **causes** higher exam scores.

🧠 EDA Checklist

Before moving to Machine Learning, ask:

Dataset

- How many rows?

- How many columns?
- What does each column represent?
- What is the target?

Data Quality

- Missing values?
- Duplicates?
- Invalid values?
- Outliers?
- Incorrect data types?

Numerical Variables

- Mean?
- Median?
- Standard deviation?
- Distribution?
- Skewness?
- Outliers?

Categorical Variables

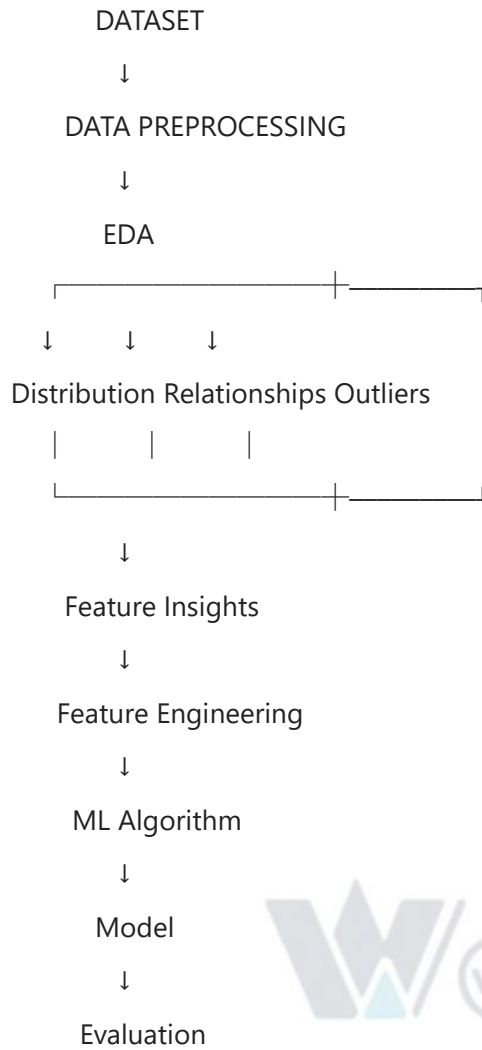
- Number of categories?
- Most common category?
- Rare categories?
- Class imbalance?

Relationships

- Correlation?
- Positive relationships?
- Negative relationships?
- Non-linear patterns?
- Potentially redundant features?

Visualization

- Histograms
- Bar charts
- Box plots
- Scatter plots
- Heatmaps
- Pair plots



Module 5 Quick Revision

Topic	Key Idea
EDA	Understand and investigate data
Univariate	Analyze one variable
Bivariate	Analyze two variables
Multivariate	Analyze multiple variables
Correlation	Measures association, commonly linear
Histogram	Shows numerical distribution
Bar Chart	Compares categories
Scatter Plot	Shows relationship between two numerical variables
Box Plot	Shows distribution and potential outliers
Heatmap	Visualizes matrix/correlation values
Pair Plot	Shows multiple pairwise relationships
Pattern Detection	Find trends, clusters, outliers, relationships
Correlation $\neq$ Causation	Association doesn't prove cause

Module 5 Interview Questions

Basic

1. What is EDA?
2. Why is EDA important in Machine Learning?
3. What is univariate analysis?
4. What is bivariate analysis?
5. What is multivariate analysis?
6. What is data visualization?
7. Why do we visualize data?

Intermediate

8. Which graph is suitable for numerical distribution?
9. Which graph is suitable for categorical data?
10. When would you use a scatter plot?
11. What is a box plot?
12. How can you identify potential outliers?
13. What is correlation?
14. What is Pearson correlation?
15. What does a correlation of +1 mean?
16. What does a correlation of -1 mean?
17. What does correlation close to 0 mean?
18. What is a correlation heatmap?

Advanced

19. What is the difference between correlation and causation?
20. Can two variables have a non-linear relationship but low Pearson correlation?
21. How would you detect class imbalance during EDA?
22. How can EDA help with feature selection?
23. Why should EDA be performed before model training?
24. How would you identify data leakage during exploratory analysis?
25. How can visualization reveal problems that summary statistics might miss?

Practice Questions

Level 1

1. Load a CSV file using Pandas.
2. Display the first 10 rows.
3. Find the number of rows and columns.
4. Display data types.

5. Find missing values.
6. Generate statistical summary.

Level 2

7. Perform univariate analysis on:

Age

Salary

8. Create histograms.
9. Create box plots.
10. Analyze a categorical column using `value_counts()`.

Level 3

11. Perform bivariate analysis on:

Experience vs Salary

12. Create a scatter plot.
13. Analyze:

Department vs Salary

using a box plot.

14. Analyze:

Gender vs Churn

using a contingency table.

Level 4

15. Calculate the correlation matrix.
16. Create a correlation heatmap.
17. Create a pair plot.
18. Identify the three strongest feature relationships.
19. Identify possible outliers.
20. Write **five business/ML insights** from your EDA.

Module 5 Mini Project — Student Performance EDA

Use:

Student_ID

Age

Gender

Study_Hours

Attendance

Assignment_Score

Previous_Score

Exam_Score

Perform the following:

1. Data Understanding

Rows

Columns

Data Types

Missing Values

Duplicates

2. Univariate Analysis

Analyze:

Age

Study_Hours

Attendance

Exam_Score

Gender

3. Bivariate Analysis

Investigate:

Study_Hours ↔ Exam_Score

Attendance ↔ Exam_Score

Previous_Score ↔ Exam_Score

Gender ↔ Exam_Score

4. Multivariate Analysis

Use:

```
sns.pairplot(df)
```

5. Correlation Analysis

Create:

```
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True
)
```

6. Pattern Finding

Write at least **5 observations**.

Example:

Students with higher study hours generally show higher exam scores in the sample.

7. Final EDA Report

Your final report should contain:

1. Dataset Overview
2. Data Quality Findings
3. Univariate Analysis
4. Bivariate Analysis
5. Multivariate Analysis
6. Correlation Analysis
7. Visualizations
8. Outlier Findings
9. Key Patterns
10. ML Recommendations

🎯 Module 5 Outcome

After completing this module, a student should be able to take a raw dataset and answer:

“What is inside this data, how is it distributed, how are the variables related, what problems exist, and what useful patterns should we consider before training an ML model?”

That is the core purpose of **EDA** and prepares students for **Module 6 — Supervised Learning**.

Module 6 — Supervised Learning

Supervised Learning is one of the most important areas of Machine Learning. It is used when we have **input data along with known output/target values** and want a model to learn the relationship between them.

1. What is Supervised Learning?

Concept

In **supervised learning**, a machine-learning algorithm learns from **labeled data**.

The dataset contains:

- **Features (X)** → inputs
- **Target (y)** → known output

The model learns:

[
X $\rightarrow$ y
]

and then uses this learned relationship to predict the target for new, unseen data.

Simple Example

Suppose we have:

Study Hours Attendance Exam Score

2	65	45
4	75	60
6	85	75
8	95	90

Here:

Features:

Study Hours

Attendance

Target:

Exam Score

The model learns from existing examples and predicts the score for a new student.

Supervised Learning Workflow

Labeled Dataset

↓

Data Preprocessing

↓

EDA

↓

Feature Engineering

↓

Train / Validation / Test Split



Choose Algorithm



Train Model



Make Predictions



Evaluate Model



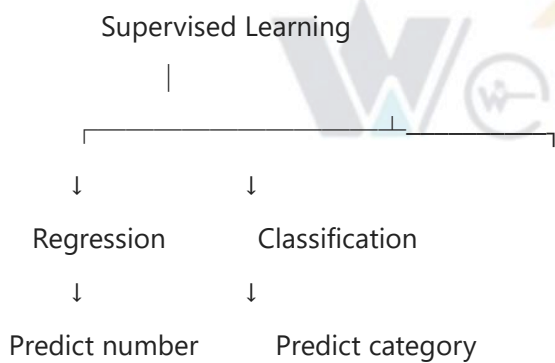
Tune Model



Final Model

2. Regression vs Classification

Supervised learning is commonly divided into:



Regression

Regression predicts a **continuous numerical value**.

Examples

- House price → ₹75,00,000
- Salary → ₹65,000
- Temperature → 32.5°C
- Exam score → 82.5
- Sales → ₹2,50,000

Common Algorithms

- Linear Regression
- Multiple Linear Regression

- Polynomial Regression

Classification

Classification predicts a **class/category**.

Examples

Spam / Not Spam

Yes / No

Pass / Fail

Disease / No Disease

Churn / No Churn

Common Algorithms

- Logistic Regression
- KNN
- Decision Tree
- Random Forest
- SVM
- Naive Bayes

Regression vs Classification

Feature	Regression	Classification
Output	Numerical	Category
Example	₹50,000	Spam
Prediction	Continuous	Discrete
Common Metric	MAE, MSE, RMSE, R^2	Accuracy, Precision, Recall, F1
Example Algorithm	Linear Regression	Logistic Regression

3. Linear Regression

Concept

Linear Regression predicts a continuous target using a linear relationship between the input features and target.

For one feature:

$$[y = mx + c]$$

where:

- y = predicted output

- x = input feature
- m = slope/coefficient
- c = intercept

In ML notation:

$$\hat{y} = w_0 + w_1x$$

Example

Suppose:

Study Hours $\rightarrow$ Exam Score

A model may learn:

$$\text{Score} = 5 + 10(\text{Hours})$$

For 6 hours:

$$\text{Score} = 5 + 10(6) = 65$$

So the predicted score is:

65

Multiple Linear Regression

When there are multiple features:

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

Example:

Exam Score =

w_0

+ $w_1(\text{Study Hours})$

+ $w_2(\text{Attendance})$

+ $w_3(\text{Assignment Score})$

Training Objective

Linear regression commonly estimates parameters by minimizing **Mean Squared Error (MSE)**:


```
[  
MSE=\frac{1}{n}\sum_{i=1}^n(y_i-\hat{y}_i)^2  
]
```

Python Example

```
import pandas as pd  
from sklearn.model_selection import train_test_split  
from sklearn.linear_model import LinearRegression  
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score  
  
data = {  
    "Study_Hours": [2, 3, 4, 5, 6, 7, 8],  
    "Exam_Score": [45, 50, 58, 65, 70, 80, 88]  
}  
  
df = pd.DataFrame(data)  
  
X = df[["Study_Hours"]]  
y = df["Exam_Score"]  
  
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42  
)  
  
model = LinearRegression()  
  
model.fit(X_train, y_train)  
  
predictions = model.predict(X_test)  
  
print("Predictions:", predictions)  
print("MAE:", mean_absolute_error(y_test, predictions))  
print("R²:", r2_score(y_test, predictions))
```

Real-World Applications

- House price prediction
- Sales forecasting
- Salary prediction

- Demand forecasting
 - Revenue prediction
-

4. Multiple Linear Regression

Concept

Multiple Linear Regression uses **multiple independent variables** to predict one continuous target.

Example

Predict house price using:

Area

Bedrooms

Location Score

Age

Parking

Mathematical form:

```
[  
Price =  
w_0 +  
w_1Area +  
w_2Bedrooms +  
w_3Location +  
w_4Age +  
w_5Parking  
]
```



Python

```
from sklearn.linear_model import LinearRegression  
  
X = df[  
[  
    "Area",  
    "Bedrooms",  
    "Age"  
]  
]  
  
y = df["Price"]  
  
model = LinearRegression()
```

```
model.fit(X, y)
```

Interpretation

Each coefficient indicates how the prediction changes with a one-unit change in that feature, **holding the other included features constant**.

5. Polynomial Regression

Concept

Linear regression assumes a linear relationship.

But real-world relationships may be curved.

Example:



Polynomial regression adds powers of the original features.

For degree 2:

```
[  
y=b_0+b_1x+b_2x^2  
]
```

For degree 3:

```
[  
y=b_0+b_1x+b_2x^2+b_3x^3  
]
```

Python

```
from sklearn.preprocessing import PolynomialFeatures  
from sklearn.linear_model import LinearRegression
```

```
poly = PolynomialFeatures(degree=2)
```

```
X_poly = poly.fit_transform(X)
```

```
model = LinearRegression()
```

```
model.fit(X_poly, y)
```

Important

Higher polynomial degrees can make the model excessively flexible and increase the risk of **overfitting**.

6. Logistic Regression

Concept

Despite its name, **Logistic Regression is primarily a classification algorithm**.

It predicts the probability of a class.

For binary classification:

0 → No

1 → Yes

The model uses the logistic/sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Its output lies between 0 and 1.

Example

Probability = 0.82

If the classification threshold is 0.5:

$$0.82 \geq 0.5$$

↓

Class = 1

Real-World Example

Customer churn:

Features

↓

Logistic Regression

↓

Probability of Churn

↓

0.87

↓

Likely Churn

Python

```
from sklearn.linear_model import LogisticRegression
```

```
model = LogisticRegression()
```

```
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

```
print(predictions)
```

Probability:

```
probabilities = model.predict_proba(X_test)
```

```
print(probabilities)
```

7. K-Nearest Neighbors (KNN)

Concept

KNN predicts a new data point based on nearby training examples.

The basic idea:

Similar data points tend to have similar labels.

Example

Suppose we want to classify a new customer.

Existing Customers

• •
• •

X ← New customer

○ ○
○ ○

If most nearby points belong to class •, the new point may be classified as •.

K Value

K represents the number of neighbors considered.

For:

K = 3

the model considers the three nearest points.

Distance

A common distance measure is Euclidean distance:

```
[
d=\sqrt{\sum_{i=1}^n(x_i-y_i)^2}
]
```

For two dimensions:

```
[
d=\sqrt{(x_1-y_1)^2+(x_2-y_2)^2}
]
```

Python

```
from sklearn.neighbors import KNeighborsClassifier
```

```
model = KNeighborsClassifier(n_neighbors=5)
```

```
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

Important

KNN is distance-based, so **feature scaling is usually important**.

8. Decision Tree

Concept

A **Decision Tree** makes predictions through a sequence of decision rules.

Example:

```

Income > 50K?
  /   \
Yes    No
  /     \
Credit Good?   Reject
  /   \
Yes    No
  /     \
Approve  Reject
    
```

It resembles a flowchart.

Components

Root

↓

Decision Node

↓

Branch

↓

Leaf

Root

The first decision.

Internal Node

A decision based on a feature.

Leaf

Final prediction.

Classification

A tree may use criteria such as:

- Gini impurity
- Entropy / information gain

Gini Impurity

[

$$\text{Gini} = 1 - \sum_{i=1}^C p_i^2$$

]

Python

```
from sklearn.tree import DecisionTreeClassifier
```

```
model = DecisionTreeClassifier(  
    max_depth=5,  
    random_state=42
```

```
)
```

```
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

Advantages

- Easy to understand
- Handles nonlinear relationships
- Little preprocessing required
- Works with numerical and appropriately encoded categorical features

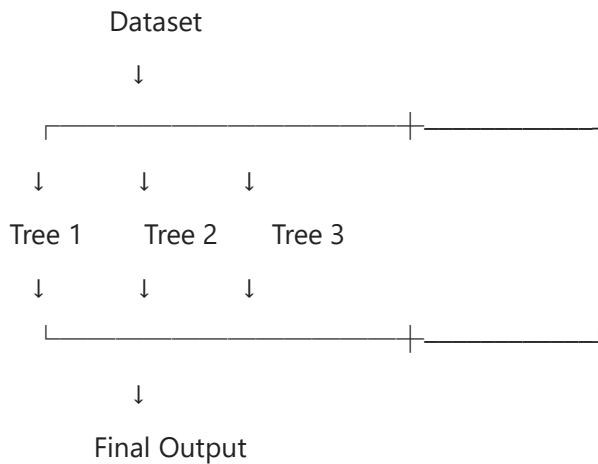
Limitation

Deep trees can overfit.

9. Random Forest

Concept

Random Forest is an ensemble learning algorithm that combines many decision trees.



For classification, trees generally vote.

For regression, their predictions are commonly averaged.

Why Multiple Trees?

A single decision tree can be unstable and overfit.

Combining many trees generally produces a more robust model.

Python

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
```

```
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

Important Parameters

`n_estimators`

`max_depth`

`min_samples_split`

`min_samples_leaf`

max_features

Applications

- Fraud detection
- Customer churn
- Credit risk
- Classification
- Regression

10. Support Vector Machine (SVM)

Concept

Support Vector Machine finds a decision boundary that separates classes.

For a simple two-class problem:

Class A Class B

• • • | ○ ○ ○
• • • | ○ ○ ○
• • • | ○ ○ ○
↑

Decision boundary



SVM tries to find a boundary with a large **margin** between classes.

Support Vectors

The data points closest to the decision boundary are called **support vectors**.

They strongly influence the boundary.

Kernel Trick

SVM can model nonlinear relationships using kernels.

Common kernels:

- Linear
- Polynomial
- RBF
- Sigmoid

Python

```
from sklearn.svm import SVC
```

```
model = SVC(
    kernel="rbf",
    probability=True
)
```

```
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

Important

SVM is sensitive to feature scale, so scaling is often important.

11. Naive Bayes

Concept

Naive Bayes is a probabilistic classification algorithm based on **Bayes' theorem**.

Bayes' theorem:

```
[
P(A|B)=
\frac{P(B|A)P(A)}{P(B)}
]
```

The "naive" assumption is that features are conditionally independent given the class.

This assumption is often unrealistic, but the method can still work well in many applications.

Example — Spam Detection

Suppose an email contains:

"Congratulations"

"Free"

"Offer"

Naive Bayes estimates:

$P(\text{Spam} | \text{Words})$

and

$P(\text{Not Spam} | \text{Words})$

The class with the larger posterior probability can be selected.

Python

For text classification:

```
from sklearn.naive_bayes import MultinomialNB
```

```
model = MultinomialNB()
```

```
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

Applications

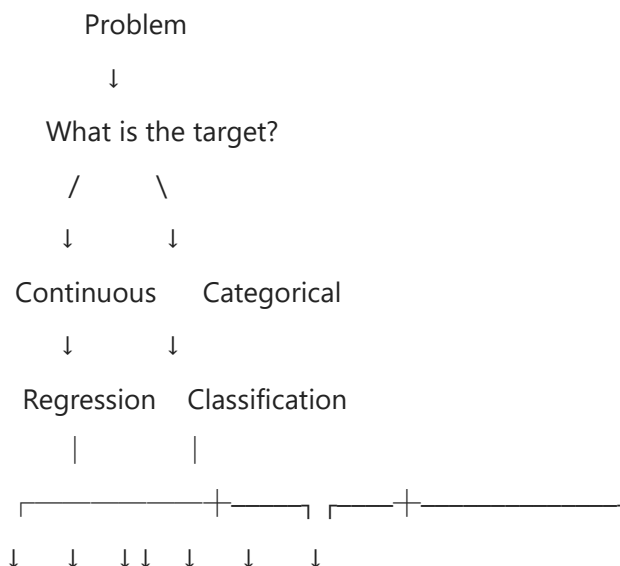
- Spam detection
- Text classification
- Sentiment classification
- Document classification
- News categorization

Algorithm Comparison

Algorithm	Type	Common Use	Scaling Usually Needed?
Linear Regression	Regression	Price prediction	Not strictly
Multiple Linear Regression	Regression	Sales/price prediction	Depends
Polynomial Regression	Regression	Nonlinear regression	Depends
Logistic Regression	Classification	Binary classification	Often helpful
KNN	Classification/Regression	Similarity-based prediction	Yes, usually
Decision Tree	Classification/Regression	Rule-based prediction	Usually not
Random Forest	Classification/Regression	General-purpose tabular ML	Usually not
SVM	Classification/Regression	Classification	Usually yes
Naive Bayes	Classification	Text classification	Depends on representation

Choosing the Right Algorithm

A practical approach:



Linear Poly RF Log KNN Tree SVM

But algorithm selection should also consider:

- Dataset size
- Feature types
- Noise
- Nonlinearity
- Interpretability
- Computational requirements
- Class imbalance
- Latency requirements
- Deployment constraints

Complete Supervised Learning Example

Let's build a small **Student Performance Classification** problem.

Suppose we want to predict whether a student will **Pass or Fail**.

Dataset

Study_Hours

Attendance

Assignment_Score

Pass



Step 1 — Create Data

```
import pandas as pd
```

```
data = {  
    "Study_Hours": [2, 3, 4, 5, 6, 7, 8, 9],  
    "Attendance": [60, 65, 70, 75, 80, 85, 90, 95],  
    "Assignment_Score": [45, 50, 55, 65, 70, 78, 85, 92],  
    "Pass": [0, 0, 0, 1, 1, 1, 1, 1]  
}
```

```
df = pd.DataFrame(data)
```

Step 2 — Separate Features and Target

```
X = df[  
    [  
        "Study_Hours",
```

```

        "Attendance",
        "Assignment_Score"
    ]
]

```

```
y = df["Pass"]
```

Step 3 — Split Data

```
from sklearn.model_selection import train_test_split
```

```

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42,
    stratify=y
)

```

For classification, stratify=y can help preserve class proportions in the split when appropriate.

Step 4 — Train Random Forest

```
from sklearn.ensemble import RandomForestClassifier
```

```

model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

```

```
model.fit(X_train, y_train)
```

Step 5 — Predict

```
predictions = model.predict(X_test)
```

```
print(predictions)
```

Step 6 — Evaluate

```
from sklearn.metrics import accuracy_score
```

```

accuracy = accuracy_score(
    y_test,
    predictions
)

```

)

```
print("Accuracy:", accuracy)
```

Detailed classification evaluation will be covered in **Module 8 — Model Evaluation**.

⚠ Overfitting vs Underfitting

These are important supervised-learning concepts.

Overfitting

The model learns the training data too closely, including noise.

Training Performance → Very High

Test Performance → Much Lower

Example

A very deep decision tree may memorize training examples.

Underfitting

The model is too simple to capture the underlying pattern.

Training Performance → Poor

Test Performance → Poor

Ideal Model

Training Performance → Good

Validation Performance → Good

Test Performance → Good

📦 Bias–Variance Intuition

Too Simple

↓

High Bias

↓

Underfitting

Balanced Model

↓

Good Generalization

Too Complex

↓

High Variance

↓

Overfitting

Model Training vs Prediction

This distinction is important.

Training

```
model.fit(X_train, y_train)
```

The model learns parameters/patterns from training data.

Prediction

```
model.predict(X_test)
```

The trained model generates predictions for unseen data.

Hyperparameters

A **hyperparameter** is a configuration chosen before/during model training rather than learned directly as a model parameter.

Examples:

KNN

```
KNeighborsClassifier(
    n_neighbors=5
)
```

Decision Tree

```
DecisionTreeClassifier(
    max_depth=5
)
```

Random Forest

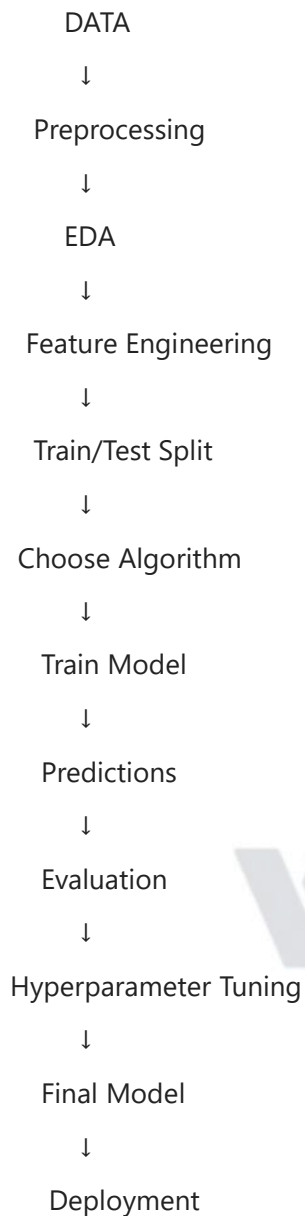
```
RandomForestClassifier(
    n_estimators=100,
    max_depth=10
)
```

SVM

```
SVC(
    C=1.0,
    kernel="rbf"
)
```

Hyperparameter tuning will connect with **cross-validation in Module 8**.

Supervised Learning Workflow



Real-World Applications

1. House Price Prediction

Area

Bedrooms

Location

Age

↓

Regression

↓

House Price

2. Customer Churn

Age

Tenure

Charges

Support Calls

↓

Classification

↓

Churn / No Churn

3. Spam Detection

Email Text

↓

NLP Features

↓

Naive Bayes / SVM

↓

Spam / Not Spam

4. Loan Approval

Income

Credit Score

Employment

Loan Amount

↓

Classification

↓

Approved / Rejected

Module 6 Quick Revision

Topic	Key Point
Supervised Learning	Learns from labeled data
Regression	Predicts continuous values
Classification	Predicts categories
Linear Regression	Fits a linear relationship
Multiple Linear Regression	Uses multiple features

Topic	Key Point
Polynomial Regression	Models polynomial relationships
Logistic Regression	Classification using probabilities
KNN	Predicts from nearby observations
Decision Tree	Uses decision rules
Random Forest	Combines many decision trees
SVM	Finds a separating boundary with a margin
Naive Bayes	Probabilistic classifier using Bayes theorem
Overfitting	Model performs too well on training data but generalizes poorly
Underfitting	Model is too simple
Hyperparameter	Configuration selected before/during training

Module 6 Interview Questions

Basic

1. What is supervised learning?
2. What is labeled data?
3. What is the difference between a feature and target?
4. What is regression?
5. What is classification?
6. Give three real-world examples of supervised learning.

Linear Regression

7. What is linear regression?
8. What is the equation of simple linear regression?
9. What is multiple linear regression?
10. What is the difference between linear and polynomial regression?
11. What is the objective of linear regression?
12. What is MSE?

Logistic Regression

13. Why is logistic regression used for classification?
14. What is the sigmoid function?
15. What does the output of logistic regression represent?
16. What is a classification threshold?

KNN

17. What is KNN?
18. What does K represent in KNN?
19. Why is scaling important for KNN?

20. What happens when K is too small?

21. What happens when K is too large?

Decision Tree & Random Forest

22. What is a decision tree?

23. What is a leaf node?

24. What is Gini impurity?

25. What is overfitting in a decision tree?

26. What is Random Forest?

27. Why does Random Forest generally perform better than a single tree?

28. What is ensemble learning?

SVM

29. What is SVM?

30. What is a support vector?

31. What is a margin?

32. What is the kernel trick?

33. Why does SVM generally benefit from feature scaling?

Naive Bayes

34. What is Naive Bayes?

35. Why is it called "naive"?

36. What is Bayes' theorem?

37. Why is Naive Bayes popular for text classification?

Advanced

38. What is overfitting?

39. What is underfitting?

40. What is bias-variance tradeoff?

41. What is a hyperparameter?

42. Why should we not tune the final model directly on the test set?

43. Why is cross-validation useful?

44. How would you choose an algorithm for a new ML problem?

45. What is the difference between model parameters and hyperparameters?

Practice Questions

Level 1

1. Explain supervised learning with a real-world example.

2. Identify whether each problem is regression or classification:

House Price

Spam Detection
 Salary Prediction
 Customer Churn
 Temperature Prediction
 Loan Approval

Level 2

3. Build a Linear Regression model for house-price prediction.
 4. Build a Logistic Regression model for customer churn.
 5. Implement KNN classification.
 6. Train a Decision Tree.
 7. Train a Random Forest.
 8. Train an SVM.
 9. Implement Naive Bayes for spam classification.
-

Level 3

10. Compare:
 Logistic Regression
 KNN
 Decision Tree
 Random Forest
 SVM

on the same classification dataset.

11. Compare their performance.
 12. Identify which model overfits.
 13. Experiment with different K values in KNN.
 14. Experiment with different tree depths.
 15. Compare a single Decision Tree with Random Forest.
-

Module 6 Industry Project

Customer Churn Prediction

Problem

A telecom company wants to predict whether a customer will leave the company.

Features

Age
 Tenure

Monthly_Charges

Contract_Type

Support_Calls

Payment_Method

Internet_Service

Target

Churn

0 → No Churn

1 → Churn

Project Pipeline

Customer Dataset



Data Preprocessing



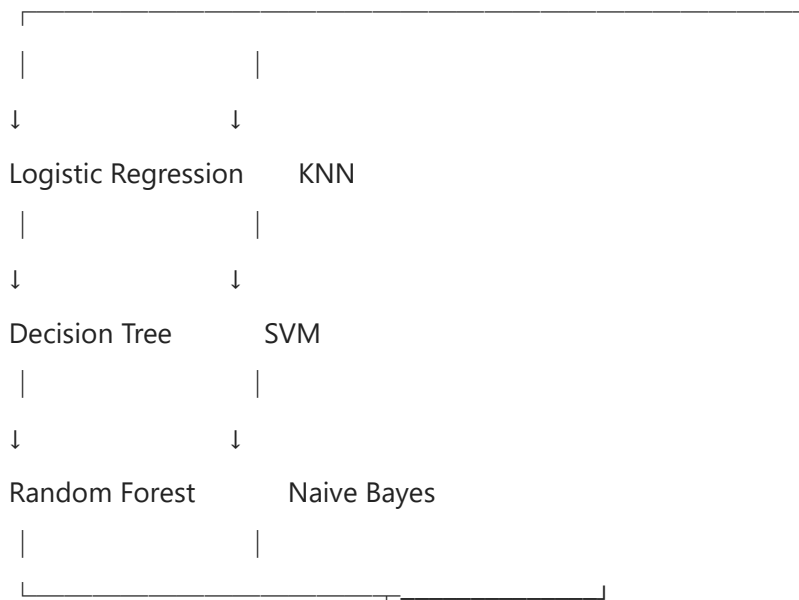
EDA



Feature Engineering



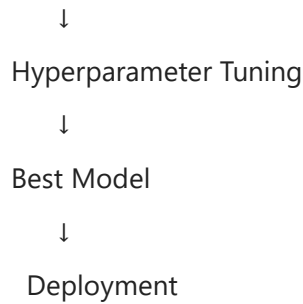
Train/Test Split



Model Comparison



Model Evaluation



Final Student Deliverables

1. Dataset
2. Data Preprocessing Notebook
3. EDA Notebook
4. Model Training Notebook
5. Model Comparison
6. Evaluation Report
7. Best Model
8. Prediction Demo
9. README Documentation

🎯 Module 6 Outcome

After completing this module, students should be able to take a **labeled dataset**, identify whether the problem is **regression or classification**, select an appropriate supervised-learning algorithm, train the model, generate predictions, recognize overfitting/underfitting, and prepare the model for proper evaluation in **Module 8 — Model Evaluation**.

📊 Module 7 — Unsupervised Learning

Overview

Unsupervised Learning is a branch of Machine Learning where the model learns patterns and structures from **unlabeled data**.

Unlike supervised learning, there is **no target/output column** provided during training.

Supervised vs Unsupervised

SUPERVISED LEARNING

Features + Target



ML Algorithm



Prediction

UNSUPERVISED LEARNING

Features only



ML Algorithm



Patterns / Groups / Structure

Example

Suppose a company has customer data:

Customer Age Income Spending Score

A	22	25K	80
B	25	28K	85
C	45	90K	30
D	48	95K	25

There is no predefined label such as:

Young Customer

High Value Customer

Low Value Customer

An unsupervised algorithm can discover groups automatically.

1. What is Unsupervised Learning?

Concept

Unsupervised learning finds **hidden patterns, structures, relationships, or groups** in data without predefined target labels.

Common applications include:

- Customer segmentation
- Anomaly detection
- Recommendation systems
- Document grouping
- Image grouping

- Data compression
- Visualization
- Pattern discovery

Simple Explanation

Imagine giving a teacher 100 photographs but no labels.

The teacher may notice:

Group 1 → Similar images

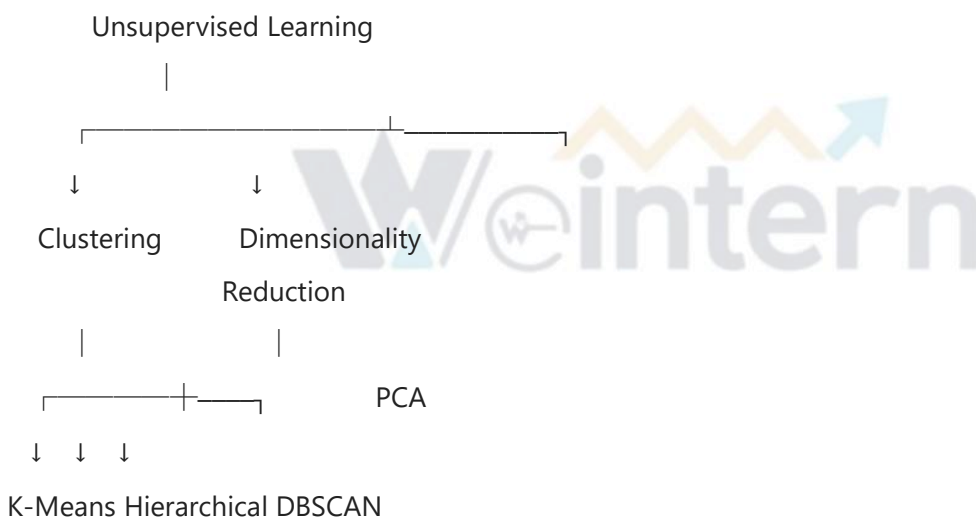
Group 2 → Similar images

Group 3 → Similar images

The groups were not provided beforehand.

This is the basic idea of unsupervised learning.

Major Types



Advantages

- Doesn't require labeled data
- Can discover hidden patterns
- Useful for exploratory analysis
- Helps with customer segmentation
- Can reduce high-dimensional data

Limitations

- Results can be difficult to interpret
- There may be no single "correct" grouping
- Sensitive to preprocessing and algorithm settings

- Choosing the number of clusters can be difficult
- Evaluation is often less straightforward than supervised learning

2. Clustering

Concept

Clustering is the process of grouping similar data points together.

The objective is generally:

Points within the same cluster should be relatively similar, while points in different clusters should be relatively different.

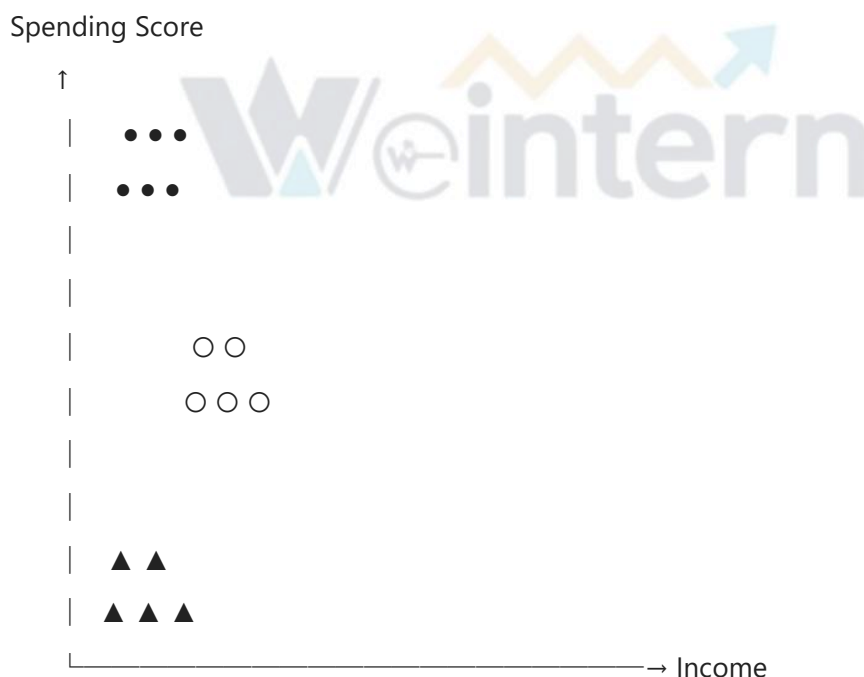
Example

Suppose we have customers based on:

Annual Income

Spending Score

The algorithm might discover:



The algorithm can identify three groups without being told their names.

We might later interpret them as:

Cluster 1 → High income / high spending

Cluster 2 → High income / low spending

Cluster 3 → Low income / high spending

The labels are **our interpretation**; the algorithm itself only produced cluster assignments.

Types of Clustering

1. Partition-based

Example:

- K-Means

2. Hierarchical

Example:

- Agglomerative Clustering

3. Density-based

Example:

- DBSCAN

3. K-Means Clustering

Concept

K-Means divides data into K clusters.

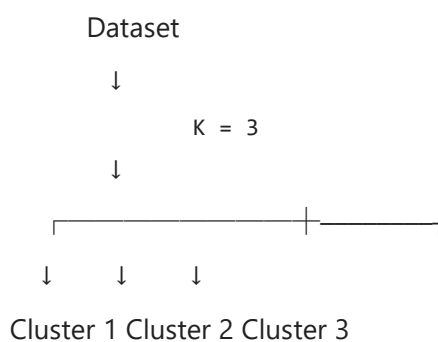
K represents the number of clusters we want to create.

Example

Suppose:

$$K = 3$$

The algorithm tries to divide the dataset into three groups.



How K-Means Works

Step 1 — Select K

Choose the number of clusters.

Example:

$$K = 3$$

Step 2 — Initialize Centroids

The algorithm starts with K initial cluster centers called **centroids**.

Step 3 — Assign Points

Each data point is assigned to the nearest centroid.

Step 4 — Recalculate Centroids

The centroid of each cluster is recalculated based on the assigned points.

Step 5 — Repeat

The assignment and centroid-update steps continue until the solution stabilizes according to the algorithm's stopping criteria.

Visual Process

Initial Data



Choose K



Initialize Centroids



Assign Points



Calculate New Centroids



Assign Again



Repeat



Final Clusters

K-Means Objective Function

K-Means minimizes the **within-cluster sum of squared distances**:

$$J = \sum_{k=1}^K \sum_{x_i \in C_k} |x_i - \mu_k|^2$$

where:

- K = number of clusters
- C_k = cluster k
- x_i = data point

- μ_k = centroid of cluster k

The goal is to make points close to their assigned centroid.

Python Example

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

data = {
    "Income": [20, 22, 25, 28, 80, 85, 90, 95],
    "Spending_Score": [85, 80, 88, 90, 25, 30, 28, 20]
}

df = pd.DataFrame(data)

model = KMeans(
    n_clusters=2,
    random_state=42,
    n_init="auto"
)

df["Cluster"] = model.fit_predict(
    df[["Income", "Spending_Score"]]
)

print(df)
```

Visualize Clusters

```
plt.scatter(
    df["Income"],
    df["Spending_Score"],
    c=df["Cluster"]
)

plt.xlabel("Income")
plt.ylabel("Spending Score")
plt.title("Customer Clusters")

plt.show()
```

Choosing K — Elbow Method

One of the common approaches for selecting K is the **Elbow Method**.

We calculate the model's inertia for different values of K.

```
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

inertia = []

for k in range(1, 11):
    model = KMeans(
        n_clusters=k,
        random_state=42,
        n_init="auto"
    )

    model.fit(df[["Income", "Spending_Score"]])

    inertia.append(model.inertia_)

plt.plot(range(1, 11), inertia, marker="o")

plt.xlabel("Number of Clusters")
plt.ylabel("Inertia")
plt.title("Elbow Method")

plt.show()
```

You look for a point where increasing K produces diminishing improvements in inertia.

Inertia



The elbow is a **heuristic**, not a guaranteed mathematically optimal choice.

Advantages of K-Means

- Simple

- Fast
- Easy to understand
- Efficient for many datasets
- Easy to implement

Limitations

- Must choose K
- Sensitive to initialization
- Sensitive to feature scaling
- Sensitive to outliers
- Works best for relatively compact, well-separated clusters
- Can struggle with non-spherical cluster shapes

4. Hierarchical Clustering

Concept

Hierarchical Clustering creates a hierarchy of clusters.

The result can be represented using a **dendrogram**.

There are two major approaches:

Agglomerative

Individual Points

↓

Merge Similar Points

↓

Small Clusters

↓

Larger Clusters

↓

One Large Cluster

Divisive

One Large Cluster

↓

Split

↓

Smaller Clusters

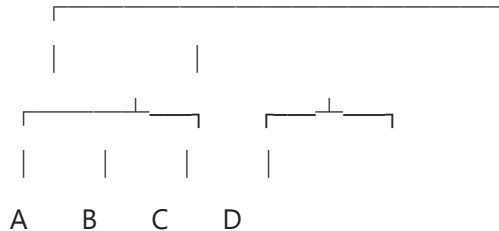
↓

Individual Points

In practice, **agglomerative clustering** is commonly used.

Dendrogram

Example:



We can choose the number of clusters by cutting the dendrogram at an appropriate height.

Python

```

from sklearn.cluster import AgglomerativeClustering

model = AgglomerativeClustering(
    n_clusters=3,
    linkage="ward"
)

clusters = model.fit_predict(
    df[["Income", "Spending_Score"]]
)

```

```
df["Cluster"] = clusters
```

Dendrogram with SciPy

```

from scipy.cluster.hierarchy import dendrogram, linkage
import matplotlib.pyplot as plt

X = df[["Income", "Spending_Score"]]

linked = linkage(X, method="ward")

dendrogram(linked)

plt.title("Hierarchical Clustering Dendrogram")
plt.xlabel("Data Points")
plt.ylabel("Distance")

plt.show()

```

Linkage Methods

Common linkage methods include:

- Single
- Complete
- Average
- Ward

They define how the distance between clusters is calculated.

Advantages

- Does not necessarily require choosing the number of clusters before building the hierarchy
- Dendrogram provides a useful visual representation
- Can reveal nested structures

Limitations

- Can be computationally expensive for large datasets
 - Sensitive to distance/linkage choices
 - Once a merge is made in standard agglomerative clustering, it generally cannot be undone
-

5. DBSCAN

Concept

DBSCAN = Density-Based Spatial Clustering of Applications with Noise

Unlike K-Means, DBSCAN does not require you to specify the number of clusters beforehand.

It groups points based on **density**.

Main Parameters

eps

Maximum neighborhood distance used to determine whether points are nearby.

min_samples

Minimum number of points required for a dense neighborhood.

Types of Points

DBSCAN classifies points conceptually as:

Core Point

Border Point

Noise / Outlier

Core Point

Has enough nearby points within the eps neighborhood.

Border Point

Is close to a core point but may not itself have enough neighbors.

Noise

Doesn't belong to a sufficiently dense region.

Example

• • •
• • • •
• • •

○
○ ○ ○
○

×

DBSCAN can identify the two dense groups while treating × as noise.

Python

```
from sklearn.cluster import DBSCAN

model = DBSCAN(
    eps=0.5,
    min_samples=5
)

clusters = model.fit_predict(
    df[["Income", "Spending_Score"]]
)

df["Cluster"] = clusters
```

```
print(df)
```

In scikit-learn's DBSCAN output, -1 represents noise points.

Advantages

- Does not require K

- Can detect irregularly shaped clusters
- Can identify noise/outliers
- Useful when cluster density is meaningful

Limitations

- Choosing eps can be difficult
- Sensitive to feature scaling
- Can struggle when clusters have very different densities
- Performance can degrade in high-dimensional spaces

⚡ K-Means vs Hierarchical vs DBSCAN

Feature	K-Means	Hierarchical	DBSCAN
Requires K beforehand	Yes	Usually no	No
Based on	Centroids	Hierarchy	Density
Handles noise	Poorly	Depends	Yes
Irregular clusters	Poor	Sometimes	Good
Outlier sensitivity	High	Can be sensitive	Relatively robust
Large datasets	Generally efficient	Can be expensive	Often suitable, depending on implementation/data
Visualization	Cluster plot	Dendrogram	Cluster plot
Main parameter	K	Linkage / distance / cluster cut	eps, min_samples

6. Dimensionality Reduction

Concept

Dimensionality reduction reduces the number of features while trying to preserve useful information or structure.

Suppose a dataset has:

500 Features

We may reduce it to:

50 Features

or even:

2 or 3 dimensions

Why Reduce Dimensions?

1. Visualization

Convert high-dimensional data into 2D or 3D.

2. Faster Processing

Fewer features can reduce computational cost.

3. Noise Reduction

Some dimensions may contain little useful information.

4. Reduce Redundancy

Highly correlated features may contain overlapping information.

5. Storage

Fewer dimensions can reduce the amount of data representation required.

Example

Imagine:

100 features

↓

Dimensionality Reduction

↓

2 components

↓

2D visualization

Important Techniques

Some dimensionality-reduction methods include:

- PCA
- t-SNE
- UMAP
- Autoencoders

For this module, we focus mainly on **PCA**.

7. PCA — Principal Component Analysis

Concept

PCA is a dimensionality-reduction technique that transforms the original features into a smaller set of new variables called **principal components**.

The first principal component captures the largest possible amount of variance in the transformed data, the second captures the largest remaining variance subject to being orthogonal to the first, and so on.

Simple Explanation

Imagine data like this:

-
-
-
-
-

The data has many features, but most of its variation may occur along one main direction.

PCA finds new axes that represent the data efficiently.

Original Features

Suppose:

X1

X2

X3

X4

X5

PCA may transform them into:

PC1

PC2

PC3

while retaining a chosen amount of variance.

PCA Process

Original Dataset

↓

Standardize Features

↓

Calculate Covariance Structure

↓

Find Principal Directions

↓

Rank Components

↓

Select Components

↓

Transform Dataset

Why Standardize Before PCA?

PCA is influenced by feature scales.

Suppose:

Age → 18–60

Salary → 20,000–2,00,000

Salary has much larger numerical magnitude.

Without appropriate preprocessing, features with larger scales can dominate the variance structure.

Therefore, standardization is often performed before PCA.

PCA Mathematical Idea

PCA is based on the covariance structure/eigendecomposition or, equivalently, singular value decomposition of the centered data matrix.

A covariance matrix can be represented as:

$$\Sigma = \frac{1}{n-1} X^T X$$

Principal directions are associated with eigenvectors of the covariance matrix, and the corresponding eigenvalues indicate the variance captured along those directions.

Explained Variance

A principal component's **explained variance ratio** tells us how much of the total variance it captures.

For example:

PC1 → 60%

PC2 → 25%

PC3 → 10%

PC4 → 5%

Then:

$$PC1 + PC2 = 85\%$$

So two components preserve 85% of the variance under this example.

PCA Python Example

```
import pandas as pd
```

```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

data = {
    "Math": [70, 75, 80, 85, 90],
    "Physics": [65, 72, 78, 84, 88],
    "Chemistry": [68, 74, 79, 83, 91]
}

df = pd.DataFrame(data)

X = df[
    [
        "Math",
        "Physics",
        "Chemistry"
    ]
]
```

Step 1 — Standardize

```
scaler = StandardScaler()
```

```
X_scaled = scaler.fit_transform(X)
```

Step 2 — Apply PCA

```
pca = PCA(n_components=2)
```

```
X_pca = pca.fit_transform(X_scaled)
```

```
print(X_pca)
```

Step 3 — Explained Variance

```
print(pca.explained_variance_ratio_)
```

Total Explained Variance

```
print(
    pca.explained_variance_ratio_.sum()
)
```

PCA Visualization

For a dataset with many features, PCA can reduce it to two components:

```
import matplotlib.pyplot as plt
```

```
plt.scatter(
```

```

X_pca[:, 0],
X_pca[:, 1]
)

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA Visualization")

plt.show()

```

This allows us to visualize high-dimensional data in 2D.

PCA vs Feature Selection

These concepts are different.

Feature Selection

Select existing features.

Example:

Age

Salary

Experience

Education

Select:

Salary

Experience

The original features remain unchanged.

PCA

Creates **new transformed features**.

Age

Salary

Experience

Education

↓

PCA

↓

PC1

PC2

So:

Feature selection chooses existing features; PCA creates new components from combinations of existing features.

⚠ Important PCA Limitation

PCA components can be difficult to interpret.

For example:

$$\begin{aligned} \text{PC1} = & \\ & 0.52(\text{Age}) \\ & + 0.61(\text{Salary}) \\ & + 0.48(\text{Experience}) \end{aligned}$$

Unlike a simple feature such as "Salary", PC1 may not have an intuitive real-world meaning.

Therefore, PCA is useful for compression and representation, but it can reduce interpretability.

🔍 Choosing the Number of PCA Components

One common method is to examine cumulative explained variance.

```
from sklearn.decomposition import PCA

pca = PCA()

X_pca = pca.fit_transform(X_scaled)

cumulative_variance = pca.explained_variance_ratio_.cumsum()

print(cumulative_variance)
```

Visualization:

```
plt.plot(
    range(1, len(cumulative_variance) + 1),
    cumulative_variance,
    marker="o"
)

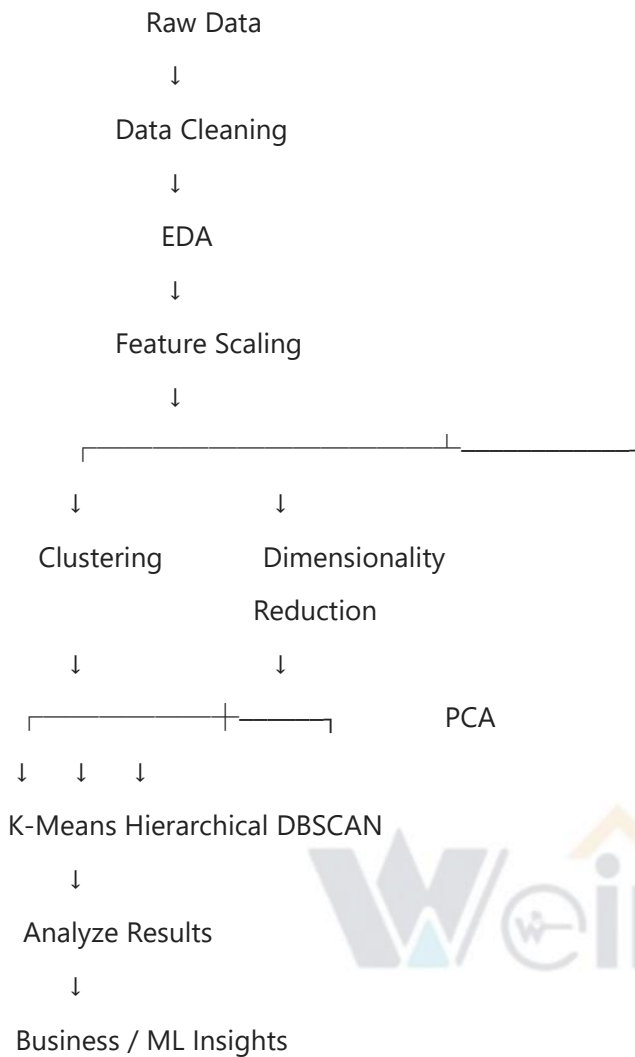
plt.xlabel("Number of Components")
plt.ylabel("Cumulative Explained Variance")
plt.title("PCA Explained Variance")

plt.show()
```

You might choose enough components to retain, for example, **90% or 95% of the variance**, depending on the problem.

This is a practical heuristic rather than a universal requirement.

Unsupervised Learning Workflow



Real-World Applications

1. Customer Segmentation

Customer Data



K-Means



Customer Groups

Example:

- Budget customers
- Regular customers
- Premium customers

2. Fraud / Anomaly Detection

Transaction Data

↓

DBSCAN / Other Methods

↓

Unusual Patterns

Anomaly detection is broader than clustering; DBSCAN's noise labels can sometimes be useful for identifying unusual observations, but dedicated anomaly-detection algorithms may be more appropriate depending on the problem.

3. Document Clustering

Thousands of Documents

↓

Text Embeddings

↓

Clustering

↓

Topic Groups

4. Image Analysis

High-dimensional image features can be reduced using PCA for visualization or preprocessing.

5. Customer Recommendation Systems

Customer behavior can be grouped into similar segments to support personalization.

Complete Example — Customer Segmentation

Dataset

```
import pandas as pd

data = {
    "Annual_Income": [20, 22, 25, 28, 80, 85, 90, 95],
    "Spending_Score": [85, 80, 88, 90, 25, 30, 28, 20]
}

df = pd.DataFrame(data)
```

```
print(df)
```

Apply K-Means

```
from sklearn.cluster import KMeans

X = df[
    "Annual_Income",
    "Spending_Score"
]

model = KMeans(
    n_clusters=2,
    random_state=42,
    n_init="auto"
)

df["Cluster"] = model.fit_predict(X)

print(df)
```

Visualize

```
import matplotlib.pyplot as plt

plt.scatter(
    df["Annual_Income"],
    df["Spending_Score"],
    c=df["Cluster"]
)

plt.xlabel("Annual Income")
plt.ylabel("Spending Score")
plt.title("Customer Segmentation")

plt.show()
```

K-Means vs PCA

These are often confused.

K-Means

Answers:

"Which group does this data point belong to?"

Data

↓

K-Means

↓

Clusters

PCA

Answers:

"Can I represent this high-dimensional data using fewer dimensions while preserving much of its variance?"

High-Dimensional Data

↓

PCA

↓

Low-Dimensional Representation

They can also be used together:

High-Dimensional Dataset

↓

PCA

↓

Reduced Features

↓

K-Means

↓

Clusters

But PCA before clustering is not automatically better; it should be validated for the particular dataset and objective.

Module 7 Quick Revision

Topic	Key Idea
Unsupervised Learning	Learns structure from unlabeled data
Clustering	Groups similar observations
K-Means	Centroid-based clustering
Centroid	Center representing a cluster
Elbow Method	Heuristic for choosing K

Topic	Key Idea
Hierarchical Clustering	Builds a hierarchy of clusters
Dendrogram	Visual representation of hierarchy
DBSCAN	Density-based clustering
eps	Neighborhood radius in DBSCAN
min_samples	Minimum neighborhood density parameter
Noise	Points not assigned to a DBSCAN cluster
Dimensionality Reduction	Reduces feature dimensions
PCA	Creates principal components
Explained Variance	Variance captured by components
Feature Selection	Selects original features
PCA	Creates transformed features

Module 7 Interview Questions

Basic

1. What is unsupervised learning?
2. How is unsupervised learning different from supervised learning?
3. What is clustering?
4. Give three real-world applications of clustering.
5. What is dimensionality reduction?

K-Means

6. What is K-Means?
7. What does K represent?
8. What is a centroid?
9. How does K-Means work?
10. What is the K-Means objective function?
11. What is inertia?
12. What is the elbow method?
13. What happens if K is too small?
14. What happens if K is too large?
15. Why is feature scaling important for K-Means?
16. What are the limitations of K-Means?

Hierarchical Clustering

17. What is hierarchical clustering?
18. What is agglomerative clustering?
19. What is divisive clustering?

20. What is a dendrogram?
21. What is linkage?
22. How do you determine clusters from a dendrogram?

DBSCAN

23. What is DBSCAN?
24. What does eps mean?
25. What does min_samples mean?
26. What is a core point?
27. What is a border point?
28. What is a noise point?
29. Why is DBSCAN useful for irregularly shaped clusters?
30. What are the limitations of DBSCAN?

PCA

31. What is PCA?
32. Why is PCA used?
33. What is a principal component?
34. What is explained variance?
35. Why is standardization often performed before PCA?
36. What is the difference between PCA and feature selection?
37. What are the limitations of PCA?
38. How do you choose the number of principal components?

Advanced

39. Compare K-Means, Hierarchical Clustering, and DBSCAN.
40. Can K-Means detect arbitrary-shaped clusters?
41. Why is DBSCAN better suited to some irregular cluster shapes?
42. Can PCA be used before K-Means?
43. Does PCA always improve model performance?
44. How would you evaluate the quality of clusters?
45. What is the difference between clustering and classification?

Practice Questions

Level 1

1. Explain unsupervised learning in your own words.
2. Give five real-world applications.
3. Explain clustering with an example.
4. Explain dimensionality reduction.

Level 2

5. Implement K-Means on customer data.
6. Find an appropriate K using the elbow method.
7. Visualize the clusters.
8. Implement hierarchical clustering.
9. Generate a dendrogram.
10. Implement DBSCAN.

Level 3

11. Compare:

K-Means

Hierarchical Clustering

DBSCAN

on the same dataset.

12. Analyze which algorithm handles outliers best.
13. Create a high-dimensional dataset and apply PCA.
14. Plot cumulative explained variance.
15. Reduce the dataset to two principal components.

Module 7 Industry Project — Customer Segmentation

Business Problem

An e-commerce company wants to divide customers into meaningful groups based on their behavior.

Dataset

Customer_ID

Age

Annual_Income

Spending_Score

Purchase_Frequency

Average_Order_Value

Website_Visits

Project Pipeline

Customer Dataset

↓

Data Cleaning

↓

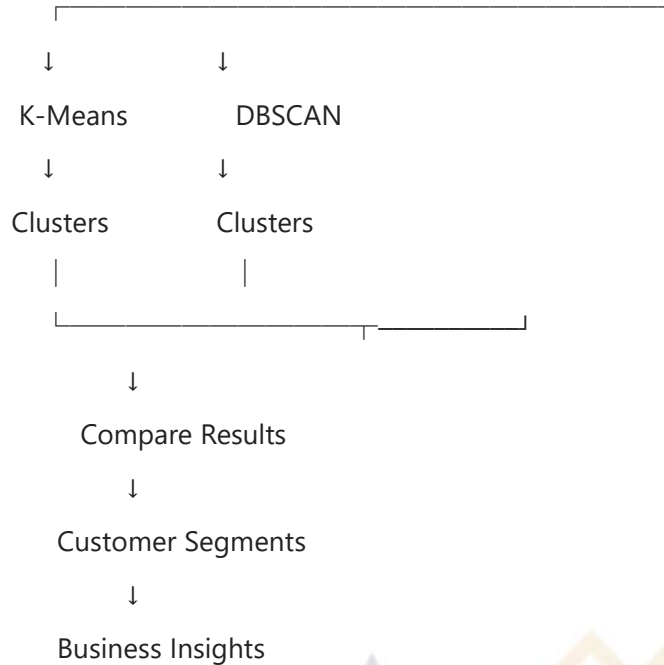
EDA

↓

Feature Selection



Feature Scaling



Example Business Output

Cluster 0 → Low income / low spending

Cluster 1 → High income / high spending

Cluster 2 → High income / low spending

Cluster 3 → Low income / high spending

The actual segments should be determined from the dataset rather than assumed in advance.

🎯 Module 7 Final Learning Outcome

After completing this module, students should be able to:

- Understand **unlabeled data**
- Explain **clustering**
- Implement **K-Means**
- Select K using methods such as the **Elbow Method**
- Understand **Hierarchical Clustering**
- Interpret a **dendrogram**
- Implement **DBSCAN**
- Understand **core, border, and noise points**
- Explain **dimensionality reduction**
- Apply **PCA**

- Interpret **explained variance**
- Distinguish **PCA from feature selection**
- Build a practical **Customer Segmentation project**

Next: Module 8 — Model Evaluation will cover **Confusion Matrix, Accuracy, Precision, Recall, F1 Score, ROC-AUC, MAE, MSE, RMSE, R^2 Score, and Cross-Validation**, which are essential for deciding whether an ML model is actually performing well.

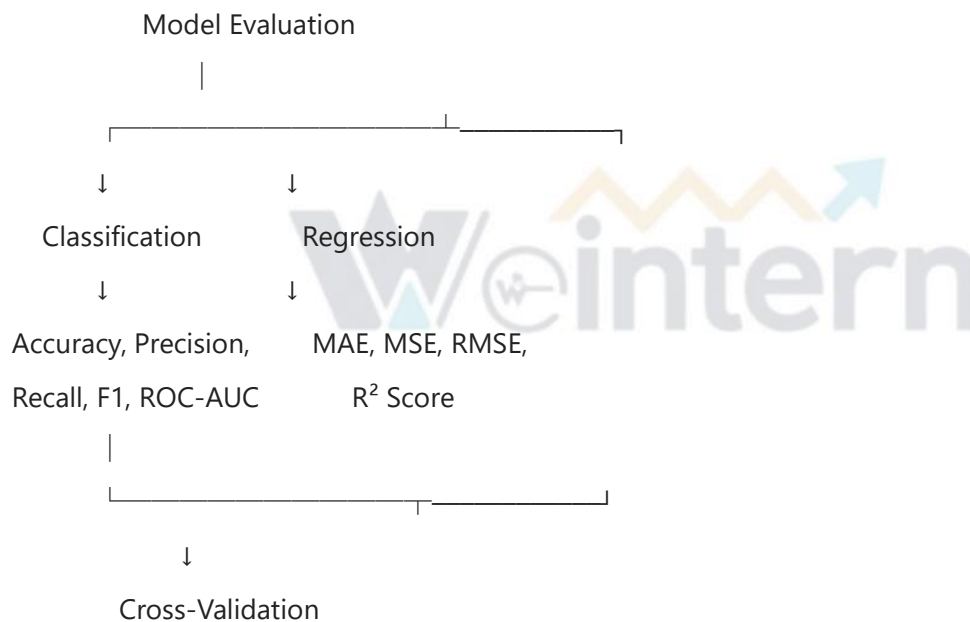
Module 8 — Model Evaluation

Model Evaluation is the process of measuring **how well a Machine Learning model performs on unseen data**.

Training a model is not enough. We need to answer:

"How accurately and reliably does this model make predictions?"

The evaluation method depends on whether the problem is **classification** or **regression**.



1. Confusion Matrix

Concept

A **Confusion Matrix** is a table used to evaluate a classification model by comparing:

- Actual values
- Predicted values

For binary classification:

	Predicted	
	Positive	Negative
Actual Positive	TP	FN

Actual Negative FP TN

Where:

Term Meaning

TP True Positive

TN True Negative

FP False Positive

FN False Negative

Simple Example

Suppose we are predicting whether an email is spam.

Actual Spam + Predicted Spam

→ True Positive

Actual Not Spam + Predicted Not Spam

→ True Negative

Actual Not Spam + Predicted Spam

→ False Positive

Actual Spam + Predicted Not Spam

→ False Negative

Confusion Matrix Example

Suppose:

TP = 80

TN = 90

FP = 10

FN = 20

The matrix is:

	Predicted Positive	Predicted Negative
Actual Positive	80	20
Actual Negative	10	90

Total observations:

[
80+90+10+20=200
]

Python

```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, predictions)

print(cm)
```

Visualization:

```
from sklearn.metrics import ConfusionMatrixDisplay
import matplotlib.pyplot as plt
```

```
ConfusionMatrixDisplay.from_predictions(
    y_test,
    predictions
)
```

```
plt.title("Confusion Matrix")
plt.show()
```

2. Accuracy

Concept

Accuracy tells us the percentage of predictions that were correct.

Formula:

```
[
Accuracy =
\frac{TP+TN}
{TP+TN+FP+FN}
]
```

Using our example:

```
[
Accuracy=
\frac{80+90}{200}
]

[
Accuracy=0.85
]
```

Therefore:

Accuracy = 85%

Python

```
from sklearn.metrics import accuracy_score
```

```
accuracy = accuracy_score(  
    y_test,  
    predictions  
)  
  
print("Accuracy:", accuracy)
```

When Accuracy Works Well

Accuracy is useful when classes are reasonably balanced and the costs of different errors are similar.

Example

Class A → 50%

Class B → 50%

Accuracy can be informative.

Accuracy Problem — Imbalanced Data

Suppose:

1000 transactions

990 → Normal

10 → Fraud

A model predicts every transaction as normal.

Accuracy:

```
[  
    \frac{990}{1000}=99%  
]
```

Looks excellent.

But:

Fraud detected = 0

So **99% accuracy can hide a useless fraud detector.**

This is why we need other metrics.

3. Precision

Concept

Precision answers:

"Of all the observations the model predicted as positive, how many were actually positive?"

Formula:

```
[
Precision=
\frac{TP}
{TP+FP}
]
```

Using:

```
TP = 80
FP = 10
```

```
[
Precision=
\frac{80}{80+10}
]
```

```
[
Precision=0.889
]
```

Approximately:

```
Precision = 88.9%
```

Example — Spam Detection

If the model predicts 100 emails as spam and 90 are actually spam:

```
Precision = 90%
```

High precision means **fewer false positives**.

Python

```
from sklearn.metrics import precision_score

precision = precision_score(
y_test,
predictions
)

print("Precision:", precision)
```

4. Recall

Concept

Recall answers:

"Of all the actual positive cases, how many did the model correctly identify?"

Formula:

```
[
Recall=
```

$$\frac{TP}{TP+FN}$$

Using:

$$TP = 80$$

$$FN = 20$$

$$\text{Recall} = \frac{80}{80+20}$$

$$\text{Recall} = 0.80$$

Therefore:

$$\text{Recall} = 80\%$$

Example — Disease Detection

Suppose:

100 people actually have a disease

The model identifies:

95 people

Then:

$$\text{Recall} = 95\%$$

For many screening problems, missing a true positive can be costly, so high recall may be particularly important.

Python

```
from sklearn.metrics import recall_score

recall = recall_score(
    y_test,
    predictions
)

print("Recall:", recall)
```

5. Precision vs Recall

This is one of the **most important interview concepts**.

Metric	Main Question	Focus
--------	---------------	-------

Metric	Main Question	Focus
Precision	How many predicted positives were correct?	Reduce FP
Recall	How many actual positives were found?	Reduce FN

Precision

Predicted Positive

↓

How many are actually positive?

Recall

Actual Positive

↓

How many did we find?

Real-World Example

Spam Filter

You may care about precision because incorrectly marking an important email as spam is undesirable.

Medical Screening

You may prioritize recall because missing a disease case can be costly.

Fraud Detection

Recall can be important for catching fraudulent transactions, but precision also matters because excessive false alarms can overwhelm investigators.

The appropriate metric depends on the business cost of FP and FN.

6. F1 Score

Concept

F1 Score combines **Precision and Recall** into a single metric using the harmonic mean.

Formula:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Using:

$$\text{Precision} = 0.889$$

$$\text{Recall} = 0.80$$

Approximately:

```
[  
F1\approx0.842  
]
```

So:

$F1 \approx 84.2\%$

Why Harmonic Mean?

The harmonic mean penalizes situations where one value is high and the other is very low.

For example:

Precision = 95%

Recall = 20%

F1 will remain relatively low.

Python

```
from sklearn.metrics import f1_score  
  
f1 = f1_score(  
y_test,  
predictions  
)  
  
print("F1 Score:", f1)
```

7. ROC-AUC

ROC Curve

ROC stands for:

Receiver Operating Characteristic

It plots:

```
[  
TPR  
]
```

against:

```
[  
FPR  
]
```

where:

True Positive Rate

$$\left[\begin{aligned} \text{TPR} &= \\ \frac{\text{TP}}{\text{TP} + \text{FN}} \end{aligned} \right]$$

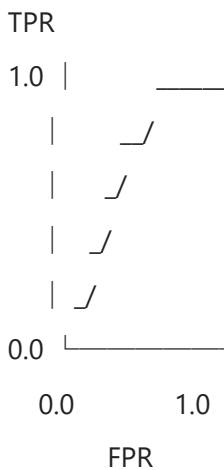
This is the same as **Recall**.

False Positive Rate

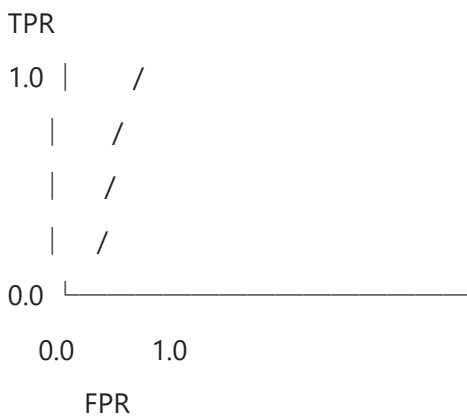
$$\left[\begin{aligned} \text{FPR} &= \\ \frac{\text{FP}}{\text{FP} + \text{TN}} \end{aligned} \right]$$

ROC Curve

Conceptually:



A random classifier roughly follows the diagonal:



AUC

AUC = Area Under the ROC Curve

It summarizes how well the model separates positive and negative classes across classification thresholds.

General intuition:

AUC Interpretation

1.0 Perfect ranking/separation

0.9 Excellent

0.8 Good

0.7 Reasonable

0.5 About random

< 0.5 Worse than random ranking

These ranges are rough guidelines, not universal quality standards.

Python

```
from sklearn.metrics import roc_auc_score

probabilities = model.predict_proba(X_test)[: , 1]

auc = roc_auc_score(
    y_test,
    probabilities
)

print("ROC-AUC:", auc)
```

8. MAE — Mean Absolute Error

Now let's move to **regression evaluation**.

Concept

MAE measures the average absolute difference between actual and predicted values.

Formula:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Example

Actual:

[100, 200, 300]

Predicted:

[110, 180, 290]

Errors:

10

20

10

MAE:

```
[  
  \frac{10+20+10}{3}  
]
```

```
[  
  MAE=13.33  
]
```

So the model's average absolute error is approximately:

13.33 units

Python

```
from sklearn.metrics import mean_absolute_error  
  
mae = mean_absolute_error(  
    y_test,  
    predictions  
)  
  
print("MAE:", mae)
```

Advantage

MAE is easy to interpret because it uses the **same units as the target**.

If predicting house prices in rupees, MAE is also expressed in rupees.

9. MSE — Mean Squared Error

Concept

MSE calculates the average of the **squared errors**.

Formula:

```
[  
  MSE=  
  \frac{1}{n}  
  \sum_{i=1}^n  
  (y_i - \hat{y}_i)^2  
]
```

Using:

Errors = 10, 20, 10

Squared errors:

100, 400, 100

Therefore:

```
[
MSE=
\frac{100+400+100}{3}
]

[
MSE=200
]
```

Why Square the Error?

Squaring makes large errors much more costly.

Example:

Error = 2
Squared = 4

Error = 10
Squared = 100

Therefore, MSE is more sensitive to large errors than MAE.

Python

```
from sklearn.metrics import mean_squared_error

mse = mean_squared_error(
y_test,
predictions
)

print("MSE:", mse)
```

10. RMSE — Root Mean Squared Error

Concept

RMSE is the square root of MSE.

Formula:

```
[
RMSE=\sqrt{MSE}
]
```

For:

```
[
MSE=200
]

[
RMSE=\sqrt{200}
]
```

Approximately:

RMSE = 14.14

Python

```
from sklearn.metrics import root_mean_squared_error

rmse = root_mean_squared_error(
y_test,
predictions
)
```

```
print("RMSE:", rmse)
```

If your scikit-learn version does not provide root_mean_squared_error, use:

```
import numpy as np
from sklearn.metrics import mean_squared_error

rmse = np.sqrt(
mean_squared_error(y_test, predictions)
)
```

```
print("RMSE:", rmse)
```

MAE vs MSE vs RMSE

Metric Large Error Sensitivity Same Unit as Target?

MAE	Lower	Yes
MSE	High	No
RMSE	High	Yes

11. R² Score

Concept

R², or coefficient of determination, measures how much of the variance in the target is explained by the regression model relative to a baseline that predicts the mean.

Formula:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

where:

- y_i = actual value
 - $\hat{y}_i$ = predicted value
 - $\bar{y}$ = mean of actual values

Example

Suppose:

$$R^2 = 0.85$$

This means the model explains about **85% of the variance in the target relative to the mean baseline**, for that evaluation dataset.

It does **not** mean the model is "85% accurate."

Python

```
from sklearn.metrics import r2_score

r2 = r2_score(
    y_test,
    predictions
)

print("R2 Score:", r2)
```

R² Interpretation

A higher R² can indicate better fit, but:

- R² alone is not enough

- It can be negative on unseen data
- A high R^2 does not guarantee useful predictions
- Business context matters

12. Cross-Validation

Concept

A single train-test split can sometimes give a misleading estimate of model performance.

Cross-validation evaluates a model across multiple train/validation splits.

The most common approach is **K-Fold Cross-Validation**.

K-Fold Cross-Validation

Suppose:

$$K = 5$$

The dataset is divided into five folds.

Fold 1 | Fold 2 | Fold 3 | Fold 4 | Fold 5

Round 1

Test/Validation → Fold 1

Training → Fold 2 + 3 + 4 + 5

Round 2

Test/Validation → Fold 2

Training → Fold 1 + 3 + 4 + 5

And so on.

Finally, we aggregate the scores.

Score 1

Score 2

Score 3

Score 4

Score 5

↓

Mean Score

Visual

5-Fold Cross Validation

Round 1 [V][T][T][T][T]

Round 2 [T][V][T][T][T]

Round 3 [T][T][V][T][T]

Round 4 [T][T][T][V][T]

Round 5 [T][T][T][T][V]

V = Validation Fold

T = Training Fold

Python

```
from sklearn.model_selection import cross_val_score
from sklearn.ensemble import RandomForestClassifier
```

```
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
```

```
scores = cross_val_score(
    model,
    X,
    y,
    cv=5,
    scoring="accuracy"
)
```

```
print("Scores:", scores)
print("Mean:", scores.mean())
```

Stratified K-Fold

For classification problems, **StratifiedKFold** can preserve the class proportions in each fold.

```
from sklearn.model_selection import StratifiedKFold, cross_val_score
```

```
cv = StratifiedKFold(
    n_splits=5,
    shuffle=True,
    random_state=42
)
```

```
scores = cross_val_score(
    model,
    X,
    y,
```



```
cv=cv,
scoring="accuracy"
)
```

```
print(scores)
```

This is especially useful when classes are imbalanced.

⚠ Important: Test Set vs Cross-Validation

A common mistake is repeatedly checking the test set while tuning the model.

A better workflow is:

Dataset

↓

Train + Validation strategy

↓

Cross-validation / Hyperparameter tuning

↓

Select final model

↓

Final evaluation on untouched Test Set

The test set should ideally be treated as an **unseen final evaluation set**.

📊 Classification Metrics Summary

Metric	Formula	Main Focus
Accuracy	$(TP+TN)/Total$	Overall correctness
Precision	$TP/(TP+FP)$	Reduce false positives
Recall	$TP/(TP+FN)$	Reduce false negatives
F1	$2PR/(P+R)$	Balance precision & recall
ROC-AUC	Area under ROC	Ranking/separation ability

📊 Regression Metrics Summary

Metric Meaning

MAE Average absolute error

MSE Average squared error

RMSE Square root of MSE

R² Variance explained relative to mean baseline

For MAE, MSE, and RMSE:

Lower is generally better.

For R^2 :

Higher is generally better, although context matters.

🤖 Which Metric Should You Use?

Balanced Classification

Use:

Accuracy

F1

ROC-AUC

depending on the objective.

Imbalanced Classification

Consider:

Precision

Recall

F1

PR-AUC

ROC-AUC

Accuracy alone can be misleading.

Medical Screening

Often prioritize:

Recall

if missing a positive case is especially costly.

Spam Filtering

Often prioritize:

Precision

if false positives are especially costly.

Regression

Use:

MAE

RMSE

R^2

and choose based on the practical meaning of errors.

🔑 Complete Classification Evaluation

```
from sklearn.metrics import (  
    accuracy_score,  
    precision_score,
```

```

recall_score,
f1_score,
confusion_matrix,
roc_auc_score
)

predictions = model.predict(X_test)

print("Accuracy:",
      accuracy_score(y_test, predictions))

print("Precision:",
      precision_score(y_test, predictions))

print("Recall:",
      recall_score(y_test, predictions))

print("F1 Score:",
      f1_score(y_test, predictions))

print("Confusion Matrix:")
print(confusion_matrix(y_test, predictions))

```

If the classifier supports probabilities:

```

probabilities = model.predict_proba(X_test)[:, 1]

print(
  "ROC-AUC:",
  roc_auc_score(y_test, probabilities)
)

```

Complete Regression Evaluation

```

from sklearn.metrics import (
  mean_absolute_error,
  mean_squared_error,
  r2_score
)

import numpy as np

predictions = model.predict(X_test)

```

```

mae = mean_absolute_error(
y_test,
predictions
)

mse = mean_squared_error(
y_test,
predictions
)

rmse = np.sqrt(mse)

r2 = r2_score(
y_test,
predictions
)

print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R²:", r2)

```

💧 Model Comparison Example

Suppose we trained three classification models:

Model	Accuracy	Precision	Recall	F1
Logistic Regression	88%	86%	82%	84%
Decision Tree	91%	90%	88%	89%
Random Forest	94%	93%	92%	92.5%

Based only on these metrics, Random Forest appears strongest.

But before choosing it for production, we should also consider:

- Cross-validation performance
- Inference speed
- Interpretability
- Model size
- Data drift
- Business costs
- Deployment constraints

⚠️ Common Evaluation Mistakes

Mistake 1 — Evaluating on Training Data

```
model.fit(X_train, y_train)
```

```
model.score(X_train, y_train)
```

A very high training score does not prove good generalization.

Use unseen validation/test data.

Mistake 2 — Using Accuracy for Everything

Accuracy can be misleading for highly imbalanced datasets.

Mistake 3 — Data Leakage

Information from validation/test data accidentally enters the training process.

Example:

Test Data

↓

Scaling entire dataset

↓

Training

Instead, fit preprocessing only on training data.

A **Pipeline** helps:

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
```

```
pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", LogisticRegression())
])
```

```
pipeline.fit(X_train, y_train)
```

```
predictions = pipeline.predict(X_test)
```

This ensures the scaler is fitted using the training data within the pipeline.

Learning Curve Concept

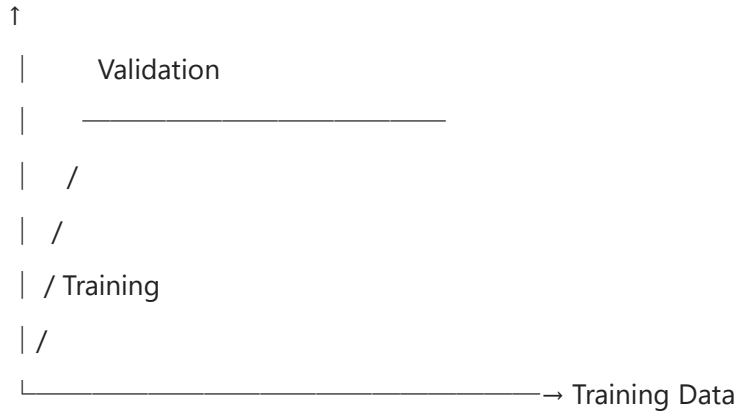
A learning curve shows how model performance changes as the amount of training data increases.

It can help diagnose:

- High bias
- High variance
- Whether more data may help

Conceptually:

Performance



Industry Example — Customer Churn

Suppose we build a churn prediction model.

Dataset

Customer Features

↓

Train/Test Split

↓

Random Forest

↓

Predictions

We calculate:

Accuracy

Precision

Recall

F1

ROC-AUC

Suppose:

Accuracy = 91%

Precision = 87%

Recall = 84%

F1 = 85.5%

ROC-AUC = 0.93

We then compare the model against Logistic Regression, Decision Tree, SVM, etc.

Finally:

Cross-validation

↓

Hyperparameter tuning

↓

Final model

↓

Untouched test set

↓

Deployment

🔗 Module 8 Interview Questions

Basic

1. What is model evaluation?
2. Why do we evaluate ML models?
3. What is the difference between training, validation, and test data?
4. What is a confusion matrix?
5. What are TP, TN, FP, and FN?
6. What is accuracy?

Classification

7. What is precision?
8. What is recall?
9. What is F1 Score?
10. Why can accuracy be misleading?
11. When would you prefer precision over recall?
12. When would you prefer recall over precision?
13. What is ROC-AUC?
14. What is TPR?
15. What is FPR?
16. What does AUC represent?
17. What is the difference between ROC-AUC and accuracy?

Regression

18. What is MAE?
19. What is MSE?
20. What is RMSE?
21. Why is RMSE sensitive to large errors?

22. What is R^2 ?
23. Can R^2 be negative?
24. What is the difference between MAE and RMSE?
25. Which regression metric is easier to interpret?

Cross-Validation

26. What is cross-validation?
27. What is K-Fold Cross-Validation?
28. Why is cross-validation useful?
29. What is Stratified K-Fold?
30. Why shouldn't we repeatedly tune a model using the test set?

Advanced

31. What is data leakage?
32. How can preprocessing cause data leakage?
33. How does a Pipeline help prevent leakage?
34. How do you evaluate an imbalanced classification model?
35. How would you select an evaluation metric for a real-world project?
36. Can a model with 95% accuracy be worse than a model with 90% accuracy?
37. Why is a high training score not enough?
38. How do you compare multiple ML models fairly?

Practice Questions

Level 1

1. Calculate accuracy from:

TP = 80

TN = 90

FP = 10

FN = 20

2. Calculate:

Precision

Recall

F1 Score

3. Explain the difference between precision and recall.
4. Explain why accuracy can be misleading for fraud detection.

Level 2

5. Train a classification model and calculate:

Accuracy

Precision

Recall

F1

ROC-AUC

6. Plot a confusion matrix.
7. Compare Logistic Regression, Decision Tree, and Random Forest.
8. Apply 5-fold cross-validation.

Level 3

9. Build a regression model and calculate:

MAE

MSE

RMSE

R^2

10. Compare two regression models.
11. Create an imbalanced classification dataset and compare accuracy with F1/recall.
12. Implement a preprocessing + model Pipeline and evaluate it using cross-validation.

Module 8 Industry Project

Customer Churn Model Evaluation

Goal

Evaluate different models and identify the most suitable model for predicting customer churn.

Models

Logistic Regression

Decision Tree

Random Forest

SVM

KNN

Evaluation Pipeline

Customer Dataset

↓

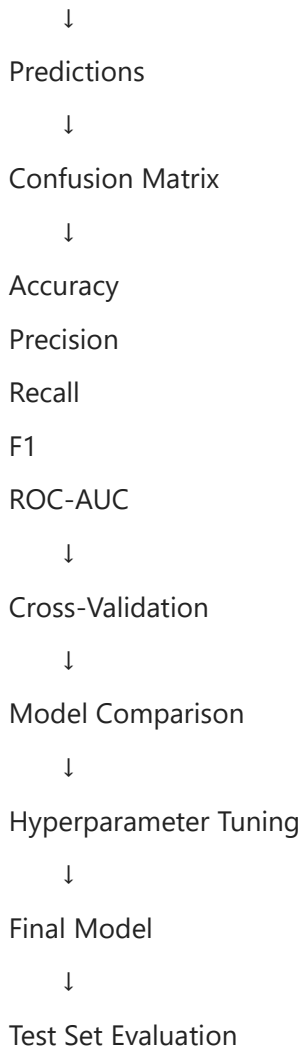
Preprocessing

↓

Train/Test Split

↓

Train Models



Final Report

Students should prepare a comparison table:

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	—	—	—	—	—
KNN	—	—	—	—	—
Decision Tree	—	—	—	—	—
Random Forest	—	—	—	—	—
SVM	—	—	—	—	—

Then answer:

Which model should be selected, and why?

The answer should not be based only on the highest accuracy; it should consider the **business objective and relevant evaluation metrics**.

🎯 Module 8 Final Learning Outcome

After completing this module, students should be able to:

- Read and interpret a **confusion matrix**

- Calculate **TP, TN, FP, FN**
- Understand **Accuracy**
- Understand **Precision**
- Understand **Recall**
- Understand **F1 Score**
- Understand **ROC and AUC**
- Evaluate regression models using **MAE, MSE, RMSE, and R^2**
- Perform **K-Fold Cross-Validation**
- Understand **Stratified K-Fold**
- Detect **data leakage**
- Use **Pipelines** for safe preprocessing
- Compare multiple ML models
- Select metrics according to the **business problem**
- Build a complete **model evaluation report**

Connection with previous modules

Module 4 → Data Preprocessing

↓

Module 5 → EDA

↓

Module 6 → Supervised Learning

↓

Module 7 → Unsupervised Learning

↓

Module 8 → Model Evaluation

↓

Module 9 → Feature Engineering

Next: Module 9 — Feature Engineering will cover **feature selection, feature extraction, feature transformation, encoding, scaling, polynomial features, imbalanced datasets, and SMOTE**.

Module 9 — Feature Engineering

Overview

Feature Engineering is the process of creating, transforming, selecting, or improving input features so that a Machine Learning model can learn useful patterns more effectively.

In simple words:

Feature Engineering = Converting raw data into useful features for ML models.

For many real-world ML projects, good features can have a bigger impact than simply choosing a more complex algorithm.

🧠 1. What is Feature Engineering?

Suppose we have raw customer data:

Customer	DOB	Purchase Date	Total Amount	Orders
A	2000-05-10	2026-07-20	₹50,000	10

Instead of directly giving DOB to the model, we can create:

Age = 26

We can also create:

Average Order Value = Total Amount / Orders

= 50000 / 10

= ₹5000

These new features may be more useful to the model.

Feature Engineering Pipeline

Raw Data

↓

Feature Selection

↓

Feature Extraction

↓

Feature Transformation

↓

Encoding

↓

Scaling

↓

New Features

↓

Machine Learning Model

2. Feature Selection

Concept

Feature Selection means choosing the most useful features from the existing dataset and removing irrelevant or redundant ones.

Example

Suppose we have:

Age

Salary

Experience

City

Customer_ID

Favorite_Color

For a salary prediction problem, we might select:

Age

Experience

Education

Job Role

and remove:

Customer_ID

if it has no predictive meaning.

Why Feature Selection?

It can:

- Reduce unnecessary features
- Reduce training time
- Reduce overfitting risk
- Improve interpretability
- Reduce noise
- Simplify the model

Common Feature Selection Methods

1. Filter Methods

Features are selected using statistical properties.

Examples:

- Correlation

- Chi-square test
- ANOVA
- Mutual information

2. Wrapper Methods

A model is used to evaluate subsets of features.

Examples:

- Forward Selection
- Backward Elimination
- Recursive Feature Elimination (RFE)

3. Embedded Methods

Feature selection happens during model training.

Examples:

- Lasso Regression
- Decision Trees
- Random Forest feature importance

Correlation-Based Selection

Suppose:

Age → 0.20

Salary → 0.85

Experience → 0.78

ID → 0.01

For a target related to income, Salary and Experience may be useful, while ID may not be.

However:

Correlation alone does not prove that a feature is useful or causal.

Python

```
import pandas as pd

correlation = df.corr(numeric_only=True)

print(correlation)
```

3. Feature Extraction

Concept

Feature extraction creates **new features from existing data**.

Example — Date

Original:

2026-08-08

Extract:

Year = 2026

Month = 8

Day = 8

Example — Text

Original:

"I love this product!"

We can extract:

Word count

Character count

TF-IDF features

Embeddings

Sentiment-related features

Example — Image

An image can be transformed into:

Pixels

Edges

Textures

CNN embeddings

Feature Selection vs Feature Extraction

Feature Selection

Selects existing features

Removes irrelevant features

Original features remain

Example: select Age, Salary

Feature Extraction

Creates new features

Transforms information

New representation is created

Example: PCA components

4. Feature Transformation

Concept

Feature transformation changes the representation or distribution of a feature.

Examples:

- Log transformation
- Square-root transformation
- Polynomial transformation
- Binning
- Scaling

Log Transformation

Suppose income is heavily skewed:

10000

20000

25000

50000

1000000

A logarithmic transformation can reduce the influence of very large values.

```
[  
X' = \log(1+X)  
]
```

Using $\log_{1p}$ is convenient when zero values may occur.

Python

```
import numpy as np  
  
df["Log_Income"] = np.log1p(  
    df["Income"]  
)
```

Why Transform Data?

Transformation can:

- Reduce skewness
- Make patterns easier for some models to learn
- Reduce the influence of extreme values
- Improve numerical behavior

But transformation is **model- and data-dependent**; it is not always necessary.

5. Encoding Categorical Data

Machine Learning algorithms generally need numerical representations.

Suppose:

Gender

Male

Female

Female

Male

We need to represent the categories numerically.

One-Hot Encoding

Creates a separate binary column for each category.

Example:

City

Pune

Mumbai

Delhi

Becomes:

	City_Pune	City_Mumbai	City_Delhi
1	0	0	
0	1	0	
0	0	1	

Python

```
import pandas as pd
```

```
df_encoded = pd.get_dummies(
```

```
df,
```

```
    columns=["City"]
```

```
)
```

```
print(df_encoded)
```

Using Scikit-Learn

```
from sklearn.preprocessing import OneHotEncoder

encoder = OneHotEncoder(
    handle_unknown="ignore"
)

X_encoded = encoder.fit_transform(
    df[["City"]]
)

handle_unknown="ignore" is useful when the test/deployment data contains a category not seen
during fitting.
```

6. Ordinal Encoding

Used when categories have a meaningful order.

Example:

Education

High School

Bachelor

Master

PhD

Could be represented as:

High School → 1

Bachelor → 2

Master → 3

PhD → 4

The important point is:

Ordinal encoding should be used only when the categories have a genuine order.

Example

```
from sklearn.preprocessing import OrdinalEncoder

encoder = OrdinalEncoder()

X_encoded = encoder.fit_transform(
    df[["Education"]]
)
```

7. Scaling

Concept

Feature scaling puts numerical features on comparable scales.

Suppose:

Age → 18–60

Salary → 20,000–2,00,000

Experience → 0–30

Salary has a much larger numerical range.

Some algorithms can be affected by these differences.

Algorithms Where Scaling Is Often Important

- KNN
- K-Means
- SVM
- Logistic Regression
- Linear Regression with regularization
- Neural Networks
- PCA

Tree-based algorithms such as Decision Trees and Random Forests are generally much less sensitive to feature scaling.

8. Standardization

Standardization transforms a feature so that it has approximately:

Mean = 0

Standard Deviation = 1

Formula:

$$Z = \frac{x - \mu}{\sigma}$$

where:

- x = original value
 - μ = mean
 - σ = standard deviation
-

Python

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)
```

Example

Suppose:

Mean = 50

Standard Deviation = 10

Value = 70

Then:

```
[
z=\frac{70-50}{10}
]

[
z=2
]
```

So the value is **2 standard deviations above the mean.**

9. Min-Max Scaling

Although Module 4 introduced normalization, it is important to connect it with feature engineering.

Min-Max scaling generally transforms values to a selected range, commonly [0, 1].

Formula:

```
[
X' =
\frac{X-X_{\min}}{X_{\max}-X_{\min}}
]
```

Python

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()

X_scaled = scaler.fit_transform(X)
```

⚠ Standardization vs Min-Max Scaling

Standardization

Mean ≈ 0

Std ≈ 1

Uses mean/std

Less directly tied to fixed bounds

Neither is universally better.

Min-Max Scaling

Usually range 0–1

Fixed range

Uses min/max

Sensitive to extreme min/max values

10. Polynomial Features

Concept

Polynomial features create additional features based on powers and interactions of existing features.

Suppose:

X

We can create:

X

X^2

X^3

With two features:

X_1

X_2

Polynomial features of degree 2 can include:

X_1

X_2

X_1^2

X_1X_2

X_2^2

Why?

Linear regression assumes a linear relationship in its features.

Polynomial features allow a linear model to represent certain nonlinear relationships.

Python

```
from sklearn.preprocessing import PolynomialFeatures
```

```
poly = PolynomialFeatures(
    degree=2,
    include_bias=False
```

```
)
```

```
X_poly = poly.fit_transform(X)
```

```
print(X_poly)
```

Example

Original:

Area

Polynomial feature:

Area²

The model can then learn a relationship such as:

```
[  
y = \beta_0 + \beta_1 X + \beta_2 X^2  
]
```

11. Handling Imbalanced Datasets

Concept

A dataset is **imbalanced** when one class contains many more observations than another.

Example

Fraud detection:

Normal → 9900

Fraud → 100

Class distribution:

Normal → 99%

Fraud → 1%

This is highly imbalanced.

Why Is It a Problem?

A model may learn to favor the majority class.

Example:

10000 transactions

If the model predicts:

Everything = Normal

Accuracy:

99%

But:

Fraud detected = 0

So accuracy is misleading.

12. Techniques for Imbalanced Data

Common approaches include:

1. Oversampling

Increase minority-class observations.

2. Undersampling

Reduce majority-class observations.

3. SMOTE

Create synthetic minority examples.

4. Class Weights

Give more importance to minority-class errors.

5. Appropriate Evaluation Metrics

Use:

- Precision
- Recall
- F1
- PR-AUC
- ROC-AUC

depending on the problem.

13. SMOTE

Concept

SMOTE = Synthetic Minority Over-sampling Technique

SMOTE generates synthetic minority-class samples instead of simply duplicating existing samples.

Simple Example

Suppose minority points are:

- A •
 - B

SMOTE can create a synthetic point between nearby minority observations:

A • • New • B

It does this based on neighboring minority samples.

How SMOTE Works

Conceptually:

Minority Samples

↓

Find Nearby Minority Neighbors

↓

Choose Neighbor

↓

Create Synthetic Point

↓

Add to Training Data

Python

You typically use the imbalanced-learn package.

```
from imblearn.over_sampling import SMOTE

smote = SMOTE(
    random_state=42
)

X_train_resampled, y_train_resampled = smote.fit_resample(
    X_train,
    y_train
)

print("Before:", y_train.value_counts())
print("After:", y_train_resampled.value_counts())
```

⚠ Very Important — SMOTE and Data Leakage

Do not apply SMOTE to the entire dataset before train/test splitting.

✗ Incorrect:

Entire Dataset

↓

SMOTE

↓

Train/Test Split

This can cause information from the test set to influence the training process.

Correct:

Dataset

↓

Train/Test Split

↓

Training Data

↓

SMOTE

↓

Model Training

↓

Untouched Test Data

↓

Evaluation

For cross-validation, use an imblearn Pipeline so SMOTE is applied separately inside each training fold.

🔧 SMOTE with Pipeline

```
from imblearn.pipeline import Pipeline
from imblearn.over_sampling import SMOTE
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
```

```
pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("smote", SMOTE(random_state=42)),
    ("model", LogisticRegression())
])
```

```
pipeline.fit(
    X_train,
    y_train
)
```

```
predictions = pipeline.predict(X_test)
```

This is a safer approach for evaluation and cross-validation.

14. Feature Engineering Example

Suppose we have:

Customer_ID

Age

Monthly_Income

Total_Purchases

City

Membership

We can engineer:

Remove

Customer_ID

if it is merely an identifier and has no predictive meaning.

Create

Average_Purchase =

Monthly_Income / Total_Purchases

Encode

City

Membership

Scale

Age

Monthly_Income

Total_Purchases

Average_Purchase

Final Dataset

Age

Monthly_Income

Total_Purchases

Average_Purchase

City_Pune

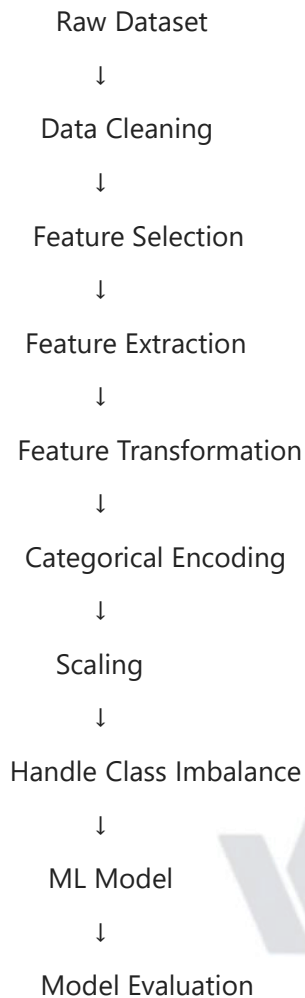
City_Mumbai

Membership_Gold

Membership_Silver

Now the data is much more suitable for many ML algorithms.

15. Complete Feature Engineering Pipeline



16. ColumnTransformer

For real-world datasets, different columns often require different preprocessing.

Example:

Numerical:

Age

Income

Experience

Categorical:

City

Education

Gender

We can process them separately.

```
from sklearn.compose import ColumnTransformer
```

```

from sklearn.preprocessing import (
    StandardScaler,
    OneHotEncoder
)

numeric_features = [
    "Age",
    "Income",
    "Experience"
]

categorical_features = [
    "City",
    "Education",
    "Gender"
]

preprocessor = ColumnTransformer([
    (
        "num",
        StandardScaler(),
        numeric_features
    ),
    (
        "cat",
        OneHotEncoder(handle_unknown="ignore"),
        categorical_features
    )
])

```

Then connect it to a model:

```

from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression

pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("model", LogisticRegression())
])

```

```
pipeline.fit(  
    X_train,  
    y_train  
)
```

This is a very important **industry-standard pattern**.

💧 17. Feature Engineering Example — House Price Prediction

Suppose the raw dataset contains:

Area

Bedrooms

Bathrooms

Location

Year_Built

Garage

Price

We can create:

Property Age

```
[  
    PropertyAge =  
    CurrentYear-YearBuilt  
]
```

Price Per Square Foot

After training/evaluation design is carefully considered:

```
[  
    PricePerSqFt =  
    \frac{Price}{Area}  
]
```

However, **do not use target-derived features like PricePerSqFt as an input feature when predicting Price**, because it directly contains the target and causes leakage.

This is a crucial feature-engineering lesson:

A feature can be mathematically valid but still invalid for prediction if it uses information unavailable at prediction time.

💧 18. Feature Engineering Example — Customer Churn

Raw:

Tenure

MonthlyCharges

TotalCharges

Contract

PaymentMethod

Create:

AverageMonthlySpend

ContractLength

IsLongTermCustomer

PaymentFrequency

Encode:

Contract

PaymentMethod

Scale:

Tenure

MonthlyCharges

TotalCharges

Then train the model.

💧 19. Feature Engineering Example — NLP

For text:

"I really love this course!"

Possible features:

Word Count

Character Count

Average Word Length

TF-IDF

Word Embeddings

Sentence Embeddings

For modern NLP systems, embeddings are often much more useful than simple handcrafted features.

💧 20. Feature Engineering Example — Date Data

Raw:

2026-08-08 14:30:00

Extract:

Year	=	2026
Month	=	8
Day	=	8
DayOfWeek	=	Saturday
Hour	=	14
Weekend	=	True

Python:

```
df["Date"] = pd.to_datetime(
    df["Date"]
)
```

```
df["Year"] = df["Date"].dt.year
df["Month"] = df["Date"].dt.month
df["Day"] = df["Date"].dt.day
df["DayOfWeek"] = df["Date"].dt.dayofweek
df["Hour"] = df["Date"].dt.hour
```

Feature Engineering Techniques Summary

Technique	Purpose
Feature Selection	Select useful existing features
Feature Extraction	Create new representations
Feature Transformation	Change feature representation/distribution
Encoding	Convert categorical data to numeric representation
Scaling	Put numerical features on comparable scales
Polynomial Features	Capture nonlinear relationships
Oversampling	Increase minority examples
Undersampling	Reduce majority examples
SMOTE	Generate synthetic minority samples
Feature Creation	Create domain-specific useful variables

Feature Selection vs Extraction vs Transformation

This is an important interview topic.

Feature Selection

A B C D E

↓

A C E

Choose existing features.

Feature Extraction

A B C D

↓

New Representation

↓

F1 F2

Create a new representation.

Feature Transformation

Original X

↓

$\log(X)$

Change the representation of an existing feature.

⚠ Common Feature Engineering Mistakes

Mistake 1 — Scaling Before Splitting

✗

Entire Dataset

↓

StandardScaler

↓

Train/Test Split

The scaler has seen information from the test set.

Correct

Train/Test Split

↓

Fit scaler on Training Data

↓

Transform Training Data

↓

Transform Test Data

A Pipeline is preferred because it automates this safely.

Mistake 2 — Applying SMOTE Before Splitting

✗

Dataset

↓

SMOTE

↓

Train/Test Split

Correct

Split

↓

SMOTE on training data only

↓

Train

↓

Evaluate on untouched test data

Mistake 3 — Keeping IDs as Features

Example:

```
Customer_ID = 100001
```

```
Customer_ID = 100002
```

An identifier usually doesn't contain meaningful predictive information.

Mistake 4 — Target Leakage

Using information derived from the target or information that would not be available at prediction time.

Example:

Predict Loan Approval

Using:

Loan_Approved_Date

as an input would be leakage if that date is known only after approval.

Mistake 5 — Too Many Features

More features don't automatically mean a better model.

Too many irrelevant features can:

- Increase complexity
 - Increase training time
 - Increase overfitting risk
 - Reduce interpretability
-

Module 9 Interview Questions

Basic

1. What is feature engineering?
2. Why is feature engineering important?
3. What is feature selection?
4. What is feature extraction?
5. What is feature transformation?

6. What is the difference between feature selection and feature extraction?

Encoding

7. What is categorical encoding?
8. What is one-hot encoding?
9. What is ordinal encoding?
10. When should ordinal encoding be used?
11. What happens if a new category appears during testing?

Scaling

12. What is feature scaling?
13. What is standardization?
14. What is normalization/min-max scaling?
15. Which ML algorithms are sensitive to feature scaling?
16. Why is scaling important for KNN?
17. Why is scaling important for K-Means?
18. Are Decision Trees sensitive to feature scaling?

Feature Creation

19. What are polynomial features?
20. Why would you use polynomial features?
21. What is a log transformation?
22. How can date features be engineered?
23. How can text features be engineered?

Imbalanced Data

24. What is an imbalanced dataset?
25. Why is class imbalance a problem?
26. What is oversampling?
27. What is undersampling?
28. What is SMOTE?
29. How does SMOTE work?
30. Why should SMOTE be applied only to training data?

Advanced

31. What is data leakage?
32. How can feature engineering cause data leakage?
33. Why should preprocessing be inside a Pipeline?
34. What is ColumnTransformer?
35. How would you handle numerical and categorical columns together?
36. How would you handle an imbalanced classification dataset?

37. How do you decide which features to keep?
38. Can feature engineering improve a weak model?
39. Can too many engineered features hurt model performance?
40. Give an example of target leakage in a real-world project.

Practice Questions

Level 1

1. Define feature engineering.
2. Explain feature selection with an example.
3. Explain one-hot encoding.
4. Explain standardization.
5. Explain SMOTE.

Level 2

6. Take a customer dataset and:

Remove irrelevant features

Create new features

Encode categorical columns

Scale numerical columns

7. Compare:

StandardScaler

MinMaxScaler

8. Create polynomial features of degree 2.
9. Convert a date column into useful ML features.
10. Create a text feature such as word count.

Level 3

11. Build a complete ColumnTransformer + Pipeline.
12. Train a classifier on an imbalanced dataset.
13. Compare:

Without SMOTE

vs

With SMOTE

using:

Precision

Recall

F1

ROC-AUC

14. Demonstrate data leakage caused by incorrect preprocessing and then fix it using a Pipeline.

Module 9 Industry Project

Customer Churn — Advanced Feature Engineering

Raw Dataset

CustomerID

Age

Tenure

MonthlyCharges

TotalCharges

Contract

PaymentMethod

InternetService

SupportCalls

Churn

Step 1 — Remove

CustomerID

if it is only an identifier.

Step 2 — Create Features

AverageMonthlySpend

SupportCallsPerMonth

IsLongTermCustomer

Step 3 — Encode

Contract

PaymentMethod

InternetService

Step 4 — Scale

Age

Tenure

MonthlyCharges

TotalCharges

SupportCalls

Step 5 — Handle Imbalance

SMOTE

on training folds only.

Step 6 — Model

Logistic Regression

Random Forest

XGBoost

Step 7 — Evaluate

Precision

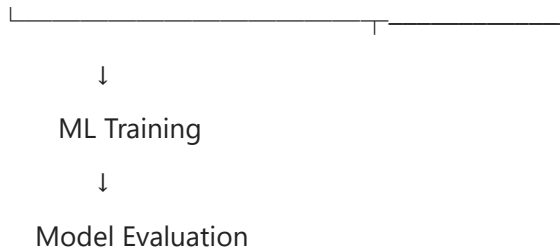
Recall

F1

ROC-AUC

🔗 Complete Module 9 Workflow





🎯 Module 9 Final Learning Outcome

After completing this module, students should be able to:

- Understand **Feature Engineering**
- Perform **Feature Selection**
- Perform **Feature Extraction**
- Perform **Feature Transformation**
- Encode categorical variables
- Apply **One-Hot Encoding**
- Apply **Ordinal Encoding**
- Apply **Standardization**
- Apply **Min-Max Scaling**
- Create **Polynomial Features**
- Engineer features from **dates, text, and domain data**
- Understand **imbalanced datasets**
- Apply **SMOTE**
- Avoid **data leakage**
- Build preprocessing using **Pipeline**
- Use **ColumnTransformer**
- Build production-oriented feature preprocessing workflows
- Improve ML models through meaningful feature design

🔗 Connection with previous modules

Module 4 → Data Preprocessing



Module 5 → EDA



Module 6 → Supervised Learning



Module 7 → Unsupervised Learning



Module 8 → Model Evaluation

↓

Module 9 → Feature Engineering

↓

Module 10 → Deep Learning

Next: Module 10 — Deep Learning will cover **Neural Networks, Artificial Neurons, Perceptrons, Activation Functions, Forward Propagation, Backpropagation, Loss Functions, Optimizers, Gradient Descent, Epoch/Batch/Iteration, Overfitting, Regularization, and Dropout.**

Module 10 — Deep Learning

Overview

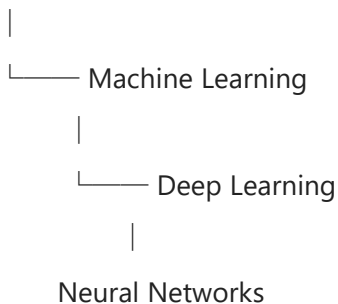
Deep Learning is a subset of Machine Learning that uses **artificial neural networks with multiple layers** to learn complex patterns from data.

Deep Learning is especially useful for:

- Image recognition
- Speech recognition
- Natural Language Processing
- Recommendation systems
- Autonomous systems
- Generative AI
- Computer vision
- Medical image analysis

Machine Learning vs Deep Learning

Artificial Intelligence



A traditional ML model often depends heavily on manually engineered features, while deep neural networks can learn useful representations directly from sufficiently large and suitable datasets.

1. Neural Networks

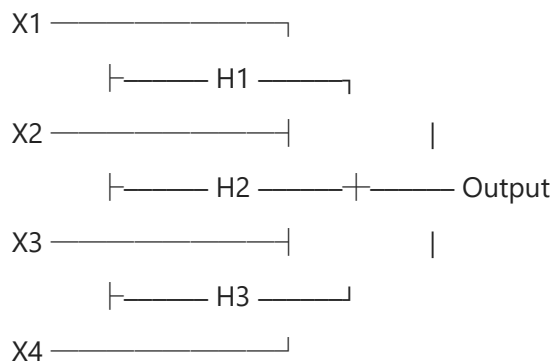
Concept

A **Neural Network** is a computational model inspired loosely by the structure of biological nervous systems.

It consists of interconnected computational units called **neurons** arranged in layers.

Basic structure:

Input Layer Hidden Layer Output Layer



Layers

Input Layer

Receives input features.

Example:

Age

Salary

Experience

Hidden Layers

Learn intermediate patterns.

Hidden Layer 1

Hidden Layer 2

Hidden Layer 3

Output Layer

Produces the final prediction.

For binary classification:

0 → No

1 → Yes

For multiclass classification:

Class A

Class B

Class C

For regression:

Predicted House Price

2. Artificial Neuron

Concept

An artificial neuron receives inputs, multiplies them by weights, adds a bias, and passes the result through an activation function.

The basic equation is:

$$z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

Then:

$$a = f(z)$$

where:

- x = input
- w = weight
- b = bias
- z = weighted sum
- f = activation function
- a = neuron output

Example

Suppose:

$$x_1 = 2$$

$$x_2 = 3$$

$$w_1 = 0.5$$

$$w_2 = 0.2$$

$$b = 1$$

Then:

$$z = (2)(0.5) + (3)(0.2) + 1$$

$$z = 1 + 0.6 + 1$$

```
[
z=2.6
]
```

The activation function then transforms 2.6.

3. Perceptron

Concept

The **Perceptron** is one of the simplest artificial neural-network models.

It was introduced as an early mathematical model of a neuron.

Basic structure:

```
X1 — w1 —
X2 — w2 —+—> Σ + Bias —> Activation —> Output
X3 — w3 —
```

The perceptron can learn a linear decision boundary.

Perceptron Equation

```
[
z = \sum_{i=1}^n w_{ix_i} + b
]
```

For a simple step activation:

```
[
output =
\begin{cases}
1 & \text{if } z \geq 0 \\
0 & \text{if } z < 0
\end{cases}
]
```

Important Limitation

A single perceptron cannot solve problems that are **not linearly separable**, such as the classic XOR problem.

This limitation motivated the development and use of multilayer neural networks.

4. Activation Functions

Concept

Activation functions introduce **non-linearity** into neural networks.

Without nonlinear activation functions, stacking many linear layers would still result in an overall linear transformation.

A. Sigmoid

Formula:

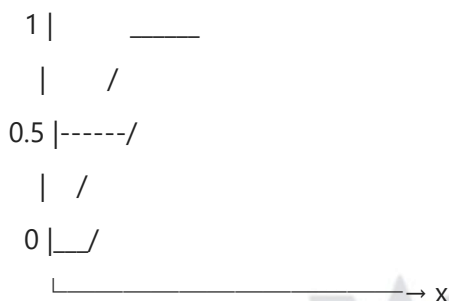
$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Output range:

0 to 1

Graphically:

Output



Common use

Binary classification output layer.

Example:

0.92 → 92% estimated probability of positive class

Python

```
import numpy as np

def sigmoid(x):
    return 1 / (1 + np.exp(-x))

print(sigmoid(2))
```

B. ReLU

ReLU = Rectified Linear Unit

Formula:

$$\text{ReLU}(x) = \max(0, x)$$

Examples:

$$\text{ReLU}(-5) = 0$$

$$\text{ReLU}(0) = 0$$

$$\text{ReLU}(3) = 3$$

It is widely used in hidden layers.

Python

```
def relu(x):  
    return max(0, x)
```

Why ReLU?

Compared with sigmoid/tanh, ReLU is computationally simple and often provides stronger gradients for positive inputs.

C. Tanh

Formula:

```
[  
tanh(x)  
]
```

Output range:

-1 to +1

It is zero-centered, unlike sigmoid.

D. Softmax

Softmax converts a vector of scores into a probability distribution.

For class i:

```
[  
Softmax(z_i)=  
\frac{e^{z_i}}  
{\sum_j e^{z_j}}  
]
```

Example:

Class A → 0.10

Class B → 0.70

Class C → 0.20

The probabilities sum to 1.

Softmax is commonly used in the output layer for **single-label multiclass classification**.

Activation Function Summary

Function	Output Range	Common Use
Sigmoid	0 to 1	Binary output
ReLU	0 to ∞	Hidden layers
Tanh	-1 to 1	Some hidden/recurrent architectures
Softmax	0 to 1, sums to 1	Multiclass output

5. Forward Propagation

Concept

Forward propagation is the process of passing input data through the neural network to generate a prediction.

Input

↓

Weights + Bias

↓

Activation

↓

Hidden Layer

↓

Weights + Bias

↓

Activation

↓

Output

Mathematical Idea

For one layer:

$$[Z = XW + b]$$

Then:

$$[A = f(Z)]$$

For multiple layers:

$$[A^{(1)} = f(W^{(1)}X + b^{(1)})]$$

```
[
A^{(2)}=f(W^{(2)}A^{(1)}+b^{(2)})
]
```

and so on.

6. Loss Functions

Concept

A **loss function** measures how different the model's prediction is from the actual target.

Actual

↓

Compare

↑

Predicted

↓

Loss

Training tries to minimize this loss.

A. Mean Squared Error

Common for regression.

```
[
MSE=
\frac{1}{n}
\sum(y-\hat{y})^2
]
```

Example:

Actual = 100

Predicted = 90

Error = 10

Squared Error = 100

B. Binary Cross-Entropy

Common for binary classification.

```
[
L=
-[y\log(p)+(1-y)\log(1-p)]
]
```

where:

- y = actual label
- p = predicted probability

If the correct class receives a high probability, the loss is low.

If the model is confidently wrong, the loss becomes much larger.

C. Categorical Cross-Entropy

Common for single-label multiclass classification.

Example:

Cat

Dog

Horse

If the actual class is Dog:

Cat $\rightarrow 0.05$

Dog $\rightarrow 0.90$

Horse $\rightarrow 0.05$

The loss is low because the correct class receives high probability.

7. Gradient Descent

Concept

Gradient Descent is an optimization algorithm used to minimize the loss function by updating model parameters in a direction that reduces the loss.

Imagine standing on a mountain:

```

      •
     /\
    /\ 
   /\ 
  /\ 
 /\ ↓ \
/\ ↓ \

```

Minimum

The algorithm tries to move toward a lower point in the loss landscape.

Basic Update Rule

[
 w_{new}

w_{old}

η

$\frac{\partial L}{\partial w}$

]

where:

- w = parameter/weight
- L = loss
 - η = learning rate

Learning Rate

The **learning rate** controls the size of each update.

Too Small

Very slow learning

Too Large

May overshoot the minimum

Appropriate

Gradually converges

8. Backpropagation

Concept

Backpropagation calculates how much each parameter contributed to the error by propagating gradients backward through the network.

Basic process:

Forward Pass

↓

Prediction

↓

Calculate Loss

↓

Backpropagation

↓

Calculate Gradients

↓

Optimizer Updates Weights

↓

Repeat

Example

Input

↓

Forward Propagation

↓

Prediction = 0.7

↓

Actual = 1

↓

Calculate Loss

↓

Backpropagation

↓

Gradients

↓

Update Weights

Important Difference

Backpropagation calculates gradients.

Gradient descent/optimizer uses those gradients to update parameters.

They are related but not the same thing.

9. Optimizers

Concept

An optimizer determines how model parameters are updated using gradients.

Common optimizers:

- SGD
 - Momentum
 - RMSprop
 - Adam
 - AdamW
-

SGD

Stochastic Gradient Descent updates parameters using batches of training examples.

```
optimizer = "sgd"
```

Adam

Adam = Adaptive Moment Estimation

Adam combines ideas related to momentum and adaptive learning rates.

It is a popular general-purpose optimizer for neural networks.

AdamW

AdamW is a variant of Adam that decouples weight decay from the gradient-based update, often providing better regularization behavior.

10. Epoch, Batch & Iteration

These three terms are extremely important.

Epoch

One complete pass through the entire training dataset.

Example:

1000 training samples

If:

epochs = 10

The model sees the training dataset 10 times.

Batch

A subset of training samples processed together.

Example:

1000 samples

Batch size = 100

Each batch contains 100 samples.

Iteration

One parameter-update step for one batch.

For:

1000 samples

Batch size = 100

approximately:

[
Iterations\ per\ epoch =
 $\frac{1000}{100}$

=10

]

Therefore:

10 iterations = 1 epoch

Visual

Dataset = 1000 samples

Batch Size = 100

Batch 1 → Update

Batch 2 → Update

Batch 3 → Update

...

Batch 10 → Update

↓

1 Epoch



11. Overfitting

Concept

Overfitting occurs when a model learns the training data too specifically and performs poorly on unseen data.

Example:

Training Accuracy = 99%

Validation Accuracy = 72%

This can indicate overfitting.

Underfitting

The model is too simple or insufficiently trained to capture the underlying pattern.

Example:

Training Accuracy = 65%

Validation Accuracy = 63%

Comparison

Situation Training Performance Validation Performance

Underfitting	Low	Low
Good Fit	High	High
Overfitting	Very High	Much Lower

12. Regularization

Concept

Regularization helps reduce overfitting by discouraging overly complex models.

Common approaches:

- L1 regularization
- L2 regularization / weight decay
- Dropout
- Early stopping
- Data augmentation
- Reducing model complexity

L1 Regularization

Adds the absolute values of weights to the loss.

```
[
Loss_{total}
Loss + \lambda \sum |w_i|
]
```

L1 can encourage some weights toward exactly zero, which can provide a form of sparsity.

L2 Regularization

Adds squared weights.

```
[
Loss_{total}
Loss + \lambda \sum w_i^2
]
```

L2 discourages excessively large weights.

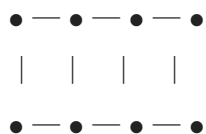
13. Dropout

Concept

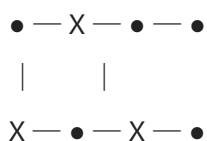
Dropout randomly deactivates a fraction of neurons during training.

Example:

Before Dropout



During one training step:



X represents temporarily dropped neurons.

Why Dropout?

It reduces reliance on specific neurons and can improve generalization.

Example:

```
from tensorflow.keras.layers import Dropout
```

Dropout(0.3)

This means approximately 30% of eligible activations are randomly dropped during each training step.

14. Building a Neural Network with TensorFlow/Keras

Example Problem

Suppose we want to classify students into:

Pass

Fail

Features:

Study Hours

Attendance

Previous Score

Model

```
import tensorflow as tf
```

```
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense

model = Sequential([
    Dense(32, activation="relu", input_shape=(3,)),
    Dense(16, activation="relu"),
    Dense(1, activation="sigmoid")
])

model.summary()
```

Compile the Model

```
model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)
```

Here:

Optimizer → Adam

Loss → Binary Cross-Entropy

Metric → Accuracy

Train the Model

```
history = model.fit(
    X_train,
    y_train,
    validation_split=0.2,
    epochs=20,
    batch_size=32
)
```

Evaluate

```
loss, accuracy = model.evaluate(
    X_test,
    y_test
)

print("Test Accuracy:", accuracy)
```

Make Predictions

```
predictions = model.predict(X_test)
```

```
print(predictions[:5])
```

For a sigmoid output, values are typically between 0 and 1.

Convert probabilities into classes:

```
predicted_classes = (
    predictions >= 0.5
```

```
).astype(int)
```

The threshold of 0.5 is common but should be chosen based on the application and validation performance.

15. Understanding the Network

Our model:

Input Layer

3 Features

↓

Dense(32, ReLU)

↓

Dense(16, ReLU)

↓

Dense(1, Sigmoid)

↓

Binary Prediction



16. Multi-Class Classification

Suppose we classify:

Cat

Dog

Horse

Use:

```
model = Sequential([
    Dense(64, activation="relu"),
    Dense(32, activation="relu"),
    Dense(3, activation="softmax")
```

```
])
```

Compile with:

```
model.compile(  
    optimizer="adam",  
    loss="sparse_categorical_crossentropy",  
    metrics=["accuracy"]  
)
```

Here the final layer contains:

3 neurons

because there are three classes.

17. Regression Neural Network

For regression, the output layer commonly uses a linear activation.

```
model = Sequential([  
    Dense(64, activation="relu"),  
    Dense(32, activation="relu"),  
    Dense(1)  
)
```

Compile:

```
model.compile(  
    optimizer="adam",  
    loss="mse",  
    metrics=["mae"]  
)
```

The final neuron outputs a continuous value.

Example:

Predicted House Price

₹58,40,000

18. Early Stopping

Concept

Training for too many epochs can cause overfitting.

Early stopping monitors validation performance and stops training when improvement stalls.

```
from tensorflow.keras.callbacks import EarlyStopping
```

```
early_stop = EarlyStopping(  
    monitor="val_loss",  
    patience=5,  
    restore_best_weights=True
```



```
)

history = model.fit(
X_train,
y_train,
    validation_split=0.2,
    epochs=100,
    callbacks=[early_stop]
)
```

19. Training vs Validation Curves

We can visualize training history.

```
import matplotlib.pyplot as plt

plt.plot(history.history["loss"])
plt.plot(history.history["val_loss"])

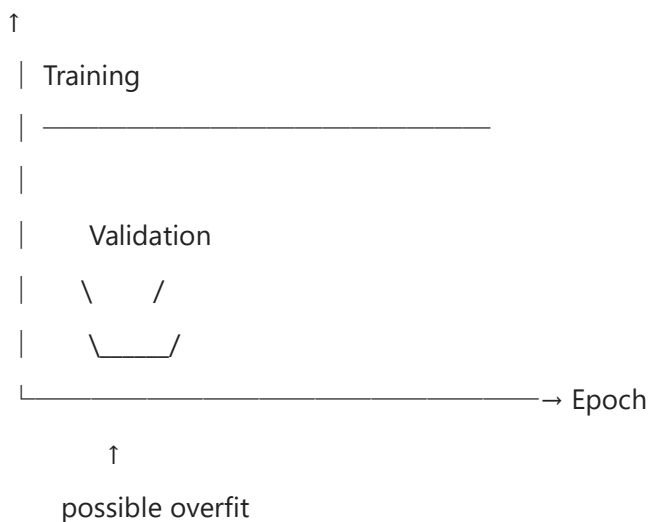
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("Training vs Validation Loss")

plt.legend([
    "Training Loss",
    "Validation Loss"
])

plt.show()
```

Overfitting Pattern

Loss



If training loss continues decreasing while validation loss begins increasing, this can indicate overfitting.

20. Deep Learning Workflow

Raw Dataset



Data Cleaning



Feature Engineering



Train / Validation / Test Split



Scaling / Preprocessing



Build Neural Network



Forward Propagation



Calculate Loss



Backpropagation



Optimizer Updates Weights



Repeat for Batches



Complete Epoch



Validation



Hyperparameter Tuning



Final Test Evaluation



Deployment



21. Hyperparameters

Hyperparameters are values set before or around training rather than learned directly as model weights.

Examples:

Learning Rate

Batch Size

Epochs

Number of Hidden Layers

Number of Neurons

Dropout Rate

Optimizer

Activation Function

Example:

```
model.fit(  
    X_train,  
    y_train,  
    epochs=50,  
    batch_size=32  
)
```

Here:

Epochs = 50

Batch Size = 32

22. Parameters vs Hyperparameters

This is a common interview question.

Parameters

Learned by the model during training:

Weights

Biases

Hyperparameters

Chosen by the practitioner:

Learning Rate

Batch Size

Number of Layers

Number of Neurons

Dropout Rate

Epochs

23. Batch Size Effects

Small Batch

Advantages:

- Can provide noisier but frequent updates
- Sometimes useful for generalization

Disadvantages:

- Can be slower computationally
- Less efficient on some hardware

Large Batch

Advantages:

- Efficient computation on suitable hardware
- More stable gradient estimates

Disadvantages:

- Requires more memory
- May sometimes generalize differently

There is no universally best batch size.

24. Learning Rate Effects

Too High

Loss



The optimization may overshoot or diverge.

Too Low

Very slow convergence

Appropriate

Loss decreases steadily

25. Deep Learning Example — Image Classification

Suppose we want to classify:

Cat

Dog

Input:

Image

Pipeline:

Image

↓

Resize

↓

Normalize

↓

Neural Network

↓

Feature Learning

↓

Classification

↓

Cat / Dog

For image data, a **CNN** is usually more appropriate than a plain fully connected network.

CNNs are covered in **Module 11**.

26. Deep Learning Example — Text Classification

Input:

"I love this product"

Pipeline:

Text

↓

Tokenization

↓

Numerical Representation

↓

Neural Network

↓

Prediction

↓

Positive

RNNs, LSTMs, GRUs, CNNs for text, and especially Transformers can be used depending on the task. NLP architectures are covered in later modules.

27. Deep Learning vs Traditional ML

Traditional ML

Often needs feature engineering

Works well with smaller datasets in many cases

Usually less computationally expensive

Easier to interpret in some models

Random Forest, SVM, KNN

Deep Learning

Learns representations automatically

Often benefits from larger datasets

Often computationally expensive

Often harder to interpret

Neural Networks, CNN, RNN, Transformers

This is a general comparison; some traditional models can also work extremely well on large datasets.

28. Common Deep Learning Problems

Vanishing Gradient

Gradients can become extremely small as they propagate through certain deep networks, making early layers difficult to train.

This was particularly associated with saturating activations such as sigmoid/tanh in deep networks.

Techniques that can help include:

- ReLU-family activations
 - Better initialization
 - Normalization
 - Residual connections
 - Appropriate architecture design
-

Exploding Gradient

Gradients become excessively large.

Possible techniques:

- Gradient clipping
 - Better initialization
 - Appropriate architecture
 - Normalization
-

29. Batch Normalization

Batch Normalization normalizes intermediate activations during training using statistics computed from the current mini-batch, with learnable scale and shift parameters.

Example:

```
from tensorflow.keras.layers import (
    Dense,
    BatchNormalization
)

model = Sequential([
    Dense(64, activation="relu"),
    BatchNormalization(),
    Dense(32, activation="relu"),
    Dense(1, activation="sigmoid")
])
```

It can help stabilize and accelerate training in some architectures.

30. Weight Initialization

Neural networks need sensible initial parameter values.

Common initialization approaches include:

- Xavier/Glorot initialization
- He initialization

For ReLU-like activations, **He initialization** is commonly appropriate.

```
from tensorflow.keras.initializers import HeNormal

Dense(
    64,
    activation="relu",
    kernel_initializer=HeNormal()
)
```

Complete Deep Learning Example

```
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping

model = Sequential([
    Dense(64, activation="relu", input_shape=(10,)),
    Dropout(0.3),
```

```
Dense(32, activation="relu"),
Dropout(0.2),

Dense(1, activation="sigmoid")
])

model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)

early_stop = EarlyStopping(
    monitor="val_loss",
    patience=5,
    restore_best_weights=True
)

history = model.fit(
X_train,
y_train,
    validation_split=0.2,
    epochs=100,
    batch_size=32,
    callbacks=[early_stop]
)

loss, accuracy = model.evaluate(
    X_test,
    y_test
)

print("Test Accuracy:", accuracy)
```

Module 10 Quick Revision

Topic	Key Idea
Neural Network	Connected layers of neurons
Neuron	Weighted sum + bias + activation
Perceptron	Basic linear binary classifier
Weight	Learned parameter controlling input influence

Topic	Key Idea
Bias	Learned offset
Activation	Introduces nonlinearity
ReLU	$\max(0, x)$
Sigmoid	Output between 0 and 1
Softmax	Multiclass probability distribution
Forward Propagation	Input → Prediction
Loss	Measures prediction error
Backpropagation	Computes gradients
Gradient Descent	Uses gradients to reduce loss
Optimizer	Updates model parameters
Epoch	One complete pass through training data
Batch	Subset of training data
Iteration	One batch update
Overfitting	Good training, poor generalization
Regularization	Helps reduce overfitting
Dropout	Randomly drops activations during training
Early Stopping	Stops training when validation performance stops improving

Module 10 Interview Questions

Basic

1. What is Deep Learning?
2. How is Deep Learning different from Machine Learning?
3. What is a neural network?
4. What is an artificial neuron?
5. What is a perceptron?
6. What are weights and biases?

Activation Functions

7. Why are activation functions required?
8. What is ReLU?
9. What is Sigmoid?
10. What is Tanh?
11. What is Softmax?
12. Why is ReLU commonly used in hidden layers?
13. When would you use sigmoid in the output layer?
14. When would you use softmax?

Training

15. What is forward propagation?
16. What is a loss function?
17. What is backpropagation?
18. What is gradient descent?
19. What is an optimizer?
20. What is learning rate?
21. What happens if the learning rate is too high?
22. What happens if the learning rate is too low?

Training Terminology

23. What is an epoch?
24. What is a batch?
25. What is an iteration?
26. If there are 1,000 samples and batch size is 100, how many iterations are there per epoch?

Overfitting

27. What is overfitting?
28. What is underfitting?
29. What is regularization?
30. What is dropout?
31. What is early stopping?
32. What is L1 regularization?
33. What is L2 regularization?

Advanced

34. What is the vanishing gradient problem?
35. What is the exploding gradient problem?
36. What is batch normalization?
37. What is weight initialization?
38. What is the difference between parameters and hyperparameters?
39. Why can neural networks require large amounts of data?
40. Why are neural networks computationally expensive?
41. How would you reduce overfitting in a neural network?
42. Why should the test set remain untouched during model tuning?

Practice Questions

Level 1

1. Explain a neural network in your own words.
2. Explain an artificial neuron with a numerical example.

3. Calculate:

```
[
z=w_1x_1+w_2x_2+b
]
```

for:

```
x1 = 2
x2 = 4
w1 = 0.5
w2 = 0.3
b = 1
```

4. Explain ReLU, Sigmoid, and Softmax.

Level 2

5. Build a binary classification neural network using Keras.

6. Train the model using:

Adam

Binary Cross-Entropy

20 epochs

Batch size = 32

7. Plot:

Training Loss

Validation Loss

8. Add Dropout and compare validation performance.

9. Add EarlyStopping.

Level 3

10. Compare:

No Regularization

L2 Regularization

Dropout

Early Stopping

11. Experiment with different learning rates:

0.1

0.01

0.001

12. Compare:

SGD

Adam

AdamW

13. Build a neural network for regression.

14. Compare its performance using:

MAE

RMSE

R^2

Module 10 Industry Project

Student Performance Prediction Using Neural Network

Problem

Predict whether a student will pass based on:

Study Hours

Attendance

Previous Marks

Assignment Score

Internal Marks

Pipeline

Student Dataset

↓

Data Cleaning

↓

Feature Engineering

↓

Train/Test Split

↓

Feature Scaling

↓

Neural Network

↓

Forward Propagation

↓

Loss Calculation

↓

Backpropagation

↓

Adam Optimizer



↓

Validation

↓

Early Stopping

↓

Test Evaluation

Suggested Architecture

Input

↓

Dense(64, ReLU)

↓

Dropout(0.3)

↓

Dense(32, ReLU)

↓

Dropout(0.2)

↓

Dense(1, Sigmoid)

↓

Pass / Fail

Evaluation

Use:

Accuracy

Precision

Recall

F1 Score

ROC-AUC

Confusion Matrix



Module 10 Final Learning Outcome

After completing this module, students should be able to:

- Understand **Deep Learning**
- Explain **Neural Networks**
- Understand **Artificial Neurons**
- Explain **Perceptrons**
- Calculate weighted sums

- Understand **Weights and Biases**
- Understand **Activation Functions**
- Implement **ReLU, Sigmoid, and Softmax**
- Understand **Forward Propagation**
- Understand **Loss Functions**
- Understand **Backpropagation**
- Understand **Gradient Descent**
- Understand **Optimizers**
- Understand **Epoch, Batch, and Iteration**
- Identify **Overfitting and Underfitting**
- Apply **L1/L2 regularization**
- Apply **Dropout**
- Use **Early Stopping**
- Build neural networks using **TensorFlow/Keras**
- Evaluate deep-learning models correctly

🔗 Connection with previous modules

Module 8 → Model Evaluation

↓

Module 9 → Feature Engineering

↓

Module 10 → Deep Learning

↓

Module 11 → CNN

↓

Module 12 → RNN & NLP

↓

Module 13 → Transformers & Generative AI

Next: Module 11 — CNN (Convolutional Neural Networks) will focus on **Computer Vision, image representation, convolution, filters/kernels, feature maps, pooling, CNN architecture, image classification, and transfer learning.**

Module 11 — CNN (Convolutional Neural Networks)

Overview

CNN (Convolutional Neural Network) is a type of Deep Learning architecture mainly designed for processing **images and other grid-like data**.

CNNs are widely used for:

- 🖼️ Image classification
- 👤 Face recognition
- 😊 Emotion detection
- 🚗 Object detection
- 🏥 Medical image analysis
- ✍️ Handwritten digit recognition
- 🛡️ Defect detection
- 🎥 Video analysis

A CNN automatically learns visual features such as:

Edges → Textures → Shapes → Parts → Objects

🧠 CNN Basic Architecture

Input Image



Convolution



Activation (ReLU)



Pooling



Convolution



Activation (ReLU)



Pooling



Flatten / Global Pooling



Dense Layer

↓

Output

Example:

Image

↓

[Conv2D]

↓

[ReLU]

↓

[MaxPooling]

↓

[Conv2D]

↓

[ReLU]

↓

[MaxPooling]

↓

[Flatten]

↓

[Dense]

↓

[Softmax]

↓

Cat / Dog / Horse



1. Computer Vision

Concept

Computer Vision is a field of AI that enables computers to interpret and analyze visual information such as images and videos.

Example

A human sees:



and understands:

"This is a dog."

A computer receives numerical pixel values and learns patterns that correspond to the object.

Applications

Healthcare

X-ray → Disease Detection

Security

Camera → Face Recognition

Transportation

Camera → Lane/Object Detection

Retail

Camera → Product Recognition

Agriculture

Plant Image → Disease Detection

2. Image Representation

A computer does not directly understand an image as a human does.

An image is represented as **numbers called pixels**.

Grayscale Image

A grayscale image has one channel.

Typical 8-bit pixel values:

0 → Black

255 → White

Example:

0 50 100

150 200 255

A grayscale image with height 28 and width 28 has:

[
28 $\times$ 28 = 784
]

pixel values.

RGB Image

A color image normally has three channels:

R → Red

G → Green

B → Blue

Each pixel contains three values.

Example:

Pixel = [255, 0, 0]

represents pure red in the standard RGB representation.

A 128×128 RGB image has:

[
128 $\times$ 128 $\times$ 3
]

values.

Image Tensor

CNN frameworks commonly represent image batches using tensors.

A typical TensorFlow/Keras image batch uses:

(batch_size, height, width, channels)

Example:

(32, 128, 128, 3)

means:

32 images

128 height

128 width

3 channels

3. Convolution

Concept

Convolution is the core operation of a CNN.

A small matrix called a **kernel/filter** slides across the image and performs mathematical operations to detect local patterns.

Example:

Input Image

1 2 3

4 5 6

7 8 9

A small kernel might be:

1 0

0 -1

The kernel moves across the image and calculates values at each location.

4. Filters / Kernels

Concept

A **kernel** is a small matrix of learnable weights.

Example:

3 × 3 Kernel

[1 0 -1]

[1 0 -1]

[1 0 -1]

A filter may learn to respond strongly to certain patterns such as:

- Vertical edges
- Horizontal edges
- Corners
- Textures
- Curves



In modern CNNs, these filters are **learned automatically during training**.

5. Convolution Operation

Suppose:

Input:

1 2 3

4 5 6

7 8 9

Kernel:

1 0

0 1

Take the top-left section:

1 2

4 5

Multiply element by element:

$$1 \times 1 + 2 \times 0$$

$$+ 4 \times 0 + 5 \times 1$$

$$[$$

$$= 1 + 0 + 0 + 5$$

$$]$$

$$[$$

$$= 6$$

$$]$$

The kernel then moves to another position.

6. Feature Map

The output generated after applying a filter to an image is called a **feature map** (also called an activation map).

Input Image

↓

Filter

↓

Feature Map

Different filters produce different feature maps.

For example:

Filter 1 → Edge Map

Filter 2 → Texture Map

Filter 3 → Corner Map

7. Why Multiple Filters?

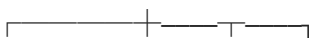
One filter cannot detect every type of visual pattern.

A convolutional layer can learn many filters.

Example:

Input Image

↓



↓ ↓ ↓ ↓

F1 F2 F3 F4

↓ ↓ ↓ ↓

Edge Texture Shape ...

Each filter learns different patterns during training.

8. Stride

Concept

Stride determines how far the kernel moves at each step.

Stride = 1

Move → 1 pixel at a time

Stride = 2

Move → 2 pixels at a time

Increasing stride generally reduces the spatial dimensions of the output.

9. Padding

Concept

Padding adds extra pixels around the border of an image.

Why?

- Preserve spatial dimensions
 - Allow edge information to be processed more fully
 - Control output size
-

Types

Valid Padding

No padding.

Output becomes smaller.

Same Padding

Padding is chosen so that, with stride 1, the output spatial dimensions are generally the same as the input dimensions.

10. Convolution Output Size

For one spatial dimension:

$$\left\lfloor \frac{N+2P-K}{S} \right\rfloor + 1$$

where:

- N = input size
- P = padding
- K = kernel size
- S = stride

Example

Suppose:

$$N = 28$$

$$K = 3$$

$$P = 0$$

$$S = 1$$

Then:

$$\left\lceil \frac{N - K}{S} + 1 \right\rceil$$

$$= \frac{28 - 3}{1} + 1$$

$$= 26$$

So:

$$28 \times 28$$

becomes:

$$26 \times 26$$

for that convolution layer and one output filter.

11. ReLU in CNN

After convolution, ReLU is commonly applied.

$$\text{ReLU}(x) = \max(0, x)$$

Example:

Before:

$$\begin{bmatrix} -2 & 5 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -4 & 8 \end{bmatrix}$$

After ReLU:

0 5 0

3 0 8

This introduces non-linearity.

12. Pooling

Concept

Pooling reduces the spatial dimensions of feature maps while retaining important information.

Common types:

- Max Pooling
- Average Pooling

Max Pooling

Consider:

1 3

2 4

Maximum:

4

So:

1 3

2 4

becomes:

4



Why Max Pooling?

It helps:

- Reduce computation
- Reduce spatial dimensions
- Retain strong activations
- Provide some robustness to small translations

Average Pooling

Instead of selecting the maximum, calculate the average.

1 3

2 4

```
[
Average=\frac{1+3+2+4}{4}
]

[
=2.5
]
```

Max Pooling vs Average Pooling

Max Pooling	Average Pooling
Selects maximum	Calculates average
Retains strongest activation	Retains average information
Very common in classic CNNs	Used in some architectures
Emphasizes strong detected features	Produces smoother summaries
Modern architectures sometimes use Global Average Pooling instead of flattening large feature maps.	

13. CNN Architecture

A basic CNN can look like:

Input Image

↓

Conv2D

↓

ReLU

↓

MaxPooling

↓

Conv2D

↓

ReLU

↓

MaxPooling

↓

Flatten

↓

Dense

↓

Output

Example:

$128 \times 128 \times 3$

↓

Conv2D(32, 3×3)

↓

$128 \times 128 \times 32$

↓

MaxPooling

↓

$64 \times 64 \times 32$

↓

Conv2D(64, 3×3)

↓

$64 \times 64 \times 64$

↓

MaxPooling

↓

$32 \times 32 \times 64$

↓

Flatten

↓

Dense

↓

Classification

The exact dimensions depend on padding, stride, and pooling configuration.

14. Flatten Layer

Concept

Convolutional layers produce multidimensional feature maps.

A Dense layer expects a vector-like input, so a traditional CNN can use **Flatten** to convert the feature maps into a one-dimensional vector.

Example:

Feature Map

2 4

1 3

Flatten:

[2, 4, 1, 3]

Python

```
from tensorflow.keras.layers import Flatten
```

Flatten()

15. Global Average Pooling

Instead of flattening the entire feature map, **Global Average Pooling (GAP)** computes the average activation of each feature map.

Example:

Feature Map 1 → Average → value 1

Feature Map 2 → Average → value 2

Feature Map 3 → Average → value 3

Result:

[value1, value2, value3]

Advantages can include:

- Fewer parameters
- Reduced overfitting risk
- Smaller model

Python

```
from tensorflow.keras.layers import GlobalAveragePooling2D
```

GlobalAveragePooling2D()

16. Image Classification

Problem

Suppose we have:

Cat

Dog

Horse

The CNN learns visual patterns from labeled images.

Training Images

↓

CNN

↓

Learn Features

↓

Classification

↓

Cat / Dog / Horse

17. CNN Image Classification Model

```
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import (
    Conv2D,
    MaxPooling2D,
    Flatten,
    Dense,
    Dropout
)
```

```
model = Sequential([
    Conv2D(
        32,
        (3, 3),
        activation="relu",
        input_shape=(128, 128, 3)
    ),
```

```
    MaxPooling2D((2, 2)),
```

```
    Conv2D(
        64,
        (3, 3),
        activation="relu"
    ),
```

```
MaxPooling2D((2, 2)),
```

```
Conv2D(
    128,
    (3, 3),
    activation="relu"
),
```

```
MaxPooling2D((2, 2)),
```

```
Flatten(),
```

```
Dense(128, activation="relu"),
```

```
Dropout(0.5),
```

```
Dense(3, activation="softmax")
])
```

```
model.summary()
```

This example assumes **three classes**.

18. Compile CNN

```
model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)
```

For integer class labels such as:

0 → Cat

1 → Dog

2 → Horse

`sparse_categorical_crossentropy` is appropriate.

If your labels are one-hot encoded, use:

```
loss="categorical_crossentropy"
```

19. Image Preprocessing

Before training, images commonly need preprocessing.

Typical steps:

Raw Image

↓

Resize

↓

Convert Color Format if Needed

↓

Normalize Pixel Values

↓

Batch

↓

CNN

Normalize Pixels

Original:

0 – 255

Convert approximately to:

0 – 1

using:

```
[  
X'=\frac{X}{255}  
]
```

Python:

```
X = X.astype("float32") / 255.0
```

20. Data Augmentation

Concept

Data augmentation creates varied training examples by applying transformations to existing images.

Common transformations:

- Rotation
- Horizontal flip
- Zoom
- Translation
- Crop

- Small contrast/brightness changes

The transformations should preserve the correct label.

Keras Example

```
from tensorflow.keras import Sequential
from tensorflow.keras.layers import (
    RandomFlip,
    RandomRotation,
    RandomZoom
)

augmentation = Sequential([
    RandomFlip("horizontal"),
    RandomRotation(0.1),
    RandomZoom(0.1)
])
```

This can help reduce overfitting when training data is limited.

21. Training CNN

```
history = model.fit(
    X_train,
    y_train,
    validation_data=(X_val, y_val),
    epochs=20,
    batch_size=32
)
```

22. CNN Prediction

```
predictions = model.predict(X_test)
```

For softmax output:

```
[
    0.05,
    0.90,
    0.05
]
```

If the class mapping is:

0 → Cat

1 → Dog

2 → Horse

prediction:

Dog

23. Transfer Learning

Concept

Transfer learning uses a model that has already learned useful visual representations on a large dataset and adapts it to a new task.

Instead of:

Train CNN from zero

we can use:

Pretrained Model

↓

Reuse Learned Features

↓

New Classification Head

↓

Train on Your Dataset

This is extremely useful when your dataset is relatively small.

24. Popular Pretrained CNN Architectures

Examples include:

- VGG
- ResNet
- Inception
- MobileNet
- EfficientNet
- DenseNet

These architectures differ significantly in structure, efficiency, and accuracy.

25. Transfer Learning Example

Using MobileNetV2:

```
import tensorflow as tf
from tensorflow.keras.applications import MobileNetV2
```

```
from tensorflow.keras import Sequential
from tensorflow.keras.layers import (
    GlobalAveragePooling2D,
    Dense
)
```

```
base_model = MobileNetV2(
    weights="imagenet",
    include_top=False,
    input_shape=(224, 224, 3)
)
```

```
base_model.trainable = False
```

```
model = Sequential([
    base_model,
    GlobalAveragePooling2D(),
    Dense(128, activation="relu"),
    Dense(3, activation="softmax")
])
```

26. Why Freeze the Base Model?

Initially:

```
base_model.trainable = False
```

This means the pretrained weights are not updated while the new classification head learns.

Benefits:

- Faster training
- Fewer trainable parameters
- Lower risk of overfitting on small datasets

After the new head has learned reasonably well, you can optionally **fine-tune** some deeper layers with a small learning rate.

27. Transfer Learning vs Fine-Tuning

Transfer Learning

Use pretrained features with the base model frozen.

Pretrained Model

↓

Frozen



New Classification Head



Training

Fine-Tuning

Unfreeze some pretrained layers and continue training them carefully.

Pretrained Model



Unfreeze Selected Layers



Small Learning Rate



Further Training

28. CNN Feature Learning

One of the most important concepts is **hierarchical feature learning**.

Early Layers

Learn simple features:

Edges

Lines

Corners

Middle Layers

Learn:

Textures

Patterns

Shapes

Deeper Layers

Learn:

Eyes

Ears

Wheels

Faces

Object parts

Final Layers

Combine these features to classify the object.

Pixels

↓

Edges

↓

Textures

↓

Shapes

↓

Parts

↓

Objects

29. CNN Parameters

For a convolution layer:

Input channels = 3

Filters = 32

Kernel = 3×3

Number of weights:

[
 $3 \times 3 \times 3 \times 32$
]

[
 =864
]

If each filter has one bias:

[
 $864 + 32 = 896$
]

So the layer has **896 trainable parameters**.

This is a common interview calculation.

30. Why CNN Instead of Fully Connected Network?

Suppose an image is:

$224 \times 224 \times 3$

Number of input values:

[
 $224 \times 224 \times 3 = 150,528$
]

A fully connected layer connected to hundreds of neurons can require a huge number of parameters.
 CNNs use:

Local Connectivity

A filter examines a small region at a time.

Weight Sharing

The same filter is applied across different locations.

Hierarchical Feature Learning

Layers progressively learn more complex visual features.

Therefore CNNs are much more suitable for many image-processing tasks.

31. Receptive Field

The **receptive field** of a neuron refers to the region of the original input that can influence that neuron's activation.

As we move deeper into a CNN:

Early Layer

↓

Small Receptive Field

Deep Layer

↓

Larger Receptive Field

This allows deeper neurons to capture more global patterns.

32. CNN vs ANN

ANN

General-purpose neural network

Often uses dense connections

Many parameters for images

Doesn't inherently exploit local spatial structure

Less suitable for raw images

CNN

Designed especially for spatial/grid data

Uses convolutional filters

More parameter-efficient for images

Exploits local spatial structure

Highly suitable for images

33. CNN Applications

Face Recognition

Face Image

↓

CNN

↓

Face Features

↓

Identity

Emotion Detection

Face

↓

CNN

↓

Features

↓

Happy / Sad / Angry / Neutral

Medical Imaging

X-Ray

↓

CNN

↓

Features

↓

Prediction

Sign Language Recognition

Hand Image

↓

CNN

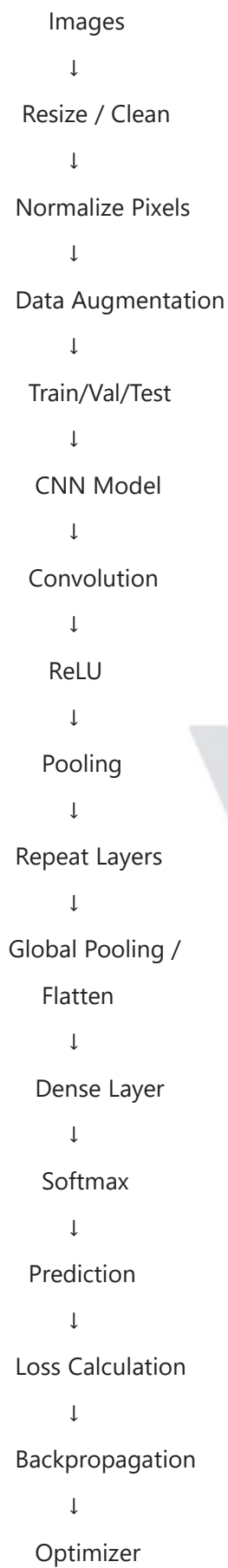
↓

Visual Features

↓

Sign Class

34. CNN Training Workflow



↓

Repeat

35. Complete CNN Example

```
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import (
    Conv2D,
    MaxPooling2D,
    Flatten,
    Dense,
    Dropout
)
```

```
model = Sequential([
    Conv2D(
        32,
        (3, 3),
        activation="relu",
        input_shape=(128, 128, 3)
    ),
```

```
    MaxPooling2D(2, 2),
```

```
    Conv2D(
        64,
        (3, 3),
        activation="relu"
    ),
```

```
    MaxPooling2D(2, 2),
```

```
    Conv2D(
        128,
        (3, 3),
        activation="relu"
    ),
```

```
    MaxPooling2D(2, 2),
```

```
    Flatten(),
```

```
Dense(
    128,
    activation="relu"
),

Dropout(0.5),

Dense(
    3,
    activation="softmax"
)
])

model.compile(
    optimizer="adam",
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

model.summary()
```

36. Output Interpretation

Suppose:

Model output:

[0.10, 0.85, 0.05]

Class mapping:

0 → Cat

1 → Dog

2 → Horse

Highest probability:

0.85

Therefore:

Prediction = Dog

Python:

```
import numpy as np
```

```
prediction = np.argmax(
    predictions[0]
)

print(prediction)
```

37. Common CNN Problems

Overfitting

Symptoms:

Training Accuracy → 98%

Validation Accuracy → 70%

Solutions:

- Data augmentation
 - Dropout
 - Early stopping
 - Regularization
 - More training data
 - Transfer learning
 - Smaller model
-

Underfitting

Possible solutions:

- Increase model capacity
 - Train longer
 - Improve features/data
 - Reduce excessive regularization
-

38. Important CNN Hyperparameters

You should understand:

Number of filters

Kernel size

Stride

Padding

Pooling size

Learning rate

Batch size

Epochs

Dropout

Number of layers

Example:

```
Conv2D(
    filters=64,
    kernel_size=(3, 3),
    strides=(1, 1),
    padding="same",
    activation="relu"
)
```

Module 11 Interview Questions

Basic

1. What is CNN?
2. Why are CNNs used for images?
3. What is Computer Vision?
4. How does a computer represent an image?
5. What is a pixel?
6. What are RGB channels?
7. What is a grayscale image?

Convolution

8. What is convolution?
9. What is a kernel?
10. What is a filter?
11. How does a CNN learn filters?
12. What is a feature map?
13. Why do we use multiple filters?
14. What is stride?
15. What is padding?
16. What is the difference between valid and same padding?
17. Calculate the output size of a convolution.

Pooling

18. What is pooling?
19. What is max pooling?

20. What is average pooling?
21. Why is pooling used?
22. What is Global Average Pooling?

Architecture

23. Explain a typical CNN architecture.
24. What is Flatten?
25. Why is ReLU used in CNNs?
26. Why is Softmax used for multiclass classification?
27. Why are CNNs better than fully connected networks for many image tasks?
28. What is a receptive field?

Transfer Learning

29. What is transfer learning?
30. Why is transfer learning useful?
31. What is fine-tuning?
32. What is the difference between transfer learning and fine-tuning?
33. Name some pretrained CNN architectures.
34. Why do we freeze the pretrained model initially?

Advanced

35. How many parameters are present in a convolutional layer?
36. What happens when kernel size increases?
37. What happens when stride increases?
38. How does padding affect output size?
39. How can you reduce CNN overfitting?
40. Why is data augmentation useful?
41. What is the difference between image classification and object detection?
42. Why might Global Average Pooling be preferred over Flatten?
43. What is the role of batch normalization in CNNs?

Practice Questions

Level 1

1. Explain CNN in simple words.
 2. Explain convolution using a 3×3 image.
 3. Explain kernel and filter.
 4. Explain max pooling.
 5. Explain RGB images.
-

Level 2

6. Calculate convolution output size:

Input = 64×64

Kernel = 3×3

Stride = 1

Padding = 0

7. Calculate parameters:

Input channels = 3

Filters = 32

Kernel = 3×3

8. Build a CNN for:

Cat vs Dog

9. Add:

Dropout

Data Augmentation

Early Stopping

10. Plot training and validation accuracy.

Level 3

11. Build an image classifier using **transfer learning**.

12. Compare:

CNN from scratch

vs

Transfer Learning

13. Compare:

Flatten

vs

GlobalAveragePooling2D

14. Experiment with:

Kernel = 3×3

Kernel = 5×5

15. Experiment with:

MaxPooling

vs

AveragePooling

Module 11 Industry Project

Sign Language Image Classification

Since sign-language recognition is a practical computer-vision application, this is a strong project for understanding CNNs.

Dataset

Example classes:

A

B

C

D

E

...

Pipeline

Hand Images



Resize



Normalize



Data Augmentation



Train / Validation / Test



CNN / Transfer Learning



Feature Extraction



Classification



Predicted Sign

Example Architecture

128×128×3 Image



Conv2D(32)



MaxPooling



Conv2D(64)

↓
 MaxPooling
 ↓
 Conv2D(128)
 ↓
 Global Average Pooling
 ↓
 Dense
 ↓
 Softmax
 ↓
 A / B / C / D / ...

Evaluation

Use:

Accuracy

Precision

Recall

F1 Score

Confusion Matrix

For a multiclass problem, also inspect **per-class performance**, because overall accuracy can hide poor performance on particular signs.

🎯 Module 11 Final Learning Outcome

After completing this module, students should be able to:

- Understand **Computer Vision**
- Understand image representation
- Understand pixels and RGB channels
- Explain **Convolution**
- Understand **Filters/Kernels**
- Understand **Feature Maps**
- Understand **Stride**
- Understand **Padding**
- Calculate convolution output dimensions
- Understand **Pooling**
- Explain Max Pooling and Average Pooling
- Understand CNN architecture

- Use **Flatten**
- Use **Global Average Pooling**
- Build CNNs with **TensorFlow/Keras**
- Perform image classification
- Apply image normalization
- Apply data augmentation
- Understand CNN overfitting
- Understand **Transfer Learning**
- Understand **Fine-Tuning**
- Use pretrained models such as **ResNet, MobileNet, EfficientNet**
- Build a practical image-classification project

🔗 Connection With Previous & Next Modules

Module 9

Feature Engineering



Module 10

Deep Learning



Module 11

CNN / Computer Vision



Module 12

RNN & NLP



Module 13

Transformers & Generative AI



Module 14

MLOps



Module 15

Industry Projects

Next → Module 12: RNN & NLP will cover **Natural Language Processing, text preprocessing, tokenization, stopwords, stemming, lemmatization, Bag of Words, TF-IDF, word embeddings, RNN, LSTM, and GRU.**

Module 12 — RNN & NLP

Overview

NLP (Natural Language Processing) is a field of AI that enables computers to understand, process, analyze, and generate human language.

RNN (Recurrent Neural Network) is a neural-network architecture designed to process sequential data, including text and time series.

This module connects traditional NLP techniques with neural-network-based language processing.

Human Language

↓

Text Preprocessing

↓

Tokenization

↓

Numerical Representation

↓

BoW / TF-IDF / Embeddings

↓

RNN / LSTM / GRU

↓

Prediction / Classification



1. Natural Language Processing

Concept

Natural Language Processing (NLP) combines AI, machine learning, and computational linguistics to work with human language.

Examples:

- Chatbots
- Sentiment analysis
- Spam detection
- Machine translation
- Text classification
- Speech-related language systems
- Search engines
- Text summarization
- Question answering

Real-World Example

Suppose a customer writes:

"I am very happy with this product."

An NLP system can determine:

Sentiment → Positive

Another example:

"Win a free iPhone! Click now!"

A spam classifier might predict:

Spam → Yes

2. NLP Pipeline

A typical traditional NLP pipeline:

Raw Text

↓

Cleaning

↓

Tokenization

↓

Stopword Handling

↓

Stemming / Lemmatization

↓

Numerical Representation

↓

Model

↓

Prediction

For modern NLP, the pipeline may instead use tokenization followed by embeddings and Transformer-based models.

3. Text Preprocessing

Concept

Raw text often contains information that needs to be normalized or cleaned before modeling.

Example:

"HELLO!!! I loved this PRODUCT 😊"

Possible preprocessing:

hello i loved this product

Common Operations

- Lowercasing
- Removing or normalizing punctuation
- Removing unwanted spaces
- Handling URLs
- Handling special characters
- Tokenization
- Stopword handling
- Stemming
- Lemmatization

⚠ Not every task requires all of these. Modern Transformer tokenizers, for example, generally do not require traditional stemming or stopword removal.

4. Tokenization

Concept

Tokenization converts text into smaller units called tokens.

Word Tokenization

Sentence:

"I love Python"

Tokens:

["I", "love", "Python"]

Sentence Tokenization

"I love Python. Python is easy."

Becomes:

```
[  
  "I love Python.",  
  "Python is easy."  
]
```

5. Token IDs

Machine-learning models cannot directly process words as text.

Words/tokens can be mapped to numerical IDs.

Example vocabulary:

hello → 1

python → 2

love → 3

AI → 4

Sentence:

"I love AI"

might become:

[5, 3, 4]

The exact IDs depend on the tokenizer and vocabulary.

6. Padding

Neural networks generally process batches of equal-length sequences.

Suppose:

Sentence 1 → [1, 2, 3]

Sentence 2 → [4, 5]

We can pad the shorter sequence:

[1, 2, 3]

[4, 5, 0]

Here 0 is a padding token.

Python

```
from tensorflow.keras.preprocessing.sequence import pad_sequences

sequences = [
    [1, 2, 3],
    [4, 5]
]

padded = pad_sequences(
    sequences,
    padding="post"
)
```

```
print(padded)
```

Output:

```
[[1 2 3]
```

```
[4 5 0]]
```

7. Stopwords

Concept

Stopwords are commonly occurring words that may carry limited information for some traditional NLP tasks.

Examples:

the

is

a

an

of

in

to

Example:

"The cat is on the table."

Possible result:

"cat table"

Important

Stopword removal is **task-dependent**.

For example:

"I do not like this"

Removing not would completely change the meaning.

Modern NLP systems often retain such information.

8. Stemming

Concept

Stemming reduces words to a simpler root-like form, often using heuristic rules.

Examples:

playing → play

played → play

studies → studi

Notice that stemming can produce words that are not proper dictionary words.

Python

```
from nltk.stem import PorterStemmer
```

```
stemmer = PorterStemmer()
```

```
print(stemmer.stem("playing"))
```

```
print(stemmer.stem("studies"))
```

Possible output:

play

studi

9. Lemmatization

Concept

Lemmatization converts a word into its **linguistic base form**, using vocabulary and linguistic information.

Examples:

running → run

better → good

cars → car

Compared with stemming, lemmatization generally produces more linguistically meaningful results, although it can be computationally more expensive.

Stemming vs Lemmatization

Stemming

Rule-based reduction

Usually faster

Can produce invalid words

Less linguistically precise

Lemmatization

Linguistic normalization

Usually more computationally involved

Usually produces valid base forms

More linguistically precise

10. Bag of Words

Concept

Bag of Words (BoW) represents text using the frequency or presence of words.

It ignores word order.

Suppose:

Document 1:

"I love Python"

Document 2:

"I love AI"

Vocabulary:

["I", "love", "Python", "AI"]

Representations:

Document 1 → [1, 1, 1, 0]

Document 2 → [1, 1, 0, 1]

Important Limitation

BoW does not understand word order or deeper semantic relationships.

These two sentences can have the same word counts:

"Dog bites man"

"Man bites dog"

but have very different meanings.

Python

```
from sklearn.feature_extraction.text import CountVectorizer

documents = [
    "I love Python",
    "I love AI"
]

vectorizer = CountVectorizer()

X = vectorizer.fit_transform(documents)

print(vectorizer.get_feature_names_out())
print(X.toarray())
```

11. N-Grams

An **n-gram** is a sequence of n consecutive tokens.

Unigram

"machine learning"

→ machine

→ learning

Bigram

"machine learning"

→ machine learning

Trigram

"I love machine learning"

→ I love machine

→ love machine learning

N-grams capture some local word-order information that a simple unigram BoW representation misses.

12. TF-IDF

Concept

TF-IDF = Term Frequency-Inverse Document Frequency

It gives higher importance to words that are frequent in a document but relatively uncommon across the overall collection of documents.

TF

Term Frequency measures how often a term appears in a document.

A common formulation is:

$$[\text{TF}(t,d) = \frac{\text{Number of occurrences of } t}{\text{Total number of terms in } d}]$$

IDF

A common formulation is:

$$[\text{IDF}(t) = \log \left(\frac{\text{Total number of documents}}{\text{Number of documents containing } t} \right)]$$

```
\frac{N}{df(t)}  
\right)  
]
```

where:

- N = number of documents
 - $df(t)$ = number of documents containing term t

Implementations may use smoothing variations.

TF-IDF

```
[  
TFIDF(t,d)=TF(t,d)\times IDF(t)  
]
```

A word appearing in many documents receives lower discriminative weight.

Python

```
from sklearn.feature_extraction.text import TfidfVectorizer  
  
documents = [  
    "Python is easy",  
    "Python is powerful",  
    "Machine learning is powerful"  
]  
  
vectorizer = TfidfVectorizer()  
  
X = vectorizer.fit_transform(documents)  
  
print(vectorizer.get_feature_names_out())  
print(X.toarray())
```

13. Word Embeddings

Concept

Traditional representations such as BoW and TF-IDF represent words using sparse vectors.

Word embeddings represent words as dense numerical vectors that can capture semantic and contextual relationships.

Example:

king → [0.21, 0.45, -0.12, ...]

queen → [0.23, 0.47, -0.10, ...]

Words with related usage patterns tend to have related vector representations.

14. Word2Vec

Word2Vec is a classic word-embedding technique.

Two main training approaches:

CBOW

Continuous Bag of Words

Predicts a target word from surrounding context.

The cat ___ on the mat.

↑

predict

Skip-Gram

Predicts surrounding words from a target word.

Target → "cat"

Predict:

the

sat

on



15. Embedding Layer

In neural networks, an embedding layer maps token IDs to dense vectors.

Example:

```
from tensorflow.keras.layers import Embedding
```

```
embedding = Embedding(  
    input_dim=10000,  
    output_dim=128  
)
```

This means:

Vocabulary size → 10,000

Embedding dimension → 128

16. What is Sequential Data?

Sequential data has an order in which previous elements can provide context for later elements.

Examples:

Sentence

Time series

Speech

Stock prices

Sensor readings

For text:

I → love → machine → learning

The order matters.

17. RNN

Concept

Recurrent Neural Network (RNN) is a neural-network architecture designed to process sequences while maintaining a hidden state that carries information from previous time steps.

Basic idea:

Input₁ → RNN → Hidden State

↓

Input₂ → RNN → Hidden State

↓

Input₃ → RNN → Output

A simplified recurrence is:

$$[h_t = f(W_{xx}x_t + W_{hh}h_{t-1} + b)]$$

where:

- (x_t) = current input
- (h_{t-1}) = previous hidden state
- (h_t) = current hidden state
- (W_x) = input weights
- (W_h) = recurrent weights
- (b) = bias

18. RNN Architecture

For:

"I love AI"

tokens may be processed as:

I
↓
RNN → h1
↓
love
↓
RNN → h2
↓
AI
↓
RNN → h3
↓

Output

The hidden state carries information forward.

19. RNN Memory

RNNs use the hidden state as a form of short-term memory.

Example:

"The movie was not very..."

The word:

"not"

is important for interpreting later words.

RNNs attempt to retain useful contextual information through their hidden state.

20. Problem with Basic RNN

Basic RNNs can suffer from:

Vanishing Gradients

Gradients become very small.

Result:

Earlier information

↓

Gets weaker

↓

Difficult to learn long-term dependencies

Exploding Gradients

Gradients become excessively large.

This can make training unstable.

21. LSTM

Concept

LSTM = Long Short-Term Memory

LSTM is a special type of RNN designed to better preserve useful information over longer sequences.

It uses a cell state and gating mechanisms.

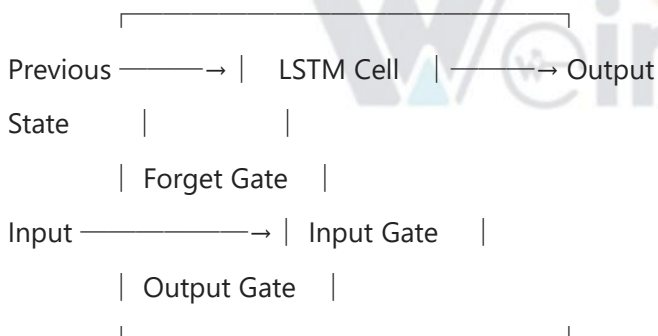
Main gates:

Forget Gate

Input Gate

Output Gate

22. LSTM Architecture



23. Forget Gate

Determines what information should be removed from the cell state.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

Output values are between:

0 and 1

Conceptually:

0 → Forget

1 → Keep

24. Input Gate

Determines which new information should be stored.

It controls how the current input contributes to the cell state.

25. Output Gate

Determines what information from the cell state should be exposed as the current hidden state.

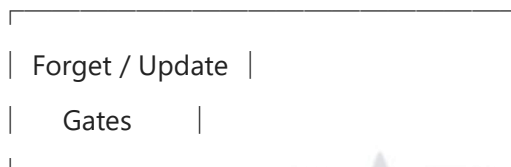
26. LSTM Cell State

The cell state acts as a pathway for information across time steps.

Conceptually:

Previous Cell State

↓



↓

Current Cell State

This helps LSTM preserve or discard information selectively.

27. GRU

Concept

GRU = Gated Recurrent Unit

GRU is another gated RNN architecture.

It is generally simpler than LSTM.

GRU uses two main gates:

Update Gate

Reset Gate

28. LSTM vs GRU

LSTM

Uses cell state + hidden state

Forget, input, output gates

GRU

Uses hidden state

Update and reset gates

LSTM

More parameters

Can model long dependencies

Often more complex

There is no universal winner; performance depends on the task and dataset.

GRU

Fewer parameters

Can also model long dependencies

Often simpler/faster

29. RNN vs LSTM vs GRU

Feature	RNN	LSTM	GRU
Basic recurrent architecture	✓	✗	✗
Handles sequence data	✓	✓	✓
Long-term dependencies	Limited	Better	Better
Gating	✗	✓	✓
Complexity	Low	Higher	Medium

30. Many-to-One Architecture

Used when a sequence produces one output.

Example:

"I love this movie"

↓

RNN

↓

Positive

Applications:

- Sentiment classification
- Spam detection
- Document classification

31. Many-to-Many Architecture

Multiple inputs produce multiple outputs.

Example:

Words → POS tags

I → Pronoun

love → Verb

Python → Noun

32. Many-to-One

x1

↓

x2 → RNN → Output

↓

x3

Example:

Review → Sentiment

33. One-to-Many

One input generates a sequence.

Example:

Image

↓

Caption Generator

↓

"The cat is sleeping"

This was historically implemented with CNN + RNN architectures, although modern systems commonly use Transformer-based architectures.

34. Sequence-to-Sequence

A sequence is transformed into another sequence.

Example:

English Sentence

↓

Encoder

↓

Decoder

↓

French Sentence

This architecture was important in early neural machine translation systems.

Modern systems commonly use Transformer architectures instead.

35. RNN with Keras

Example for text classification:

```
import tensorflow as tf
from tensorflow.keras import Sequential
from tensorflow.keras.layers import (
    Embedding,
    SimpleRNN,
    Dense
)

model = Sequential([
    Embedding(
        input_dim=10000,
        output_dim=128
    ),

    SimpleRNN(64),

    Dense(
        1,
        activation="sigmoid"
    )
])

model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)

model.summary()
```

36. LSTM with Keras

```
from tensorflow.keras import Sequential
from tensorflow.keras.layers import (
    Embedding,
    LSTM,
```

```

Dense
)

model = Sequential([
    Embedding(
        input_dim=10000,
        output_dim=128
    ),

    LSTM(64),

    Dense(
        1,
        activation="sigmoid"
    )
])

model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)

```

37. GRU with Keras

```

from tensorflow.keras import Sequential
from tensorflow.keras.layers import (
    Embedding,
    GRU,
    Dense
)

model = Sequential([
    Embedding(
        input_dim=10000,
        output_dim=128
    ),

    GRU(64),

```



```
Dense(  
    1,  
    activation="sigmoid"  
)  
])  
  
model.compile(  
    optimizer="adam",  
    loss="binary_crossentropy",  
    metrics=["accuracy"]  
)
```

38. Text Classification Workflow

Example: **Spam Detection**

Email

↓

Text Cleaning

↓

Tokenization

↓

Token IDs

↓

Padding

↓

Embedding

↓

LSTM / GRU

↓

Dense

↓

Sigmoid

↓

Spam / Not Spam



39. Complete NLP Example

```
import tensorflow as tf
```

```

from tensorflow.keras import Sequential
from tensorflow.keras.layers import (
    Embedding,
    LSTM,
    Dense
)

from tensorflow.keras.preprocessing.sequence import pad_sequences

texts = [
    "I love this product",
    "This product is amazing",
    "I hate this product",
    "This is a terrible product"
]

labels = [1, 1, 0, 0]

tokenizer = tf.keras.preprocessing.text.Tokenizer(
    num_words=1000
)

tokenizer.fit_on_texts(texts)

sequences = tokenizer.texts_to_sequences(texts)

X = pad_sequences(
    sequences,
    padding="post"
)

y = tf.convert_to_tensor(labels)

model = Sequential([
    Embedding(
        input_dim=1000,
        output_dim=32
    ),

    LSTM(32),

```

```
Dense(  
    1,  
    activation="sigmoid"  
)  
])  
  
model.compile(  
    optimizer="adam",  
    loss="binary_crossentropy",  
    metrics=["accuracy"]  
)  
  
model.fit(  
    X,  
    y,  
    epochs=10  
)
```

This is a small educational example; a useful production model requires a substantially larger and properly split dataset.

40. Sentiment Analysis

Concept

Sentiment analysis determines the emotional polarity or sentiment expressed in text.

Example:

"I really enjoyed this movie."

→ Positive

"The service was horrible."

→ Negative

"The product arrived today."

→ Neutral

Pipeline

Review

↓

Preprocessing

↓

Tokenization

↓

Embedding

↓

LSTM / GRU / Transformer

↓

Classification

↓

Positive / Negative / Neutral

41. Spam Detection

Input:

"Congratulations! You won a free prize!"

Output:

Spam

Input:

"Your meeting is scheduled for 3 PM."

Output:

Not Spam

42. Important NLP Metrics

For classification:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

For language-generation tasks, metrics can differ depending on the task; BLEU, ROUGE, and perplexity are examples with different purposes and limitations.

43. Perplexity

Perplexity is commonly used to evaluate language models.

It measures how well a probabilistic language model predicts a sequence.

Lower perplexity generally indicates that the model assigns higher probability to the observed sequence, although comparisons should be made carefully across compatible settings and tokenizers.

44. Word Embedding vs One-Hot Encoding

One-Hot

Suppose vocabulary:

["cat", "dog", "fish"]

Then:

cat → [1, 0, 0]

dog → [0, 1, 0]

fish → [0, 0, 1]

Properties:

- Sparse
- High-dimensional for large vocabularies
- No inherent semantic similarity

Embedding

cat → [0.21, 0.73, ...]

dog → [0.25, 0.70, ...]

Dense vectors can encode useful relationships.

45. Why Word Order Matters

Compare:

"The dog chased the cat."

and:

"The cat chased the dog."

The same words are present, but the meaning changes because the order changes.

BoW may represent both similarly.

Sequential models attempt to preserve order information.

46. Context

Consider:

"I went to the bank to deposit money."

versus:

"I sat near the river bank."

The word:

bank

has different meanings depending on context.

Traditional word embeddings such as Word2Vec generally assign one vector per word, whereas contextual language models can generate representations influenced by surrounding words.

This distinction becomes especially important in **Module 13 — Transformers & Generative AI**.

47. RNN Limitations

Traditional RNNs have several limitations:

1. Sequential Computation

Time steps depend on previous states, making training less parallelizable.

2. Long-Term Dependencies

Basic RNNs struggle with long-range dependencies.

3. Vanishing/Exploding Gradients

Training can become difficult for long sequences.

4. Slow for Long Sequences

Processing is inherently sequential.

These limitations helped motivate architectures such as LSTM/GRU and eventually Transformers.

48. Bidirectional RNN

A **Bidirectional RNN** processes a sequence in both directions.

Forward:

$A \rightarrow B \rightarrow C \rightarrow D$

Backward:

$D \rightarrow C \rightarrow B \rightarrow A$

The model can use information from both earlier and later positions.

Example:

```
from tensorflow.keras.layers import (
    Bidirectional,
    LSTM
)
```

```
layer = Bidirectional(  
    LSTM(64)  
)
```

This can be useful for tasks where the full sequence is available, such as text classification.

49. Text Classification Model Architecture

Text

↓

Tokenizer

↓

Padding

↓

Embedding

↓

Bidirectional LSTM

↓

Dense

↓

Sigmoid

↓

Prediction

Example:

```
from tensorflow.keras import Sequential  
from tensorflow.keras.layers import (  
    Embedding,  
    Bidirectional,  
    LSTM,  
    Dense  
)
```

```
model = Sequential([  
    Embedding(  
        input_dim=10000,  
        output_dim=128  
    ),  
)
```

```
Bidirectional(  
    LSTM(64)  
),  
  
Dense(  
    1,  
    activation="sigmoid"  
)  
])
```

50. NLP Project Workflow

Dataset

↓

Understand Text

↓

Clean / Normalize

↓

Train / Validation / Test Split

↓

Tokenizer

↓

Convert Text → IDs

↓

Padding

↓

Embedding

↓

RNN / LSTM / GRU

↓

Train

↓

Validate

↓

Evaluate

↓

Deploy

Module 12 Quick Revision

Topic	Key Idea
NLP	Processing human language
Tokenization	Text → tokens
Token ID	Token → integer
Padding	Equalize sequence lengths
Stopwords	Common words that may be filtered for some tasks
Stemming	Heuristic word reduction
Lemmatization	Linguistic base-form conversion
BoW	Word-frequency representation
N-gram	Sequence of N tokens
TF-IDF	Importance based on term/document frequency
Embedding	Dense numerical representation
Word2Vec	Classic word-embedding method
RNN	Sequence-processing neural network
Hidden State	Carries information across time
LSTM	Gated RNN for longer dependencies
GRU	Simpler gated RNN
Bidirectional RNN	Processes sequence in both directions
Perplexity	Language-model prediction metric

Module 12 Interview Questions

NLP Basics

1. What is NLP?
2. What are the applications of NLP?
3. What is text preprocessing?
4. What is tokenization?
5. What is padding?
6. Why do we convert text into numbers?
7. What are stopwords?
8. What is stemming?
9. What is lemmatization?
10. What is the difference between stemming and lemmatization?

Text Representation

11. What is Bag of Words?

12. What are the limitations of Bag of Words?
13. What are n-grams?
14. What is TF-IDF?
15. Explain TF-IDF mathematically.
16. What are word embeddings?
17. What is Word2Vec?
18. Explain CBOW and Skip-Gram.
19. What is the difference between one-hot encoding and embeddings?
20. Why are embeddings useful?

RNN

21. What is an RNN?
22. Why are RNNs useful for sequential data?
23. What is hidden state?
24. Explain forward recurrence in an RNN.
25. What is the vanishing-gradient problem?
26. What is the exploding-gradient problem?
27. What are the limitations of basic RNNs?

LSTM

28. What is LSTM?
29. Why was LSTM introduced?
30. What is the cell state?
31. What is a forget gate?
32. What is an input gate?
33. What is an output gate?
34. How does LSTM handle long-term dependencies?

GRU

35. What is GRU?
36. What are the gates in GRU?
37. LSTM vs GRU?
38. Why can GRU be computationally cheaper than LSTM?

Advanced

39. What is a bidirectional RNN?
40. What is many-to-one architecture?
41. What is many-to-many architecture?
42. What is sequence-to-sequence learning?
43. What is sentiment analysis?

44. What is perplexity?
45. Why is word order important in NLP?
46. Why can traditional word embeddings have difficulty with words having multiple meanings?
47. Why are Transformers often preferred over RNNs for modern large-scale NLP?

Practice Questions

Level 1

1. Explain NLP in simple words.
2. Tokenize:

"Machine learning is interesting"

3. Explain stemming vs lemmatization.
4. Explain Bag of Words.
5. Explain TF-IDF.
6. Explain word embeddings.
7. Explain RNN using a simple diagram.

Level 2

8. Create a spam-classification dataset.
9. Apply:

Tokenization

Padding

Embedding

10. Build an RNN-based spam classifier.
11. Replace RNN with LSTM.
12. Replace LSTM with GRU.
13. Compare:

RNN

LSTM

GRU

using validation performance and training time.

Level 3

14. Build a sentiment-analysis system.
15. Compare:

TF-IDF + Logistic Regression

against:

Embedding + LSTM

16. Add a Bidirectional LSTM.
17. Analyze the confusion matrix.
18. Investigate whether class imbalance affects precision, recall, and F1 score.

Module 12 Industry Project

Sentiment Analysis System

Problem

Build an AI system that predicts whether a customer review is:

Positive

Negative

Example

Input:

"The service was excellent and the staff were very helpful."

Output:

Positive

Complete Pipeline

Customer Reviews

↓

Data Cleaning

↓

Train/Validation/Test Split

↓

Tokenization

↓

Padding

↓

Embedding

↓

LSTM / GRU

↓

Dense Layer

↓

Sigmoid

↓

Positive / Negative

Suggested Architecture

Text

↓

Embedding(128)

↓

Bidirectional LSTM(64)

↓

Dropout

↓

Dense(32, ReLU)

↓

Dense(1, Sigmoid)

↓

Sentiment

Evaluation

Accuracy

Precision

Recall

F1 Score

Confusion Matrix



🎯 Module 12 Final Learning Outcome

After completing this module, students should be able to:

- Understand **NLP**
- Perform basic text preprocessing
- Understand tokenization
- Convert tokens into IDs
- Apply padding
- Understand stopwords
- Explain stemming and lemmatization
- Implement **Bag of Words**
- Implement **TF-IDF**
- Understand n-grams
- Understand word embeddings
- Explain Word2Vec, CBOW, and Skip-Gram

- Understand sequential data
- Explain RNN architecture
- Understand hidden states
- Understand vanishing/exploding gradients
- Explain **LSTM**
- Explain LSTM gates
- Explain **GRU**
- Compare RNN, LSTM, and GRU
- Build NLP classifiers using TensorFlow/Keras
- Perform sentiment analysis
- Build spam-detection systems
- Understand bidirectional RNNs
- Understand why modern NLP moved toward Transformers

Connection With the Next Module

Module 10

Deep Learning



Module 11

CNN → Computer Vision



Module 12

RNN & NLP → Sequential Language Processing



Module 13

Transformers & Generative AI



Attention

Self-Attention

BERT

GPT

LLMs

Prompt Engineering

Embeddings

RAG

Vector Databases

Fine-Tuning

AI Agents

Module 13 is especially important for modern AI development because it moves from traditional RNN/LSTM-based NLP to **Transformers, LLMs, Generative AI, RAG, embeddings, vector databases, fine-tuning, and AI agents**.

Module 13 — Transformers & Generative AI

This module explains how modern AI systems such as **BERT, GPT, LLMs, RAG systems, and AI agents** work, with a focus on concepts useful for Machine Learning students.

1. Transformers

Concept

A **Transformer** is a deep learning architecture designed to process sequential data using the **attention mechanism** instead of relying mainly on recurrence.

Transformers are widely used in:

- Natural Language Processing (NLP)
- Machine translation
- Text generation
- Question answering
- Summarization
- Code generation
- Multimodal AI

Simple Explanation

Suppose a sentence is:

"The student submitted the project because she completed it."

To understand "**she**", the model needs to connect it with "**student**".

Transformers use attention to determine which words are important to each other.

Basic Architecture

Input Text

↓

Tokenization

↓

Embeddings

↓

Positional Information

↓

Transformer

↓

Attention

↓

Feed Forward Network

↓

Output

Real-World Example

When you ask a chatbot:

"Explain Machine Learning in simple words."

A transformer-based model processes the words and generates an appropriate response based on learned patterns.

2. Attention Mechanism

Concept

Attention allows a model to focus on the most relevant parts of the input while processing information.

Simple Explanation

Think about reading a paragraph.

You don't give equal importance to every word. Some words are more important for understanding the meaning.

Attention works similarly.

Input Sentence

↓

Identify Important Words

↓

Calculate Relationships

↓

Focus on Relevant Information

↓

Generate Representation

Real-World Example

Sentence:

"The dog chased the ball because it was excited."

The model needs to understand what "it" refers to.
Attention helps establish relationships between words.

3. Self-Attention

Concept

Self-attention allows each token in a sequence to look at other tokens in the same sequence and determine their importance.

The mechanism uses three concepts:

- Query (Q)
- Key (K)
- Value (V)

Formula

The scaled dot-product attention formula is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where:

- Q = Query
- K = Key
- V = Value
 - d_k = dimension of the key vectors

Simple Explanation

Query + Key

↓

Similarity Score

↓

Softmax

↓

Attention Weights

↓

Weighted Values

4. Encoder and Decoder

Transformers can use **encoders**, **decoders**, or both depending on the architecture.

Encoder

The encoder converts input information into meaningful representations.

Input

↓

Embedding

↓

Self-Attention

↓

Feed Forward

↓

Representation

Decoder

The decoder generates output sequentially while using available context.

Context

↓

Attention

↓

Prediction

↓

Next Token

↓

Next Token



Examples

BERT → primarily encoder-based.

GPT → decoder-based.

Original Transformer for translation → encoder + decoder.

5. BERT

Concept

BERT stands for:

Bidirectional Encoder Representations from Transformers

BERT is a transformer-based language model designed primarily for understanding text.

Key Idea

BERT considers context from both directions.

Previous Context ← WORD → Following Context

Applications

- Text classification
- Sentiment analysis
- Question answering
- Named Entity Recognition
- Search
- Text understanding

Example

Input:

"The bank is near the river."

BERT can use surrounding words to understand that **bank** refers to a riverside bank rather than a financial institution.

6. GPT

Concept

GPT stands for:

Generative Pre-trained Transformer

GPT-style models are decoder-based transformer models designed for generating text.

Basic Process

Input Prompt

↓

Tokenization

↓

Transformer

↓

Predict Next Token

↓

Generate Next Token

↓

Continue

Example

Prompt:

"Machine Learning is"

Possible generated continuation:

"a field of AI that enables computers to learn patterns from data."

7. Large Language Models (LLMs)

Concept

A **Large Language Model (LLM)** is a machine learning model trained on large amounts of text and other data to understand and generate language.

Common Capabilities

- Text generation
- Summarization
- Question answering
- Translation
- Classification
- Coding assistance
- Information extraction
- Conversational interaction

Basic LLM Pipeline

Large Dataset



Pretraining



Transformer Model



Fine-tuning / Alignment



LLM



User Prompt



Generated Response

Important Point

An LLM does not simply "look up an answer" in the way a traditional database query does. It generates output based on patterns and representations learned during training and the information available in its context.

8. Prompt Engineering

Concept

Prompt engineering means designing instructions and input context so that an AI model is more likely to produce the desired output.

Poor Prompt

Tell me about ML.

Better Prompt

Explain Machine Learning to a beginner in simple language with 3 real-world examples and a short Python example.

Useful Prompt Components

Role

+

Task

+

Context

+

Constraints

+

Output Format

Example

You are an ML instructor.

Explain Random Forest to a beginner.

Include:

1. Definition

2. Simple example

3. Advantages

4. Python code

Use simple language.

9. Embeddings

Concept

An **embedding** is a numerical representation of data such as text, where similar meanings can be represented by vectors that are close in an embedding space.

Example

"Machine Learning"

↓

[0.21, -0.14, 0.78, 0.35, ...]

Another sentence with similar meaning may have a nearby vector representation.

Applications

- Semantic search
- Recommendation systems
- Document similarity
- RAG
- Clustering
- Retrieval systems

Simple Representation

Text

↓

Embedding Model

↓

Vector

↓

Compare / Search / Retrieve

10. RAG — Retrieval-Augmented Generation

Concept

RAG combines information retrieval with generative AI.

Instead of depending only on information learned during model training, a RAG system retrieves relevant external information and provides it as context to the language model.

RAG Architecture

User Question

↓

Create Query Embedding

↓

Vector Database Search



Retrieve Relevant Documents



Add Context to Prompt



LLM



Answer

Real-World Example

Suppose a company has:

Company Policies

Employee Handbook

Project Documents

FAQs

Training Material

A user asks:

"What is our internship leave policy?"

The RAG system retrieves the relevant policy document and gives it to the LLM as context.

11. Vector Databases

Concept

A **vector database** stores and searches numerical vector representations called embeddings.

Instead of searching only for exact words, vector search can find information based on semantic similarity.

Traditional Search

"ML course"



Keyword matching

Vector Search

"How can I learn machine learning?"



Embedding



Similar vectors

↓

Relevant ML documents

Common Use Cases

- RAG
- Semantic search
- Recommendation systems
- Document retrieval
- Knowledge assistants

12. Fine-Tuning

Concept

Fine-tuning means continuing the training of a pretrained model on a more specific dataset for a particular task or behavior.

Process

Pretrained Model

↓

Task-Specific Dataset

↓

Fine-Tuning

↓

Specialized Model

Example

A general language model can be adapted for:

- Customer support
- Medical text classification
- Legal document classification
- Domain-specific language tasks
- Company-specific workflows

Important Difference

Prompting:

Same model

+

Better instructions

Fine-tuning:

Pretrained model

+

Additional training

=

Adapted model

13. AI Agents

Concept

An **AI agent** is a system that can use a model together with tools, memory/context, and a workflow to accomplish a task.

Basic Architecture

User Goal

↓

AI Agent

↓

Reason / Plan

↓

Choose Tool

↓

Execute Action

↓

Observe Result

↓

Continue / Finish

Example

A learning-platform AI agent could:

Student asks question

↓

Understand question

↓

Search learning material

↓

Retrieve relevant content

↓

Generate explanation

↓

Give answer

Possible Tools

- Web search
- Database
- Calculator
- Python execution
- APIs
- File search
- CRM
- Email systems

14. Generative AI

Concept

Generative AI refers to AI systems capable of generating new content based on learned patterns.

Types

Type	Generated Content
Text Generation	Articles, answers, summaries
Image Generation	Images, designs
Audio Generation	Speech, music
Video Generation	Videos
Code Generation	Program code

Traditional ML vs Generative AI

Traditional ML

Data → Model → Prediction

Generative AI

Prompt → Model → New Content

15. Interview Questions

Beginner

Q1. What is a Transformer?

A Transformer is a deep learning architecture that uses attention mechanisms to process relationships between elements in sequential data.

Q2. What is attention?

Attention allows a model to assign different importance to different parts of its input.

Q3. What is self-attention?

Self-attention allows tokens within the same sequence to interact and determine their relevance to each other.

Q4. What is BERT?

BERT is an encoder-based transformer model designed mainly for language understanding.

Q5. What is GPT?

GPT is a decoder-based transformer family designed for generative language tasks.

Q6. What is an LLM?

An LLM is a large language model trained on large datasets to understand and generate language.

Q7. What are embeddings?

Embeddings are numerical vector representations that capture useful relationships or semantic information.

Q8. What is RAG?

RAG combines retrieval of relevant external information with generation by an AI model.

Q9. What is fine-tuning?

Fine-tuning adapts a pretrained model by training it further on a task- or domain-specific dataset.

Q10. What is an AI agent?

An AI agent is a system that uses an AI model and potentially tools and context to accomplish tasks.

Practice Questions

1. Explain the Transformer architecture.
2. What is the attention mechanism?
3. Explain Query, Key and Value.
4. What is self-attention?
5. Differentiate BERT and GPT.
6. What is an LLM?
7. What are embeddings?
8. Explain semantic search.
9. What is RAG?
10. Explain the RAG pipeline.
11. What is a vector database?
12. What is fine-tuning?
13. Prompt engineering vs fine-tuning.
14. What is Generative AI?
15. What is an AI agent?
16. Design a RAG-based chatbot for a learning platform.
17. Explain how embeddings are used in RAG.
18. Explain a real-world application of AI agents.

🎯 Module 13 Quick Revision

TRANSFORMERS

↓

ATTENTION

↓

SELF-ATTENTION

↓

BERT / GPT

↓

LLMs

↓

PROMPT ENGINEERING

↓

EMBEDDINGS

↓

VECTOR DATABASE

↓

RAG

↓

FINE-TUNING

↓

AI AGENTS

Key takeaway: Modern Generative AI systems are largely built around **transformers, attention, language models, embeddings, retrieval, and tool-using workflows**.

📊 Module 15 — Industry Projects

Module 15 is the **practical/project-based part of Machine Learning**. The goal is to take everything learned in Modules 1–14 and build complete ML applications.

For every project, follow:

Problem → Dataset → Features/Target → Preprocessing → EDA → Feature Engineering → Model → Evaluation → Prediction → Deployment

1. 🏠 House Price Prediction

Concept

House Price Prediction is a **Regression** problem where the model predicts a continuous value: the price of a house.

Real-World Example

Given:

Area Bedrooms Location Age Price

800	2	Pune	5	₹50 L
1200	2	Pune	3	₹72 L
1500	3	Mumbai	4	₹95 L

The model learns the relationship between the features and house price.

Features

- Area
- Bedrooms
- Bathrooms
- Location
- Property age
- Parking

Target

House Price

Algorithms

- Linear Regression
- Random Forest Regressor
- Decision Tree Regressor
- Gradient Boosting

Evaluation

- MAE
- MSE
- RMSE
- R^2 Score

Basic Python

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score
```

```
data = pd.read_csv("house_prices.csv")
```

```
X = data[["area", "bedrooms"]]
y = data["price"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

```
print("MAE:", mean_absolute_error(y_test, predictions))
print("R2:", r2_score(y_test, predictions))
```

Interview Questions

1. Why is house-price prediction a regression problem?
2. Which features can influence house price?
3. Why might categorical location need encoding?
4. Which metrics are suitable for this project?

2. 🎓 Student Performance Prediction

Concept

This project predicts a student's performance based on academic and behavioral information.

Features

- Study hours
- Attendance
- Previous scores
- Assignment completion
- Sleep hours
- Participation

Possible Targets

Regression:

Predict final marks.

Classification:

Predict Pass / Fail.

Algorithms

- Linear Regression
- Random Forest
- Decision Tree
- Logistic Regression

Example

Study Hours = 6

Attendance = 90%

Previous Score = 78

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ML Model

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Predicted Score = 84

Real-World Application

Educational platforms can use ML to identify students who may need additional support.

Practice Questions

1. What are the features?
2. What is the target?
3. Is this regression or classification?
4. What preprocessing is required?
5. Which metric would you use?

3. Customer Churn Prediction

Concept

Customer churn means a customer stops using a company's product or service.

The ML model predicts whether a customer is likely to leave.

Target

Churn = 1 → Customer will leave

Churn = 0 → Customer will stay

Features

- Customer age
- Tenure
- Monthly charges
- Contract type
- Number of support calls
- Payment method
- Usage frequency

Algorithm

This is usually a **classification problem**.

Possible algorithms:

- Logistic Regression
- Decision Tree
- Random Forest
- SVM

Example

Customer:

Tenure = 3 months

Monthly Charges = ₹999

Support Calls = 8

Contract = Monthly

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Churn Prediction

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High Churn Risk

Important Metrics

For churn prediction, **recall** can be particularly important because missing a customer who is likely to churn may be costly.

Interview Questions

- What is customer churn?
- Why is churn prediction classification?
- Why can accuracy be misleading?
- What is the difference between precision and recall?

4. 🏠 Loan Approval Prediction

Concept

This project predicts whether a loan application should be approved based on applicant information.

Features

- Applicant income
- Loan amount

- Credit history
- Employment status
- Education
- Property area
- Dependents

Target

Approved

Rejected

ML Type**Supervised Classification****Workflow**

Loan Dataset



Data Cleaning



Missing Values



Categorical Encoding



Feature Scaling



Train/Test Split



Classification Model



Evaluation



Loan Prediction

Possible Algorithms

- Logistic Regression
- Decision Tree
- Random Forest
- SVM

Important Consideration

Real financial applications require careful handling of **fairness, privacy, data quality, and regulatory requirements**. A model prediction should not automatically be treated as the final decision.

5. 📧 Spam Detection

Concept

Spam Detection classifies messages or emails as:

Spam

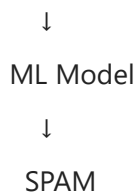
OR

Not Spam

It is a **classification problem**.

Example

"Congratulations! You won ₹10,00,000!"



NLP Pipeline

Text



Cleaning



Tokenization



Stopword Handling



TF-IDF / Embeddings



ML Model



Prediction

Algorithms

- Naive Bayes
- Logistic Regression
- SVM

Example Code

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
```

```
messages = [
    "Win a free prize now",
    "Meeting at 10 AM",
    "Congratulations you won money",
    "Please submit your assignment"
]

labels = ["spam", "ham", "spam", "ham"]

vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(messages)

model = MultinomialNB()
model.fit(X, labels)

new_message = vectorizer.transform(
    ["You won a free prize"]
)

print(model.predict(new_message))
```

Expected Output

```
['spam']
```

6. Movie Recommendation System

Concept

A recommendation system suggests movies that a user may like.

Example

A user watches:

Inception

Interstellar

The Matrix

The system can recommend movies with similar characteristics.

Types

Content-Based Filtering

Recommends items similar to items the user already likes.

Collaborative Filtering

Uses preferences from multiple users to identify patterns.

Content-Based Workflow

Movie Data

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Movie Features

↓

Vectorization

↓

Similarity Calculation

↓

Similar Movies

↓

Recommendation

Similarity

A common measure is **cosine similarity**:

$$\frac{A \cdot B}{\|A\| \|B\|}$$

Example

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

movies = [
    "science fiction space",
    "science fiction future",
    "romantic comedy",
    "action superhero"
]

vectorizer = TfidfVectorizer()
vectors = vectorizer.fit_transform(movies)

similarity = cosine_similarity(vectors)

print(similarity)
```

7. 👤 Customer Segmentation

Concept

Customer segmentation groups customers based on similar characteristics.

This is an **Unsupervised Learning** problem.

Example

A company may discover:

Cluster 1 → High-value customers

Cluster 2 → Occasional customers

Cluster 3 → Discount-focused customers

Features

- Annual income
- Spending score
- Purchase frequency
- Age
- Total purchases

Algorithm

K-Means Clustering

Workflow

Customer Data



Data Cleaning



Feature Selection



Scaling



K-Means



Customer Clusters



Business Strategy

Business Applications

- Personalized marketing
- Customer targeting
- Product recommendations

- Loyalty programs

Interview Questions

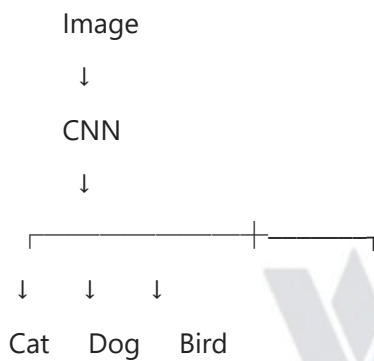
1. Why is segmentation unsupervised?
2. What is K in K-Means?
3. How do you select K?
4. Why is scaling important?

8. Image Classification

Concept

Image classification assigns an image to a predefined category.

Example



Example Applications

- Animal classification
- Plant classification
- Defect detection
- Medical image analysis
- Object recognition

CNN Workflow

Image

↓

Resize

↓

Normalize

↓

Convolution

↓

Pooling

↓

Feature Extraction

↓

Classification

↓

Prediction

Tools

- Python
- TensorFlow
- Keras
- OpenCV

Important Concepts

- Convolution
- Filters
- Feature maps
- Pooling
- Activation functions
- CNN
- Transfer learning

9. 😊 Face / Emotion Detection

Concept

A computer vision system can detect faces and, where an appropriately trained model is available, classify visible facial-expression categories.

Pipeline

Camera / Image

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Face Detection

↓

Face Region

↓

Preprocessing

↓

CNN / ML Model

↓

Expression Classification

Possible Expression Classes

Happy

Sad

Angry

Neutral

Surprised

Tools

- OpenCV
- TensorFlow/Keras
- NumPy
- Python

Example

```
import cv2

image = cv2.imread("person.jpg")

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

print("Image processed successfully")
```

Important Note

Expression classification is not the same as reliably determining a person's internal emotional state. Real-world systems should be designed and evaluated carefully.

10. 🤖 Chatbot

Concept

A chatbot is a software system that communicates with users using text or voice.

Basic ML/NLP Chatbot

User Message

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Text Preprocessing

↓

Intent Detection

↓

Find Response

↓

Response Generation

Example

User:

"What courses are available?"

↓

Intent:

Course Information

↓

Response:

"We offer AI/ML, Python,
Data Science and Full Stack courses."

Chatbot Approaches

Rule-Based

IF user says "hello"
THEN respond "Hello!"

ML-Based

The model predicts the user's intent.

LLM-Based

A language model generates a response using the conversation context.

11. 📄 RAG-Based AI Assistant

Concept

A **Retrieval-Augmented Generation (RAG)** system combines document retrieval with an LLM.

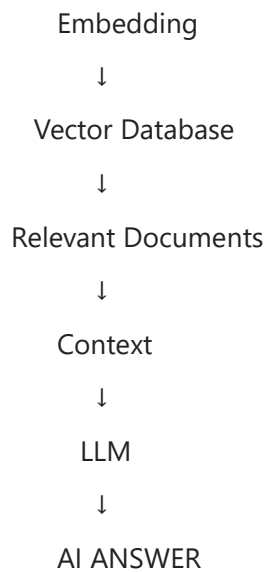
It is especially useful for a learning platform containing:

- Course notes
- PDFs
- FAQs
- Documentation
- Training material
- Internal knowledge

Architecture

USER QUESTION

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Example

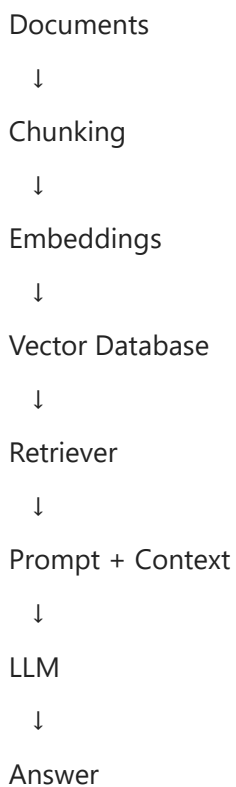
Student asks:

"Explain K-Means with an example."

The system:

1. Converts the query into an embedding.
2. Searches the vector database.
3. Retrieves relevant ML notes.
4. Sends the retrieved context to the LLM.
5. Generates a grounded explanation.

Main Components



12. 🚀 End-to-End ML Project Structure

A professional ML project can follow this structure:

ml-project/

```

|
├── data/
|   ├── raw/
|   └── processed/
|
├── notebooks/
|   └── EDA.ipynb
|
├── src/
|   ├── preprocessing.py
|   ├── features.py
|   ├── train.py
|   ├── evaluate.py
|   └── predict.py
|
├── models/
|   └── model.pkl
|
├── app/
|   └── app.py
|
├── requirements.txt
├── README.md
└── .gitignore
  
```

Complete Workflow

Problem Definition

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Data Collection

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Data Cleaning

↓

EDA



Feature Engineering



Train/Test Split



Model Training



Evaluation



Hyperparameter Tuning



Save Model



API / Application



Deployment



Monitoring



Industry Project Template

For **every ML project**, students should document:

1. Problem Statement

What problem are we solving?

2. Objective

What should the model achieve?

3. Dataset

Where does the data come from?

4. Features

What inputs are used?

5. Target

What does the model predict?

6. Preprocessing

What cleaning and transformation is required?

7. EDA

What patterns were discovered?

8. Model Selection

Which algorithms were tested and why?

9. Training

How was the model trained?

10. Evaluation

Which metrics were used?

11. Error Analysis

Where does the model make mistakes?

12. Deployment

How will users access the model?

13. Monitoring

How will model performance be tracked after deployment?

Common Technology Stack

Area	Technologies
Programming	Python
Data Handling	NumPy, Pandas
Visualization	Matplotlib, Seaborn
ML	Scikit-learn
Deep Learning	TensorFlow, Keras, PyTorch
Computer Vision	OpenCV
NLP	NLTK, spaCy, Transformers
API	FastAPI, Flask
Database	MySQL, PostgreSQL, MongoDB
Vector Search	Vector databases
Deployment	Docker, Cloud platforms
Version Control	Git, GitHub
Development	VS Code, Jupyter Notebook

🎯 Module 15 Interview Questions

Basic

1. What is an ML project?
2. Explain the complete ML lifecycle.
3. What is a feature?
4. What is a target variable?
5. Why is data preprocessing important?
6. What is train-test split?
7. Why do we perform EDA?
8. What is model evaluation?
9. What is overfitting?
10. Why is model deployment required?

Intermediate

11. How would you build a house-price prediction system?
12. How would you detect customer churn?
13. How does a recommendation system work?
14. How would you handle imbalanced data?
15. How would you deploy an ML model?
16. What is the difference between an ML model and an API?
17. How would you monitor a deployed model?
18. How does a RAG system work?
19. Why are embeddings used in RAG?
20. How would you build an AI assistant for a learning platform?

Advanced

21. How would you prevent data leakage?
22. How would you select the best ML algorithm?
23. How would you perform error analysis?
24. How would you handle model drift?
25. How would you design an end-to-end production ML system?

📄 Module 15 Practice Assignment

Choose **one** project and build it completely:

Beginner

House Price Prediction

Intermediate

Customer Churn Prediction

NLP

Spam Detection

Recommendation

Movie Recommendation System

Unsupervised ML

Customer Segmentation

Computer Vision

Image Classification

Generative AI

RAG-Based Learning Assistant

Final Project

RAG-Based AI Learning Assistant

Features:

PDF / Notes



Document Processing



Chunking



Embeddings



Vector Database



Retriever



LLM



Chatbot

This project combines **ML + NLP + Transformers + Embeddings + RAG + APIs + Deployment**, making it a strong final project for an industrial-oriented AI/ML learning path.